

1
2
3
4

Draft standard P802[®]
(Revision of IEEE Std 802[®]-2014 as amended by
IEEE Std 802d[™]-2017, IEEE Std 802c[™]-2017
and IEEE Std 802f[™]-2023)

5

6 **Draft Standard for Local and** 7 **Metropolitan Area Networks:** 8 **Overview and Architecture**

9 Sponsor

10 **LAN/MAN Standards Committee**
11 **of the**
12 **IEEE Computer Society**
13

Important Notice

This document is an unapproved draft of a proposed IEEE Standard. IEEE hereby grants the named IEEE SA Working Group or Standards Committee Chair permission to distribute this document to participants in the receiving IEEE SA Working Group or Standards Committee, for purposes of review for IEEE standardization activities. No further use, reproduction, or distribution of this document is permitted without the express written permission of IEEE Standards Association (IEEE SA). Prior to any review or use of this draft standard, in part or in whole, by another standards development organization, permission must first be obtained from IEEE SA (stds-copyright@ieee.org). This page is included as the cover of this draft, and shall not be modified or deleted.

IEEE Standards Association
445 Hoes Lane
Piscataway, NJ 08854, USA

Abstract: This standard provides an overview to the family of IEEE 802[®] standards. It describes the reference models for the IEEE 802 standards and explains the relationship of these standards to the higher layer protocols; it specifies the structure of IEEE 802 MAC addresses; it specifies identification of public, private, prototype, and standard protocols; it specifies an object identifier hierarchy used within IEEE 802 for uniform allocation of object identifiers used in IEEE 802 standards; and it specifies a method for higher layer protocol identification.

Keywords: BANs, body area networks, EtherTypes, IEEE 802[®], IEEE 802 architecture, IEEE 802 reference model, LANs, local area networks, MANs, metropolitan area networks, object identifiers, PANs, personal area networks, RANs, regional area networks, protocol types

The Institute of Electrical and Electronics Engineers, Inc.
3 Park Avenue, New York, NY 10016-5997, USA

Copyright © 2024 by The Institute of Electrical and Electronics Engineers, Inc.
All rights reserved. Published xx Month 20xx. Printed in the United States of America.

IEEE and 802 are registered trademarks in the U.S. Patent & Trademark Office, owned by The Institute of Electrical and Electronics Engineers, Incorporated.

Object Management Group[®], OMG[®], UML[®] and Unified Modeling Language[™] are either registered trademarks or trademarks of Object Management Group, Inc. in the United States and/or other countries.

PDF: ISBN 978-0-7381-9219-2 STD98723
Print: ISBN 978-0-7381-9220-8 STDPD98723

IEEE prohibits discrimination, harassment, and bullying.

For more information, visit <http://www.ieee.org/web/aboutus/whatis/policies/p9-26.html>.

No part of this publication may be reproduced in any form, in an electronic retrieval system or otherwise, without the prior written permission of the publisher.

1 Important Notices and Disclaimers Concerning IEEE Standards Documents

2 IEEE documents are made available for use subject to important notices and legal disclaimers. These notices
3 and disclaimers, or a reference to this page (<https://standards.ieee.org/ipr/disclaimers.html>), appear in all
4 standards and may be found under the heading “Important Notice” or “Important Notices and Disclaimers
5 Concerning IEEE Standards Documents.”

6 Notice and Disclaimer of Liability Concerning the Use of IEEE Standards 7 Documents

8 IEEE Standards documents are developed within IEEE Societies and subcommittees of IEEE Standards
9 Association (IEEE SA) Board of Governors. IEEE develops its standards through an accredited consensus
10 development process, which brings together volunteers representing varied viewpoints and interests to
11 achieve the final product. IEEE Standards are documents developed by volunteers with scientific, academic,
12 and industry-based expertise in technical working groups. Volunteers are not necessarily members of IEEE
13 or IEEE SA and participate without compensation from IEEE. While IEEE administers the process and
14 establishes rules to promote fairness in the consensus development process, IEEE does not independently
15 evaluate, test, or verify the accuracy of any of the information or the soundness of any judgments contained
16 in its standards.

17 IEEE makes no warranties or representations concerning its standards, and expressly disclaims all
18 warranties, express or implied, concerning this standard, including but not limited to the warranties of
19 merchantability, fitness for a particular purpose and non-infringement. In addition, IEEE does not warrant or
20 represent that the use of the material contained in its standards is free from patent infringement. IEEE
21 standards documents are supplied “AS IS” and “WITH ALL FAULTS.”

22 Use of an IEEE standard is wholly voluntary. The existence of an IEEE standard does not imply that there
23 are no other ways to produce, test, measure, purchase, market, or provide other goods and services related to
24 the scope of the IEEE standard. Furthermore, the viewpoint expressed at the time a standard is approved and
25 issued is subject to change brought about through developments in the state of the art and comments
26 received from users of the standard.

27 In publishing and making its standards available, IEEE is not suggesting or rendering professional or other
28 services for, or on behalf of, any person or entity nor is IEEE undertaking to perform any duty owed by any
29 other person or entity to another. Any person utilizing any IEEE Standards document, should rely upon his
30 or her own independent judgment in the exercise of reasonable care in any given circumstances or, as
31 appropriate, seek the advice of a competent professional in determining the appropriateness of a given IEEE
32 standard.

33 IN NO EVENT SHALL IEEE BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL,
34 EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO:
35 PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR
36 BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY,
37 WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR
38 OTHERWISE) ARISING IN ANY WAY OUT OF THE PUBLICATION, USE OF, OR RELIANCE
39 UPON ANY STANDARD, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE AND
40 REGARDLESS OF WHETHER SUCH DAMAGE WAS FORESEEABLE.

41

42

1 Translations

2 The IEEE consensus development process involves the review of documents in English only. In the event
3 that an IEEE standard is translated, only the English version published by IEEE should be considered the
4 approved IEEE standard.

5 Official statements

6 A statement, written or oral, that is not processed in accordance with the IEEE-SA Standards Board
7 Operations Manual shall not be considered or inferred to be the official position of IEEE or any of its
8 committees and shall not be considered to be, or be relied upon as, a formal position of IEEE. At lectures,
9 symposia, seminars, or educational courses, an individual presenting information on IEEE standards shall
10 make it clear that the presenter's views should be considered the personal views of that individual rather
11 than the formal position of IEEE, IEEE SA, the Standards Committee, or the Working Group. Statements
12 made by volunteers may not represent the formal position of their employer(s) or affiliation(s).

13 Comments on standards

14 Comments for revision of IEEE Standards documents are welcome from any interested party, regardless of
15 membership affiliation with IEEE or IEEE SA. However, **IEEE does not provide interpretations,**
16 **consulting information or advice pertaining to IEEE Standards documents.**

17 Suggestions for changes in documents should be in the form of a proposed change of text, together with
18 appropriate supporting comments. Since IEEE standards represent a consensus of concerned interests, it is
19 important that any responses to comments and questions also receive the concurrence of a balance of
20 interests. For this reason, IEEE and the members of its Societies and subcommittees of the IEEE SA Board
21 of Governors are not able to provide an instant response to comments, or questions except in those cases
22 where the matter has previously been addressed. For the same reason, IEEE does not respond to
23 interpretation requests. Any person who would like to participate in evaluating comments or in revisions to
24 an IEEE standard is welcome to join the relevant IEEE working group. You can indicate interest in a
25 working group using the Interests tab in the Manage Profile & Interests area of the [IEEE SA myProject](#)
26 [system](#).¹ An IEEE Account is needed to access the application.

27 Comments on standards should be submitted using the Contact Us form.²

28 Laws and regulations

29 Users of IEEE Standards documents should consult all applicable laws and regulations. Compliance with the
30 provisions of any IEEE Standards document does not imply compliance to any applicable regulatory
31 requirements. Implementers of the standard are responsible for observing or referring to the applicable
32 regulatory requirements. IEEE does not, by the publication of its standards, intend to urge action that is not
33 in compliance with applicable laws, and these documents may not be construed as doing so.

34 Data privacy

35 Users of IEEE Standards documents should evaluate the standards for considerations of data privacy and
36 data ownership in the context of assessing and using the standards in compliance with applicable laws and
37 regulations.

¹Available at: <https://development.standards.ieee.org/myproject-web/public/view.html#landing>.

²Available at: <https://standards.ieee.org/content/ieee-standards/en/about/contact/index.html>.

1 Copyrights

2 IEEE draft and approved standards are copyrighted by IEEE under US and international copyright laws.
3 They are made available by IEEE and are adopted for a wide variety of both public and private uses. These
4 include both use, by reference, in laws and regulations, and use in private self-regulation, standardization,
5 and the promotion of engineering practices and methods. By making these documents available for use and
6 adoption by public authorities and private users, neither IEEE or its licensors waive any rights in copyright
7 to the documents.

8 Photocopies

9 Subject to payment of the appropriate licensing fees, IEEE will grant users a limited, non-exclusive license
10 to photocopy portions of any individual standard for company or organizational internal use or individual,
11 non-commercial use only. To arrange for payment of licensing fees, please contact Copyright Clearance
12 Center, Customer Service, 222 Rosewood Drive, Danvers, MA 01923 USA; +1 978 750 8400; [https://](https://www.copyright.com/)
13 www.copyright.com/. Permission to photocopy portions of any individual standard for educational
14 classroom use can also be obtained through the Copyright Clearance Center.

15 Updating of IEEE Standards documents

16 Users of IEEE Standards documents should be aware that these documents may be superseded at any time
17 by the issuance of new editions or may be amended from time to time through the issuance of amendments,
18 corrigenda, or errata. An official IEEE document at any point in time consists of the current edition of the
19 document together with any amendments, corrigenda, or errata then in effect.

20 Every IEEE standard is subjected to review at least every 10 years. When a document is more than 10 years
21 old and has not undergone a revision process, it is reasonable to conclude that its contents, although still of
22 some value, do not wholly reflect the present state of the art. Users are cautioned to check to determine that
23 they have the latest edition of any IEEE standard.

24 In order to determine whether a given document is the current edition and whether it has been amended
25 through the issuance of amendments, corrigenda, or errata, visit [IEEE Xplore](#) or [contact IEEE](#).³ For more
26 information about the IEEE SA or IEEE's standards development process, visit the IEEE-SA Website.

27 Errata

28 Errata, if any, for all IEEE standards can be accessed on the [IEEE SA Website](#).⁴ Search for standard number
29 and year of approval to access the web page of the published standard. Errata links are located under the
30 Additional Resources Details section. Errata are also available in [IEEE Xplore](#). Users are encouraged to
31 periodically check for errata .

32 Patents

33 IEEE Standards are developed in compliance with the [IEEE SA Patent Policy](#).⁵

34 Attention is called to the possibility that implementation of this standard may require use of subject matter
35 covered by patent rights. By publication of this standard, no position is taken by the IEEE with respect to the

³Available at: <https://ieeexplore.ieee.org/browse/standards/collection/ieee>.

⁴Available at: <https://standards.ieee.org/standard/index.html>.

⁵Available at: <https://standards.ieee.org/about/sasb/patcom/materials.html>.

1 existence or validity of any patent rights in connection therewith. If a patent holder or patent applicant has
2 filed a statement of assurance via an Accepted Letter of Assurance, then the statement is listed on the IEEE-
3 SA Website at <http://standards.ieee.org/about/sasb/patcom/patents.html>. Letters of Assurance may indicate
4 whether the Submitter is willing or unwilling to grant licenses under patent rights without compensation or
5 under reasonable rates, with reasonable terms and conditions that are demonstrably free of any unfair
6 discrimination to applicants desiring to obtain such licenses.

7 Essential Patent Claims may exist for which a Letter of Assurance has not been received. The IEEE is not
8 responsible for identifying Essential Patent Claims for which a license may be required, for conducting
9 inquiries into the legal validity or scope of Patents Claims, or determining whether any licensing terms or
10 conditions provided in connection with submission of a Letter of Assurance, if any, or in any licensing
11 agreements are reasonable or non-discriminatory. Users of this standard are expressly advised that
12 determination of the validity of any patent rights, and the risk of infringement of such rights, is entirely their
13 own responsibility. Further information may be obtained from the IEEE Standards Association.

14 **IMPORTANT NOTICE**

15 IEEE Standards do not guarantee or ensure safety, security, health, or environmental protection, or ensure
16 against interference with or from other devices or networks. IEEE Standards development activities consider
17 research and information presented to the standards development group in developing any safety
18 recommendations. Other information about safety practices, changes in technology or technology
19 implementation, or impact by peripheral systems also may be pertinent to safety considerations during
20 implementation of the standard. Implementers and users of IEEE Standards documents are responsible for
21 determining and complying with all appropriate safety, security, environmental, health, and interference
22 protection practices and all applicable laws and regulations.

1 Participants

2 At the time this standard was completed, the IEEE 802.1 Working Group had the following membership:

3 **Glenn Parsons, *Chair***
4 **Jessy Rouyer, *Vice Chair***
5 **Jessy Rouyer, *Recording Secretary***
6 **James P. K. Gilb, *802-REVc Technical Editor***

7 In addition to the members of the IEEE 802.1 Working Group, significant contributions were received from
8 the following individuals:

¹ The following members of the individual balloting committee voted on this standard. Balloters may have
² voted for approval, disapproval, or abstention.

1 Historical participants

2 When the IEEE Std 802-1990 was approved on 31 May 1990, the IEEE 802.1 Working Group had the
3 following officer:

4 **William P. Lidinsky, *Chair***

5 When the IEEE Std 802-2001 was approved on 6 December 2001, the IEEE 802.1 Working Group had the
6 following officers:

7 **William P. Lidinsky, *Chair***
8 **Tony Jeffree, *Vice Chair and Editor***
9 **Alan Chambers, Tony Jeffree, *Editors***

10 When the IEEE Std 802a-2003 was approved on 12 June 2003, the IEEE 802.1 Working Group had the
11 following officers:

12 **Tony Jeffree, *Chair and Editor***
13 **Neil Jarvis, *Vice Chair***

14 When the IEEE Std 802b-2004 was approved on 25 March 2004, the IEEE 802.1 Working Group had the
15 following officers:

16 **Tony Jeffree, *Chair and Editor***
17 **Neil Jarvis, *Vice Chair***

18 When the IEEE Std 802-2014 was approved on 12 June 2014, the IEEE 802.1 Working Group had the
19 following officers:

20 **Glenn Parsons, *Chair***
21 **John Messenger, *Vice Chair***
22 **Eric Gray, *Recording Secretary***
23 **James P. K. Gilb, *IEEE 802-2014 Technical Editor***
24

25 When the IEEE Std 802c-2017 was approved on 15 June 2017, the IEEE 802.1 Working Group had the
26 following officers:

27 **Glenn Parsons, *Chair***
28 **John Messenger, *Vice Chair***
29 **Pat Thaler, *Chair, Data Center Bridging Task Group***
30 **Roger B Marks, *IEEE 802c Technical Editor***
31

32 When the IEEE Std 802d-2017 was approved on 14 February 2017, the IEEE 802.1 Working Group had the
33 following officers:

34 **Glenn Parsons, *Chair***
35 **John Messenger, *Vice Chair***
36 **János Farkas, *Chair, Time-Sensitive Networking Task Group***
37 **Tony Jeffree, *Editor***
38

39 The following individuals participated in the IEEE 802.1 working group during various stages of the
40 standard's development. Since the initial publication, many IEEE standards have added functionality or
41 provided updates to material included in this standard. The following is a historical list of participants who
42 have dedicated their valuable time, energy, and knowledge to the creation of this standard:

Steve Adams	Paul Congdon	Mogens Hansen
Fumio Akashi	Glenn Connery	Harold Harrington
Paul D. Amer	Jim Corrigan	John Hart
Charles Arnold	Paul Cowell	Mike Harvey
Floyd Backes	David Cullerot	Richard Hausman
Ann Ballard	Ted Davies	David Head
Richard Bantel	Peter Dawe	Deepak Hegde
John Bartlett	Stan Degen	Ariel Hendel
Sy Bederman	Fred Deignan	Bob Herbst
Les Bell	David Delaney	Steve Horowitz
Amatzia Ben-Artzi	Ron Dhondy	Robert W. Hott
Michael Berger	Jeffrey Dietz	Jack R. Hung
James S. Binder	Eiji Doi	Altaf Hussain
Robert Bledsoe	Barbara J. Don Carlos	Thomas Hytry
Kwame Boakye	Peter Ecclesine	Ran Ish-Shalom
Paul Bottorff	J. J. Ekstrom	Jay Israel
Laura Bridge	Hesham Elbakoury	Vipin K. Jain
Juan Bulnes	Walder Eldon	Neil Jarvis
Bill Bunch	Norman W. Finn	Tony Jeffree
Fred Burg	David Frattura	Shyam Kaluve
Jim Burns	Lars Henrik Frederiksen	Toyoyuki Kato
Peter Carbone	Eldon D. Feist	Hal Keen
Paul Carroll	Len Fishler	Kevin Ketchum
Jeffrey Catlin	Kevin Flanagan	Alan Kirby
Dirceu Cavendish	Anoop Ghanwani	Kimberly Kirkpatrick
Alan Chambers	James Gilb	Keith Klamm
David W. Chang	Pat Gonia	Steve Kleiman
Ken Chapman	Gerard Goubert	Bruce Kling
Alice Chen	Richard Graham	Dan Krent
Jade Chien	Michael A. Gravel	
Hon Wah Chin	Steve Haddock	
Chris Christ	Sharam Hakimi	

James Kristof	Dinel Pitt	Lennart Swartz
H. Eugene Latham	Ron L. G. Prince	Kazuo Takagi
Bing Liao	Steve Ramberg	Kenta Takumi
William P. Lidinsky	Nigel Ramsden	Robin Tasker
George Lin	Shlomo Reches	Angus Telfer
Paul Lachapelle	Frank Reichstein	Pat Thaler
Bill Lane	Trudy Reusser	Dave Thompson
Paul Langille	James Richmond	Geoffrey O. Thompson
Roger Lapuh	Anil Rijasinghani	Michel Thorsen
Loren Larsen	Eduoard Rocher	Nathan Tobol
Johann Lindmeyr	John Roesse	Wendell Turner
Andy Luque	Allyn Romanow	Steve Van Seters
Philip Magnuson	Dan Romascanu	Dono van-Mierop
Bruce McClure	Paul Rosenblum	Paul Videcrantz
Tom McGowan	Dolors Sala	Dennis Volpano
Milan Merhar	John Salter	Paul Wainright
Margaret A. Merrick	Alan Sarsby	John Wakerly
John Messenger	Ayman Sayed	Peter Wang
Colin Mick	Susan Schannning	Y. C. Wang
Dinesh Mohan	Mick Seaman	Trevor Warwick
John Montrose	Gerry Segal	Scott Wasson
Bob Moskowitz	Rich Seifert	Daniel Watts
Yaron Nachman	Lee Sendelbach	Karl Weber
Krishna Narayanaswamy	Himanshu Shah	Alan Weissberger
Lawrence Ng	Howard Sherry	Deborah Wilbert
Henry Ngai	Wu-Shi Shung	Keith Willette
Satoshi Obara	Phil Simmons	Michael Witkowski
Don O'Connor	Curtis Simonson	Edward Wong
Jerry O'Keefe	Paramjeet Singh	Michael D. Wright
Toshio Ooka	Rosemary V. Slager	Michele Wright
Jorg Ottensmeyer	Alexander Smith	Allen Yu
Richard Patti	Andrew Smith	Wayne Zakowski
Luc Pariseau	M. Soha	Igor Zhovnirovosky
Glenn Parsons	Larry Stefani	Carolyn Zimmer
Roger Pfister	Dan Stokesberry	Nick Zuccherro
Thomas L. Phinney	Stuart Soloway	
John Pickens	Sundar Subramaniam	

1 Introduction

1 This introduction is not part of IEEE Std 802-20xx, IEEE Standard for Local and Metropolitan Area Networks
2 Overview and Architecture.

3

4 This document is the third major revision of the IEEE 802[®] overview and architecture. This revision
5 integrates into the previous revision of the standard, IEEE Std 802[®]-2014, the three subsequent
6 amendments:

- 7 — IEEE Std 802d[™]-2017: Specifying a Uniform Resource Name (URN) root identifier for YANG data
8 models
- 9 — IEEE Std 802c[™]-2017: Specifying an optional local medium access control (MAC) address space
10 structure
- 11 — IEEE Std 802f[™]-2023: Specifying a YANG module that contains EtherType information

12

1 Contents

2	1.	Overview.....	18
3	1.1	Scope.....	18
4	1.2	Purpose.....	18
5	1.3	Word usage	18
6	2.	Normative references.....	20
7	3.	Definitions, acronyms and abbreviations.....	21
8	3.1	Definitions	21
9	3.2	Acronyms and abbreviations	23
10	4.	Family of IEEE 802 standards	26
11	4.1	Key concepts.....	26
12	4.2	Application and support.....	27
13	4.3	An international family of standards	28
14	4.4	IEEE 802 standards.....	28
15	5.	Reference models (RMs)	30
16	5.1	Introduction.....	30
17	5.2	RM description for end stations.....	31
18	5.2.1	SAPs.....	32
19	5.2.2	LLC sublayer	32
20	5.2.3	MAC sublayer.....	32
21	5.2.4	PHY	33
22	5.2.5	Layer and sublayer management	33
23	5.3	Interconnection and interworking.....	33
24	5.3.1	Interconnection at the PHY.....	34
25	5.3.2	MAC-sublayer interconnection: Bridges.....	34
26	5.3.3	Network-layer interconnection: Routers.....	37
27	6.	General requirements for an IEEE 802 network.....	39
28	6.1	Services supported	39
29	6.2	Error ratios	39
30	7.	IEEE 802 network management	40
31	7.1	General.....	40
32	7.2	General-purpose IEEE 802 network management.....	40
33	7.2.1	Management functions.....	40
34	7.2.2	Management architecture.....	40
35	7.2.3	Managed object definitions.....	41
36	7.3	Special-purpose IEEE 802 network management standards	41
37	8.	MAC addresses	42
38	8.1	Terms and notational conventions	42
39	8.2	Universal addresses.....	42

1	8.2.1	Concept and overview	42
2	8.2.2	Assignment of universal addresses	43
3	8.2.3	Assignment by organizations.....	45
4	8.2.4	Uniqueness of address assignment	45
5	8.3	Interworking with 48-bit and 64-bit MAC addresses	45
6	8.4	Local MAC addresses	46
7	8.4.1	Concept and overview	46
8	8.4.2	Local MAC address assignment protocols	46
9	8.4.3	Structured Local Address Plan (SLAP)	46
10	8.4.4	SLAP local identifier types.....	48
11	8.4.5	Network Unique Identifier (NUI).....	51
12	8.5	Group MAC addresses.....	51
13	8.6	Bit-ordering within octets	51
14	9.	Protocol identifiers and context-dependent identifiers	52
15	9.1	Introduction.....	52
16	9.2	EtherTypes and E-Type protocol identifiers.....	53
17	9.2.1	Format, function, and administration.....	53
18	9.2.2	Public EtherType assignments subset.....	53
19	9.2.3	EtherType sub-protocol encoding.....	54
20	9.2.4	Local Experimental EtherTypes	54
21	9.3	LSAP addresses and L-Type protocol identifiers	55
22	9.4	O-Type protocol identifiers	56
23	9.5	PIF Encoding	56
24	9.5.1	Type 2 PIF encoding.....	56
25	9.5.2	Type 3 PIF encoding.....	57
26	9.5.3	Encoding type and PIF length.....	59
27	9.6	Context-dependent identifiers.....	59
28	10.	Allocation of OID values in IEEE 802 standards	60
29	10.1	General.....	60
30	10.2	OIDs and ISO standards	60
31	10.3	The OID hierarchy for IEEE 802 standards.....	61
32	10.4	The OID hierarchy under iso(1) std(0) iso8802(8802).....	62
33	10.5	Migration from previous OID allocations	62
34	11.	Allocation of Uniform Resource Name (URN) values in IEEE 802 standards.....	63
35	11.1	Introduction.....	63
36	11.2	The IEEE Namespace ID and Namespace Specific String.....	63
37	11.3	ResourceSpecificString values in IEEE 802 standards.....	63
38	Annex A	Bibliography	65
39	Annex B	Reference Models for IEEE 802 standards.....	67
40	B.1	IEEE 802.3 RMs.....	67
41	B.2	IEEE 802.11 RM.....	69
42	B.3	IEEE 802.15 TM RMs	71
43	B.3.1	IEEE 802.15.3 TM RM.....	71
44	B.3.2	IEEE 802.15.4 TM RM.....	73
45	B.3.3	IEEE 802.15.6 TM RM.....	73

1	B.3.4	IEEE 802.15.7™ RM.....	73
2	B.4	IEEE 802.16™ RM.....	74
3	B.4.1	Protocol RM.....	74
4	B.4.2	Network RM	75
5	B.5	IEEE 802.21™ RM.....	76
6	B.6	IEEE 802.22™ RM.....	77
7	B.6.1	Data plane	78
8	B.6.2	Management/control plane	78
9	B.6.3	Cognitive plane	78
10		Annex C Examples of bit ordering for addresses	79
11		Annex D List of IEEE 802 standards.....	80
12		Annex E History	82
13	E.1	Universal addresses.....	82
14	E.2	IEEE RA address block products.....	82
15	E.3	Local MAC addresses	83
16		Annex F EtherType Listing Subset.....	84
17	F.1	Introduction.....	84
18	F.2	Tabular format	84
19	F.3	YANG module for EtherType subset	87
20	F.3.1	YANG Framework	87
21	F.3.2	Definition for ieee802-ethertype YANG module'	88
22		Annex G Wake-on-LAN.....	100
23			
24			
25			
26			
27			
28			
29			
30			
31			
32			
33			
34			

1

2

P802 IEEE *draft* Standard for Local and Metropolitan Area Networks: Overview and Architecture

IMPORTANT NOTICE: IEEE Standards documents are not intended to ensure safety, security, health, or environmental protection, or ensure against interference with or from other devices or networks. Implementers of IEEE Standards documents are responsible for determining and complying with all appropriate safety, security, environmental, health, and interference protection practices and all applicable laws and regulations.

This IEEE document is made available for use subject to important notices and legal disclaimers. These notices and disclaimers appear in all publications containing this document and may be found under the heading “Important Notice” or “Important Notices and Disclaimers Concerning IEEE Documents.” They can also be obtained on request from IEEE or viewed at <http://standards.ieee.org/IPR/disclaimers.html>.

1. Overview

1.1 Scope

This standard contains descriptions of the IEEE 802[®] standards published by the IEEE for frame-based data networks as well as a reference model (RM) for protocol standards. A specification for the identification of public, private, and standard protocols is included.

1.2 Purpose

This standard serves as the foundation for the family of IEEE 802 standards published by IEEE for local area networks (LANs), metropolitan area networks (MANs), personal area networks (PANs), and regional area networks (RANs), etc.

1.3 Word usage

The word *shall* indicates mandatory requirements strictly to be followed in order to conform to the standard and from which no deviation is permitted (*shall* equals *is required to*).

The word *should* indicates that among several possibilities one is recommended as particularly suitable, without mentioning or excluding others; or that a certain course of action is preferred but not necessarily required (*should* equals *is recommended that*).¹

¹The use of the word *must* is deprecated and cannot be used when stating mandatory requirements, *must* is used only to describe unavoidable situations.

¹ The word *may* is used to indicate a course of action permissible within the limits of the standard (*may* equals *is permitted to*).²

³ The word *can* is used for statements of possibility and capability, whether material, physical, or causal (*can* equals *is able to*).

²The use of *will* is deprecated and cannot be used when stating mandatory requirements, *will* is only used in statements of fact.

2. Normative references

The following referenced documents are indispensable for the application of this document (i.e., they must be understood and used; therefore, each referenced document is cited in text and its relationship to this document is explained). For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments or corrigenda) applies.

IEEE Std 802.1Q™, IEEE Standard for Local and metropolitan area networks—Bridges and Bridged Networks.^{3, 4}

IEEE Std 802.1AC™, IEEE Standard for Local and metropolitan area networks—Media Access Control (MAC) Service Definition.

IETF RFC 2578, Structure of Management Information Version 2 (SMIv2).⁵

IETF RFC 3406, Uniform Resource Names (URN) Namespace Definition Mechanisms, October 2002.

IETF RFC 8069, URN Namespace for IEEE, February 2017.

ISO/IEC/IEEE 8802-2:1998, International Standard for Information technology—Telecommunications and information exchange between systems—Local and metropolitan area networks—Specific requirements—Part 2: Logical link control.⁶

ITU-T Recommendation X.660, Information technology—Procedures for the operation of object identifier registration authorities: General procedures and top arcs of the international object identifier tree.⁷

18

³The IEEE standards referred to in Clause 2 are trademarks owned by The Institute of Electrical and Electronics Engineers, Incorporated.

⁴IEEE publications are available from The Institute of Electrical and Electronics Engineers (<https://standards.ieee.org/>).

⁵IETF documents (i.e., RFCs) are available the Internet Engineering Task Force (<https://www.rfc-archive.org/>).

⁶ISO/IEC publications are available from the International Organization for Standardization (<https://www.iso.org/standards.html>) and the International Electrotechnical Commission (<https://www.iec.ch/>). ISO/IEC publications are also available in the United States from the American National Standards Institute (<https://www.ansi.org/>).

⁷ITU-T publications are available from the International Telecommunications Union (<https://www.itu.int/>).

1 3. Definitions, acronyms and abbreviations

2 3.1 Definitions

3 For this document, the following terms and definitions apply. *The IEEE Standards Dictionary Online*⁸
4 should be consulted for terms not defined in this clause.⁹

5 **access domain:** A set of stations in an IEEE 802[®] network together with interconnecting data transmission
6 media and functional units (e.g., repeaters), in which the stations use the same medium access control
7 (MAC) protocol to access a common communications channel to exchange information.

8 **bridge:** In the general sense, a functional unit that interconnects two or more access domains. In the context
9 of IEEE Std 802, this is narrowed to interconnecting two or more bridgeable IEEE 802[®] networks that use
10 the same data link layer (DLL) protocols above the medium access control (MAC) sublayer, but can use
11 different MAC protocols. Forwarding and filtering decisions are made on the basis of Layer 2 information.

12 **bridgeable network:** A communication resource that provides the medium access control (MAC) service
13 specified in IEEE Std 802.1AC, between two or more MAC service access points (MSAPs), supporting the
14 MAC Internal Sublayer Service.

15 **end station:** A functional unit in an IEEE 802[®] network that acts as a source of, and/or destination for, link
16 layer data traffic carried on the network.

17 **Ethernet:** A communication protocol specified by IEEE Std 802.3[™].

18 **EtherType:** A 2-octet value assigned by the IEEE Registration Authority (RA) that provides data field
19 context for medium access control (MAC) frame interpretation (protocol identification).

20 **Extended Unique Identifier (EUI):** A 48-bit or 64-bit identifier intended to be globally unique and sub-
21 assigned, by the assignee of a block of such identifiers assigned by the IEEE Registration Authority, to a
22 hardware device instance or other object that requires unique identification.

23 **filtering:** A function in a bridge that is used to determine if a received medium access control (MAC) frame
24 is to be forwarded or discarded on any given outbound port.

25 **forwarding:** A function in a bridge that transfers a received medium access control (MAC) frame to one or
26 more outbound ports.

27 **handover:** A process by which a mobile node obtains facilities and preserves traffic flows when traffic is
28 switched from one link to another.

29 **interworking:** The use of interconnected stations in a network for the exchange of data, by means of
30 protocols operating over the underlying data transmission paths.

31 **local area network (LAN):** A network of devices, whether indoors or outdoors, covering a limited
32 geographic area, e.g., a building or campus.

⁸The IEEE Standards Dictionary Online is not a dictionary but rather is a compendium of balloted definitions from individual approved standards.

⁹The *IEEE Standards Dictionary Online* is available at <https://dictionary.ieee.org>. An IEEE Account is required for access to the dictionary, and one can be created at no charge on the dictionary sign-in page.

1 **local medium access control (MAC) address:** A medium access control (MAC) address with the
2 universally or locally administered (U/L) bit set to one.

3 **logical link:** A logical communication connection between two devices.

4 **logical link control (LLC) entity:** The instantiation of an active element embodying LLC-specific
5 capabilities at a single station.

6 **logical link control (LLC) protocol data unit (LPDU):** An LLC protocol data unit carried in the data field
7 of a frame as an medium access control (MAC) service data unit (MSDU).

8 **medium access control (MAC) entity:** The instantiation of an active element embodying MAC-specific
9 capabilities at a single station.

10 **medium access control (MAC) frame:** A data structure consisting of fields in accordance with a MAC
11 protocol, for the communication of user data and/or control information in a network.

12 **medium access control (MAC) protocol:** A protocol that governs access to the transmission medium in a
13 network, to enable the exchange of data between stations in a network.

14 **medium access control (MAC) service data unit (MSDU):** Data sent within a MAC frame by a MAC
15 entity for delivery to one or more MAC entities.

16 **media-independent control function:** A parallel control plane that provides control functions for different
17 medium access control (MAC) and physical layer (PHY) sublayers and provides a media-independent
18 abstraction to higher layer protocols.

19 **media-independent handover function:** A function that provides the ability to relocate traffic flows
20 between different medium access technologies and associated physical media.

21 **metropolitan area network (MAN):** A network of devices, extending over a large geographical area such
22 as an urban area, often providing integrated communication services such as data, voice, and video.

23 **Network Unique Identifier (NUI):** An identifier that is unique within the IEEE 802[®] local area network
24 (LAN).

25 **personal area network (PAN):** A network of devices extending over a region generally smaller than about
26 ten meters, used to convey information among a group of participant stations.

27 **private protocol:** A protocol whose use and specification are controlled by a private organization.

28 **protocol data unit:** A unit of information transmitted among peer instances of a layer or sublayer
29 comprising protocol-specific control information and user data.

30 **public protocol:** A protocol whose specification is published and known to the public, but controlled by an
31 organization other than a formal standards body.

32 **regional area network (RAN):** A network of devices that generally covers a service area that is larger than
33 metropolitan area networks (MANs), typically in sparsely populated areas.

34 **repeater:** A device used to interconnect segments of the physical communications media, for example, to
35 extend the range of a network when the physical specifications of the technology would otherwise be
36 exceeded, while providing a one access domain for the attached stations.

1 **standard protocol:** A protocol whose specification is published and known to the public and is controlled
2 by a standards body.

3 **station:** An end station or bridge. *See also:* **bridge; end station.**

4 **Structured Local Address Plan (SLAP):** An optional standardized specification to differentiate local
5 medium access control (MAC) addresses.

6 **universal address:** A 48-bit Extended Unique Identifier (EUI-48) or 64-bit Extended Unique Identifier
7 (EUI-64) that is used as a unique address.

8 3.2 Acronyms and abbreviations

9	AAI	Administratively Assigned Identifier
10	AAI-48	48-bit AAI
11	AAI-64	64-bit AAI
12	AN	auto negotiation
13	BAN	body area network
14	BS	base station
15	BSS	basic service set
16	CID	Company ID
17	CGMII	100 Gb/s media-independent interface
18	CPE	customer-premises equipment
19	CS	convergence sublayer
20	C-SAP	control service access point
21	DCN	data center networking
22	DLL	data link layer
23	ELI	Extended Local Identifier
24	ELI-48	48-bit ELI
25	ELI-64	64-bit ELI
26	EPON	Ethernet passive optical networks
27	EUI	Extended Unique Identifier
28	EUI-48	48-bit EUI
29	EUI-64	64-bit EUI
30	FEC	forward error correction
31	HLPDE	higher layer protocol discrimination entity
32	IANA	Internet Assigned Numbers Authority ¹⁰
33	IEEE RA	Institute of Electrical and Electronic Engineers Registration Authority
34	IEEE RAC	Institute of Electrical and Electronic Engineers Registration Authority Committee
35	I/G	individual/group
36	IM	implementation model
37	IoT	Internet of things
38	IP	Internet Protocol
39	LAN	local area network
40	LLC	logical link control
41	LPDU	logical link control protocol data unit

¹⁰<https://www.iana.org>

1	LSAP	link service access point
2	LSB	least significant bit
3	LMSC	Local Area Network/Metropolitan Area Network (LAN/MAN) Standards Committee
4	MAC	medium access control, media access control ¹¹
5	MA-L	MAC Address Block Large
6	MA-M	MAC Address Block Medium
7	MAN	metropolitan area network
8	MA-S	MAC Address Block Small
9	MCPS	MAC common part sublayer
10	MCPS-SAP	MAC common part sublayer data service access point
11	MDI	medium dependent interface
12	MIB	management information base
13	MICF	media-independent control function
14	MICLSAP	media-independent control link service access point
15	MICPSAP	media-independent control physical service access point
16	MICSAP	media-independent control service access point
17	MIH	media-independent handover
18	MIHF	media-independent handover function
19	MLME	MAC sublayer management entity
20	MSAP	MAC service access point
21	M-SAP	management service access point
22	MSB	most significant bit
23	MSDU	medium access control protocol data unit
24	MSTP	multiple spanning tree protocol
25	NCMS	network control and management system
26	NETCONF	Network Configuration Protocol
27	NID	Namespace Identifier
28	NUI	Network Unique Identifier
29	NUI-48	48-bit NUI
30	NUI-64	64-bit NUI
31	OAM	operations, administration, and maintenance
32	OID	object identifier
33	OLT	optical line terminal
34	OMG®	Object Management Group®
35	ONU	optical network unit
36	OSI/RM	Open Systems Interconnection basic reference model
37	OUI	Organizationally Unique Identifier
38	PAN	personal area network
39	PCS	physical coding sublayer
40	PDU	protocol data unit
41	PIF	protocol identification field
42	PHY	physical layer (OSI reference model and IEEE 802® reference model)
43	PHY	physical layer device or entity (IEEE 802.3™ reference model)
44	PICS	protocol implementation conformance statement

¹¹Both forms are used, with the same meaning. This standard uses medium.

1	PLME	physical layer management entity
2	PMA	physical medium attachment
3	PMD	physical medium dependent
4	PSAP	physical service access point
5	RAN	regional area network
6	RM	reference model
7	RS	reconciliation sublayer
8	RSTP	rapid spanning tree protocol
9	SAP	service access point
10	SAI	Standard Assigned Identifier
11	SAI-48	48-bit Standard Assigned Identifier
12	SAI-64	64-bit Standard Assigned Identifier
13	SLAP	Structured Local Address Plan
14	SNAP	Subnetwork Access Protocol
15	SNMP	Simple Network Management Protocol
16	SPB	shortest path bridging
17	SSF	spectrum sensing function
18	SS/MS	subscriber station/mobile subscriber station
19	TSN	Time-Sensitive Networking
20	U/L	universally or locally administered
21	UML™	unified modeling language®
22	URN	Uniform Resource Name
23	VLAN	virtual local area network
24	VOIP	voice over Internet protocol
25	WAN	wide area network
26	WLAN	wireless local area network
27	WPAN	wireless personal area network
28	WRAN	wireless regional area network
29	YANG	The name of the data modeling language specified in IETF RFC 6020 [B16] ¹² and IETF RFC 7950 [B18].
30		

¹²The numbers in brackets correspond to the numbers of the bibliography in Annex A.

1 4. Family of IEEE 802 standards

2 4.1 Key concepts

3 The family of IEEE 802 standards is the collection of standards that have been developed under the IEEE
4 802 LMSC; see 4.4 for a list of the IEEE 802 standards current as of the publication of this standard. The
5 scope of IEEE 802 standards is not limited to the physical layers (PHYs) and data link layers (DLLs). IEEE
6 802 networks use frame-based communications with source and destination addressing over a variety of
7 media to connect various digital apparatus regardless of computer technology and data type.

8 User and management data flowing within IEEE 802 networks can be secured by a variety of authentication,
9 secure key exchange, and encryption mechanisms that are described in the various IEEE 802 standards. In
10 addition, IEEE 802 standards specify mechanisms by which a station is able to discover neighboring
11 networks information that may include IEEE 802 and non-IEEE 802 technologies. IEEE 802 standards also
12 specify mechanisms to achieve service discovery (e.g., support for Internet or virtual private network
13 service) and session continuity [e.g., a voice over Internet Protocol (VOIP) or multimedia session] in a
14 heterogeneous networking environment when stations, while either stationary or in motion, have a choice of
15 connecting to multiple access networks.

16 The early IEEE 802 local area network (LAN) wired technologies used shared-medium communication,
17 with information broadcast for all stations to receive. That approach has evolved over the years, but in ways
18 that preserve the appearance of simple peer-to-peer communications behavior for end stations. In particular,
19 the use of bridges, as described in 5.3.2, for interconnecting bridgeable IEEE 802 networks is now
20 widespread. These bridges allow the construction of networks with much larger numbers of end stations and
21 much higher aggregate throughput than would be achievable with a single shared medium. End stations
22 attached to such a bridged IEEE 802 network can communicate with each other just as though they were
23 attached to a single shared medium; however, the ability to communicate with other stations can be limited
24 by use of management facilities in the bridges, particularly where broadcast or multicast transmissions are
25 involved. A further stage in this evolution has led to the use of point-to-point full duplex communication in
26 LANs, either between an end station and a bridge or between a pair of bridges.

27 Some IEEE 802 technologies, in particular wireless-based technologies, are inherently shared-medium
28 communication systems. They too have been augmented over time. Many wireless local area networks
29 (WLANs) support mobile node mobility and hence dynamic topologies. These additional facilities may,
30 depending on the IEEE 802 technology in use, restrict bridged LAN interconnects to the static topology
31 nodes within the wireless portion of a heterogeneous technology LAN. Additionally, it is common to
32 interconnect individual networks and bridged networks at layers above the DLL with devices called routers.
33 The specification of routers is not provided in IEEE 802 standards.

34 Personal area networks (PANs) are used to convey information among a small group of participant stations.
35 Unlike a LAN, a connection made through a PAN typically involves little or no infrastructure or direct
36 connectivity to the world outside the connection. This approach allows small, power-efficient, inexpensive
37 solutions to be implemented. In the context of the family of IEEE 802 standards, PANs are implemented
38 with wireless technology and are, therefore, sometimes referred to as wireless personal area networks
39 (WPANs).

40 A body area network (BAN) is a short-range, wireless communication network in the vicinity of, or inside, a
41 body (typically a human body). BANs are used for a variety of uses, including entertainment, medical and
42 other healthcare services. Considerations for the design of BAN devices typically include effects on antenna
43 patterns due to the presence of the wide variety of bodies, changes in the RF channel due to user motions,
44 and radiation pattern shaping to minimize specific absorption rate into the body.

1 A LAN is a peer-to-peer communication network that enables stations to communicate directly on a point-
2 to-point, or point-to-multipoint, basis without requiring them to communicate with any intermediate stations
3 that perform forwarding or filtering above the PHY. LAN communication takes place at moderate to high
4 data rates and with short transit delays, on the order of a few milliseconds or less.

5 A LAN is generally owned, used, and operated by a single organization. This is in contrast to wide area
6 networks (WANs) that interconnect communication facilities in different parts of a country or are used as a
7 public utility. LANs are useful for deployment on a variety of scales, whether indoors or outdoors, including
8 covering a scale up to a large building or campus environment.

9 A metropolitan area network (MAN) is optimized for a larger geographical area than the area of a LAN,
10 ranging from several blocks of buildings to entire cities. As with local networks, MANs can also depend on
11 communications channels of moderate to high data rates. A MAN might be owned and operated by a single
12 organization, but it is usually used by many individuals and organizations. MANs might also be owned and
13 operated as public utilities. They often provide means for internetworking of local networks.

14 Regional area networks (RANs) generally cover a service area that is larger than the MANs. A RAN is
15 similar to a MAN in that it is typically owned and operated by a single organization, but it is usually used by
16 many individuals and organizations. For wireless regional area networks (WRANs), the unique propagation
17 characteristics of the frequency bands in which they operate, typically from 30 MHz to 1 GHz, require a
18 specialized design of the PHY and the medium access control (MAC) that can absorb long channel impulse
19 responses and large propagation delays. In some cases, operation in these bands is subject to coordination
20 with existing users, e.g., television broadcast.

21 IEEE 802 networks can also be used to perform the task of an access network, i.e., to connect end stations to
22 a larger, heterogeneous network, e.g., the Internet.

23 The early IEEE 802 standards for LAN and MAN technologies were all based on the use of copper or optical
24 fiber cables as the physical transmission medium. However, in addition to the use of cable-based media,
25 today's IEEE 802 standards include technologies, radio and optical, that use free space as the physical
26 transmission medium. IEEE 802 standards for wireless networks include wireless LANs, MANs, RANs, and
27 PANs. These technologies also target usage scenarios for both fixed and mobile wireless. These IEEE 802
28 network solutions address the challenges of mobility, higher error rates, and potentials for signal loss and
29 interference that are inherent to using wireless media.

30 4.2 Application and support

31 IEEE 802 networks are intended to have wide applicability in many environments. The primary aim is to
32 provide for low-cost devices and networks, suitable for consumer, commercial, educational, governmental,
33 and industrial applications. The following lists are intended to show some applications and devices and, as
34 such, are not intended to be exhaustive, nor do they constitute a set of required items. IEEE 802 networks
35 can be found in the following environments:

- 36 — Client/server applications
- 37 — Database access
- 38 — Desktop publishing
- 39 — Electronic mail
- 40 — File transfer
- 41 — Graphics
- 42 — Multimedia
- 43 — Office automation

- 1 — Vehicular communication
- 2 — Process control
- 3 — Robotics
- 4 — Telecommunication
- 5 — Text processing
- 6 — Transaction processing

7 IEEE 802 networks are intended to support communication, for example, between:

- 8 — Networking equipment, such as bridges, routers, and gateways
- 9 — Desktop, laptop and tablet computers
- 10 — Video, audio and multimedia equipment
- 11 — Cloud computing services, including web servers and data storage
- 12 — Monitoring and control equipment
- 13 — Scanners and printers
- 14 — Mobile phones and VOIP phones (desk phones)
- 15 — Internet of Things (IoT) devices, such as: thermostats, switches, and light bulbs

16 4.3 An international family of standards

17 The terms *PAN*, *BAN*, *LAN*, *MAN*, and *RAN* encompass a number of data communications technologies and
18 applications of these technologies. So it is with the IEEE 802 standards. In order to provide a balance
19 between the proliferation of a very large number of different and incompatible local and metropolitan
20 networks, on the one hand, and the need to accommodate rapidly changing technology and to satisfy certain
21 applications or cost goals, on the other hand, several types of medium access technologies are currently
22 specified in the family of IEEE 802 standards. In turn, these MAC standards are specified for a variety of
23 physical media. A secure data exchange standard and MAC bridging standards are intended to be used in
24 conjunction with the MAC standards.

25 The IEEE 802 standards have been developed and applied in the context of a global data communications
26 industry. IEEE 802 standards are recognized to be international standards in their own right. In addition,
27 some IEEE 802 standards have progressed to become standards within Joint Technical Committee 1 of the
28 International Organization for Standardization and the International Electrotechnical Commission (ISO/IEC
29 JTC 1), International Telecommunication Union Telecommunication Standardization Sector (ITU-T),
30 International Telecommunication Union Radiocommunication Sector (ITU-R), and a wide variety of
31 national body standards development organizations.

32 4.4 IEEE 802 standards

33 The IEEE 802 LAN/MAN Standards Committee sponsors a large number of standards projects. The current
34 family of IEEE 802 standards¹³ as of the approval of this standard is illustrated in Figure 1. IEEE 802
35 standards are under continuous development; for the latest status refer to <https://www.ieee802.org>.

¹³Throughout this standard, the term IEEE 802 standard is interpreted to include standards, recommended practices and guides.

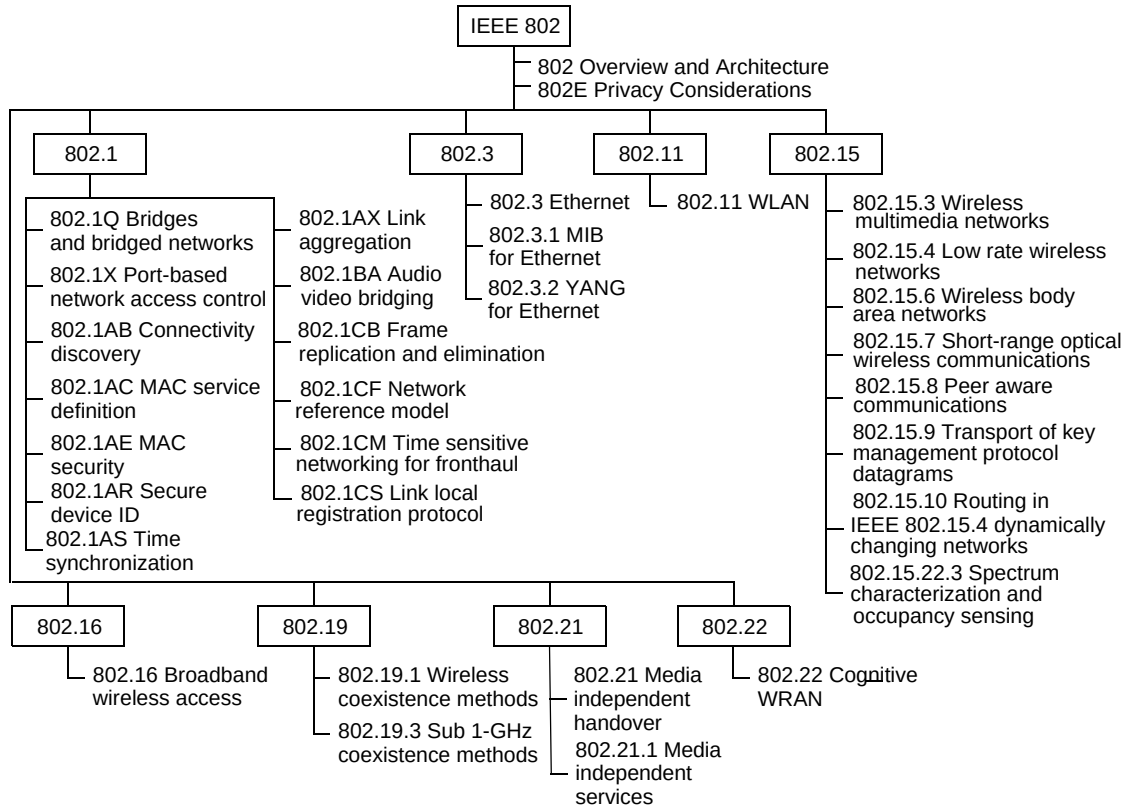


Figure 1—Current family of IEEE 802 standards

1 At any given time, an IEEE 802 standard may have one or more amendments related to it. Each amendment,
2 once approved, is considered to be part of the base standard. At a future time, through the periodic IEEE-SA
3 revision process, these amendments are incorporated into the base standard so that a new single document
4 can be issued. This process is illustrated in Figure 2 for IEEE Std 802.15.4TM-2011,¹⁴ which incorporated
5 the amendments IEEE Std 802.15.4aTM-2007, IEEE Std 802.15.4cTM-2009, and IEEE Std 802.15.4dTM-2009
6 into the base standard IEEE Std 802.15.4-2006.

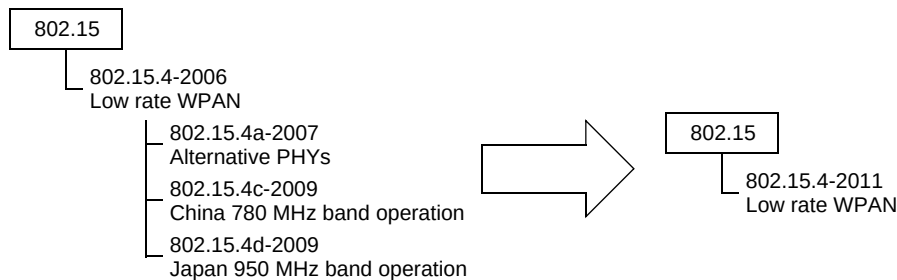


Figure 2—Issuance of IEEE Std 802.15.4-2011 from previous base standard and amendments

¹⁴See Annex D for a list of approved IEEE 802 standards that were current when this standard was completed.

5. Reference models (RMs)

5.1 Introduction

The IEEE 802 RM is derived from the Open Systems Interconnection basic reference model (OSI/RM), ISO/IEC 7498-1:1994 [B20]¹⁵. It is assumed that the reader has some familiarity with the OSI/RM and its terminology. The IEEE 802 standards emphasize the functionality of the lowest two layers of the OSI/RM, i.e., PHY and DLL, and the higher layers as they relate to network management. The IEEE 802 RM is similar to the OSI/RM in terms of its layers and the placement of its service boundaries. Figure 3 shows the architectural view of IEEE 802 RM for end stations and its relation to the OSI/RM. A variation of the model applies within bridges, as described in 5.3.2.

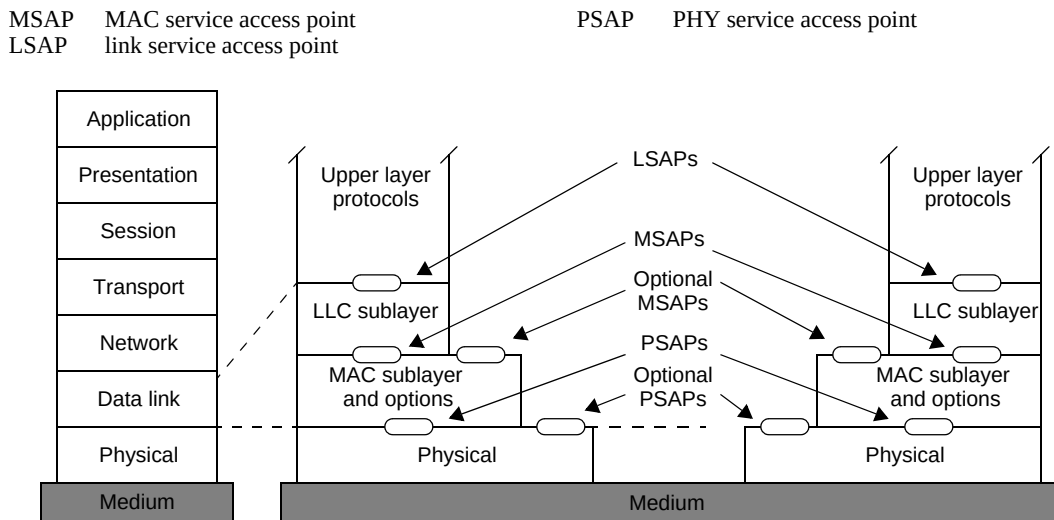


Figure 3—IEEE 802 RM for end stations

For the mandatory data services supported by all IEEE 802 networks, the DLL is structured as two sublayers, with the logical link control (LLC) sublayer, described in 5.2.2, operating over a MAC sublayer, described in 5.2.3.

Each IEEE 802 standard has RMs that are more detailed in order to describe the structure for that specific standard. The RMs for the IEEE 802 standards are given in Annex B.

IEEE 802 standards also provide implementation models (IMs), which are more specific than the IEEE 802 RMs, allowing differentiation between implementation approaches (e.g., different MAC protocols and PHYs). Figure 4 illustrates an IEEE 802.3 IM and its relation to the IEEE 802 RM.

Considerations of management, security, and media-independent handover (MIH) in IEEE 802 networks are also covered by IEEE 802 standards; these optional features lead to an elaboration of the RM, as illustrated in Figure 5. IEEE 802 network management provides protocols for exchange of management information between stations. The media-independent control function (MICF) is a parallel control plane that provides control functions for different MAC and PHY sublayers. Some examples of this MICF are the media-independent handover function (MIHF) of IEEE Std 802.21™ and the control functions proposed in the IEEE 802.19.1 Task Group and IEEE Std 802.22™. IEEE Std 802.1X™ forms part of the LLC sublayer and provides a secure, connectionless service immediately above the MAC sublayer.

¹⁵The numbers in brackets correspond to those of the bibliography in Annex A.

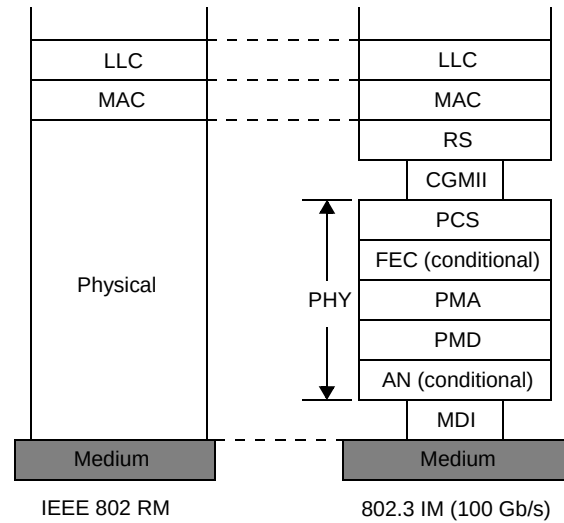


Figure 4—An IEEE 802 IM (100 Gb/s) and its relation to the IEEE 802 RM

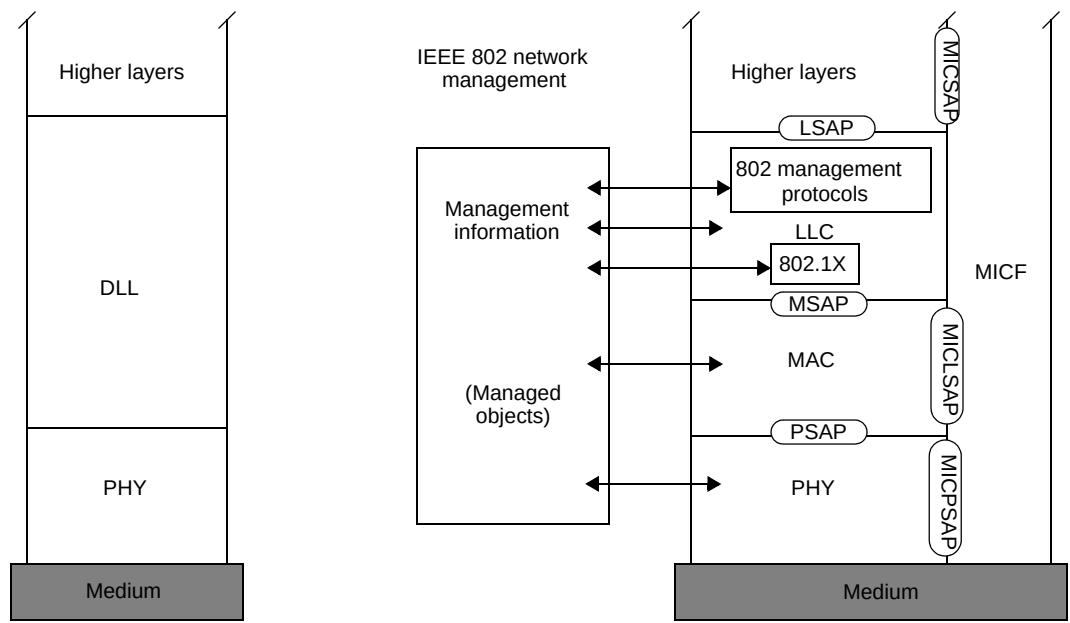


Figure 5—IEEE 802 RM with end-station management, security, and MIH

1 5.2 RM description for end stations

2 The IEEE 802 RM maps to the OSI/RM as shown in Figure 3. The applicable part of the OSI/RM consists of
3 the lowest two layers: the DLL and the PHY. These map onto the same two layers in the IEEE 802 RM. The
4 MAC sublayer of the IEEE 802 RM exists between the PHY and the LLC sublayer to provide a service for
5 the LLC sublayer (certain MAC types provide additional MAC service features that can be used by LLC
6 sublayer, in addition to the common core features). Service access points (SAPs) for connecting the layers
7 and sublayers are shown in Figure 3.

1 5.2.1 SAPs

2 One or more link service access points (LSAPs) provide interface ports to support one or more higher layer
3 users above the LLC sublayer.

4 In addition, the end station optionally provides one or more media-independent control service access points
5 (MICSAPs) that interface between one or more higher layers and the control and management planes
6 enabling higher layer information to pass to the MICF and vice versa.

7 The MAC sublayer provides one or more MAC service access points (MSAPs) as interfaces to the LLC
8 sublayer in an end station. Clause 8 provides details of how broadcast and group addresses are constructed.
9 The MAC sublayer optionally provides a media-independent control link service access point (MICLSAP),
10 which is used to provide an interface to support control of the MAC by the MICF.

11 The PHY provides a PHY service access point (PSAP). In addition, the PHY optionally provides a media-
12 independent control PHY service access point (MICPSAP), which is used to provide an interface port for the
13 control of the PHY by the MICF.

14 5.2.2 LLC sublayer

15 The LLC sublayer is between the MAC sublayer and the Network layer (Layer 3) in the OSI model. In IEEE
16 802, the functions in the LLC are specified in IEEE 802.1 and in aspects of ISO/IEC/IEEE 8802-2:1998
17 standards. Among the functions that can be present in the LLC are functions from the following:

- 18 1) IEEE Std 802.1AE™ provides MAC security allowing authorized systems that attach to and
19 interconnect LANs in a network to maintain confidentiality of transmitted data and to take
20 measures against frames transmitted or modified by unauthorized devices.
- 21 2) IEEE Std 802.1AX™ provides the ability to aggregate two or more links together to form a
22 single logical link at a higher data rate.
- 23 3) IEEE Std 802.1X provides authentication, authorization, and cryptographic key agreement
24 mechanisms to support secure communication between end stations connected by IEEE 802
25 networks.

26 NOTE: Prior to the 2014 revision of this standard, the LLC layer was specified as only the functions described in IEEE
27 Std 802.2. However, this did not allow for the other end-station functions developed by various IEEE 802.1 standards.
28 Since 2014, the LLC layer has been specified to include all of the relevant functions that occur between the MAC sub-
29 layer and the Network Layer.

30 5.2.3 MAC sublayer

31 The MAC sublayer provides a data transfer service to the LLC sublayer; a data unit received by a MAC
32 entity from its associated LLC entity is transferred to zero or more peer MAC entities for delivery to each
33 peer MAC entity's associated LLC entity. The unit that carries the data for transfer between MAC sublayer
34 entities is referred to as a MAC frame. In some MAC types, MAC frames are also used to support other
35 MAC sublayer functionality, such as the transfer of control or management information.

36 The principal functions of the MAC sublayer comprise the following:

- 37 — MAC frame delimiting and recognition
- 38 — Addressing of destination stations (both as individual stations and as groups of stations)
- 39 — Conveyance of source-station addressing information
- 40 — Transparent data transfer of PDUs between peer next higher sublayers
- 41 — Protection against errors, generally by means of generating and checking MAC frame check
42 sequences
- 43 — Control of access to the physical transmission medium

1 — Delivery to the next higher sublayer of data frames received with suitable destination addresses

2 Other functions of the MAC sublayer—applicable particularly when the supporting implementation includes
3 interconnection devices such as bridges—include flow control between an end station and an
4 interconnection device, as described in 5.3, and forwarding of MAC frames according to their destination
5 addresses to reduce the extent of propagation of MAC frames in parts of an IEEE 802 network that do not
6 contain communication paths leading to the intended destination end station(s).

7 The functions listed are those of the MAC sublayer as a whole. Responsibility for performing them is
8 distributed across the transmitting and receiving end stations and any interconnection devices such as
9 bridges. Devices with different roles, therefore, can behave differently in support of a given function. For
10 example, the basic transmission of a MAC frame by a bridge is very similar to transmission by an end
11 station, but not identical. Principally, the handling of source-station addressing is different.

12 The various MAC specifications all specify MAC frame formats in terms of a serial transmission model for
13 the service provided by the supporting PHY. This model supports concepts such as “first bit (e.g., of a
14 particular octet) to be transmitted” and a strict order of octet transmission in a uniform manner. However,
15 the ways in which the model has been applied in different MAC specifications are not completely uniform
16 with respect to bit-ordering within octets (see Clause 8, and particularly 8.6, for examples and explanation).
17 The serial transmission model does not preclude current or future MAC specifications from using partly or
18 wholly octet-oriented specifications of MAC frame formats or of the interface to the PHY.

19 5.2.4 PHY

20 MAC entities use their respective PHY entities to exchange bits with their peers. The PHY provides the
21 capability to transmit and receive modulated signals assigned to one or more channels for broadband.

22 Whereas the service offered to the MAC sublayer is expressed as the transfer of bits (in sequences
23 representing MAC frames), the symbols that are encoded for transmission do not always represent
24 individual bits. In many cases the PHY can map blocks of multiple bits to different multi-element symbols.
25 In some PHY encodings, these symbols are subject to further transformation before transmission, and in
26 some cases, the transmission is spread over multiple physical data paths.

27 5.2.5 Layer and sublayer management

28 The LLC, MAC, and PHY standards also include a management component that specifies managed objects
29 and aspects of the protocol machine that provides the management view of these resources. See Clause 7 for
30 further information.

31 5.3 Interconnection and interworking

32 In some cases, the end stations in an IEEE 802 network have no need to communicate with end stations on
33 other networks. However, there are many cases in which end stations on an IEEE 802 network need to
34 communicate with end stations on other networks; therefore, devices that interconnect the IEEE 802
35 network with other kinds of networks are required. In addition, several standard methods have been
36 developed that permit a variety of interconnection devices to operate transparently to end stations on a
37 network in order to extend the capabilities available to end stations, particularly in terms of the geographical
38 extent and/or total number of end stations that can be supported.

39 Standard methods of interworking fall into the following three general categories, depending on the layer at
40 which the corresponding interconnection devices operate:

41 — PHY interconnection, using devices usually termed *repeaters*, as described in 5.3.1

- 1 — MAC interconnection, using devices termed *bridges*, as described in 5.3.2
- 2 — Network-layer interconnection, using devices usually termed *routers*, as described in 5.3.3

3 5.3.1 Interconnection at the PHY

4 The original IEEE 802 standards were for end stations attached to a shared communication medium. This
5 basic configuration is referred to as an access domain; the domain consists of the set of stations such that, at
6 most, only one can transmit at a given time, with all other stations acting as (potential) receivers. In this
7 situation, the function of handling the “one-at-a-time” access arbitration is performed by the set of MAC
8 entities on a shared medium.

9 A repeater is a device used to interconnect segments of the physical communications media, for example, to
10 extend the range of a network when the physical specifications of the technology would otherwise be
11 exceeded, while providing one access domain for the attached stations.

12 5.3.2 MAC-sublayer interconnection: Bridges

13 5.3.2.1 IEEE 802 bridged networks

14 Bridges, in the general sense, are stations that interconnect multiple access domains. In this standard, the
15 term bridge is restricted to a functional unit that provides IEEE Std 802.1Q¹⁶ capabilities for bridge
16 interworking among bridgeable IEEE 802 networks. A bridged IEEE 802 network consists of one or more
17 bridges together with the complete set of access domains that they interconnect. A bridged IEEE 802
18 network provides end stations belonging to any of its access domains with the connectivity of a network that
19 contains the whole set of attached end stations.

20 An IEEE 802 bridged network can provide for the following:

- 21 — Communication between stations attached to networks of different MAC types that conform to the
- 22 Internal Sublayer Service as specified in IEEE Std 802.1AC.
- 23 — An increase in the total throughput of a network over that of a purely shared media network
- 24 — An increase in the physical extent of, or number of permissible attachments to, a network
- 25 — Partitioning of the physical network for administrative or maintenance reasons
- 26 — Virtual local area networks (VLANs)
- 27 — Provider bridging
- 28 — Transfer of priorities between peer MAC entities
- 29 — Support of latency, loss, and delay variation guarantees
- 30 — Support for traffic management and virtualization within data center networks

31 NOTE—The term switch is sometimes used in the industry to refer to products that include a bridging capability, such as
32 an IEEE 802.1Q bridging capability, often with other interconnection functions. IEEE 802 Standards do not use the
33 term switch to refer to IEEE 802.1Q bridging functions or capabilities..

34 5.3.2.2 Bridge relaying and filtering

35 A bridge processes protocols in the MAC sublayer and is functionally transparent to LLC sublayer and
36 higher layer protocols. MAC frames are forwarded between access domains, or filtered (i.e., not forwarded
37 to certain access domains), on the basis of addressing and protocol information contained in the MAC frame.
38 Figure 6 shows the position of the bridging functions within the MAC sublayer; note particularly that
39 relaying and filtering are considered to belong entirely within the MAC sublayer.

¹⁶Information on normative references can be found in Clause 2.

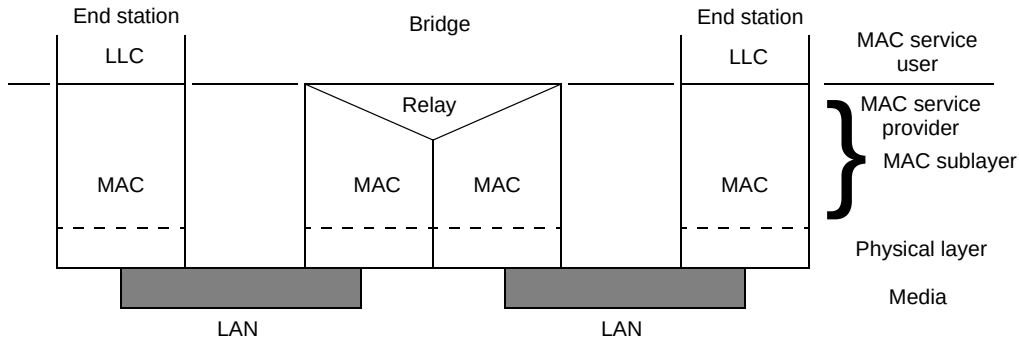


Figure 6—Internal organization of the MAC sublayer with bridging

1 Filtering by bridges tends to confine traffic to only the parts of the bridged network that lie between
2 transmitting end stations and the intended receivers. This permits a bridged network to support several
3 transmitting end stations at any given time (up to the total number of access domains present).

4 5.3.2.3 Resolving topologies with multiple paths

5 A key aspect of IEEE Std 802.1Q is the specification of the spanning tree protocols and shortest path
6 bridging (SPB) which are used by bridges to configure their interconnections in order to prevent looping
7 data paths in the bridged IEEE 802 network.

8 If the basic interconnection topology of bridges and networks contains multiple possible paths between
9 certain points, use of the rapid spanning tree protocol (RSTP) blocks some paths in order to produce a
10 simply connected active topology for the flow of MAC user traffic between end stations. For each point of
11 attachment of a bridge to a network, the RSTP selects whether MAC user traffic is to be received and
12 transmitted by the bridge at that point of attachment.

13 The RSTP adapts to changes in the configuration of the bridged IEEE 802 network, maintaining
14 connectivity while avoiding data loops. Some configuration changes can cause temporary interruptions of
15 connectivity between parts of the bridged IEEE 802 network, typically lasting for a few tens of milliseconds
16 at most. IEEE Std 802.1Q specifies a variant of RSTP, the multiple spanning tree protocol (MSTP), that can
17 configure multiple, independent spanning trees within a bridged network.

18 In addition, IEEE Std 802.1Q specifies SPB, which allows the use of shortest path communication within
19 administratively specified network regions, while retaining concurrent support for all existing spanning tree
20 protocols. The use of SPB, for unicast and multicast, allows multiple paths to be used simultaneously.

21 5.3.2.4 Transparent bridging

22 IEEE Std 802.1Q specifies transparent bridging operation, so called because the MAC bridging function
23 does not require the MAC frames transmitted and received to carry any additional information relating to the
24 operation of the bridging functions; end-station operation is unchanged by the presence of bridges.

25 5.3.2.5 Virtual and provider bridging

26 IEEE Std 802.1Q specifies the method by which the MAC service is supported by virtual bridged LANs, the
27 principles of operation of those networks, and the operation of VLAN-aware bridges, including
28 management, protocols, and algorithms. The standard also enables a service provider to use the architecture
29 and protocols specified in order to offer the equivalent of separate LANs, bridged LANs, or virtual bridged

1 LANs to a number of customers, while requiring no cooperation between the customers and minimal
2 cooperation between each customer and the service provider.

3 Provider backbone bridging further extends the concept of provider bridging by allowing a backbone
4 network, under the administrative control of a single backbone service provider, to support multiple service
5 providers, each administering its own distinct provider-bridged network to support distinct sets of
6 customers.

7 **5.3.2.6 Time-Sensitive Networking (TSN)**

8 Some IEEE 802 standards specify network protocols and mechanisms for applications that need TSN
9 capabilities such as data transport from one end station to one or more other end stations with low and
10 bounded latency, low and bounded latency variation, and low packet loss. Network protocols and
11 mechanisms specified in IEEE 802 standards for applications that need TSN capabilities include the
12 following:

- 13 a) Timing and Synchronization for Time-Sensitive Applications (IEEE Std 802.1AS-2020 [B3])
- 14 b) Credit-Based Shaper (IEEE Std 802.1Q-2022, 5.4.1.5)
- 15 c) Frame Preemption (IEEE Std 802.3-2022 [B8] Clause 99 and IEEE Std 802.1Q-2022, 6.7.2)
- 16 d) Scheduled Traffic (IEEE Std 802.1Q-2022, 8.6.8.4)
- 17 e) Cyclic Queuing and Forwarding (IEEE Std 802.1Q-2022, 5.4.1.9)
- 18 f) Asynchronous Traffic Shaping (IEEE Std 802.1Q-2022, 5.4.1.10)
- 19 g) Per-Stream Filtering and Policing (IEEE Std 802.1Q-2022, 5.4.1.8)
- 20 h) Frame Replication and Elimination for Reliability (IEEE Std 802.1CB-2017 [B5])
- 21 i) Stream Reservation Protocol (IEEE Std 802.1Q-2022, clause 35.)
- 22 j) Link-local Registration Protocol (IEEE Std 802.1CS-2020 [B7])
- 23 k) Path Control and Reservation (IEEE Std 802.1Q-2022, 5.4.6)
- 24 l) TSN Configuration (IEEE Std 802.1Q-2022, 5.29)
- 25 m) Configuration Enhancements for Time-Sensitive Networking (IEEE Std 802.1Qdj-2023)

26 NOTE—The application area or actual deployment can determine which network protocols and mechanisms in items a)
27 through m) are appropriate (e.g., whether or not time synchronization is used). TSN profile standards (e.g., IEEE Std
28 802.1BA [B4] and IEEE Std 802.1CM [B6]) select network protocols and mechanisms and give guidelines on their use
29 in a particular application area.

30 **5.3.2.7 Data center networking bridging**

31 The IEEE 802.1 Working Group develops standards that support data center networking (DCN), including
32 bridging enhancements. A data center is a facility composed of compute and storage servers interconnected
33 by a high bandwidth network and located in a small area, typically not exceeding 100 m in diameter. DCN
34 standards target network congestion for data centers and data center network virtualization. The DCN
35 features can provide networks free of congestion loss and support for in server virtualized networking for
36 attachment of containers and virtual machines. DCN features evolve and new capabilities are added as part
37 of IEEE 802 standardization efforts. Therefore, the following list is incomplete and just provides a snapshot
38 of DCN features:

- 39 a) Congestion Notification (IEEE Std 802.1Q-2022, Clause 30, Clause 31, Clause 32, Clause 33)
- 40 b) Priority-based Flow Control (IEEE Std 802.1Q-2022, Clause 36)
- 41 c) Enhanced Transmission Selection (IEEE Std 802.1Q-2022, Clause 37)
- 42 d) Congestion Isolation (IEEE Std 802.1Qcz-2023)
- 43 e) Edge Virtual Bridging (IEEE Std 802.1Q-2022, Clause 40, Clause 41, Clause 42, Clause 43)

1 5.3.2.8 Bridging example

2 Some bridges are used to interconnect access domains that each contain a very small number of end stations
3 (often, a single end station). Other bridges interconnect multiple access domains that contain principally
4 other bridges. These bridges and links are referred to as an IEEE 802 backbone network. Bridged IEEE 802
5 network configurations that involve these kinds of interconnection have become widespread as the
6 technologies have developed. These configurations allow the construction of networks with much larger
7 numbers of end stations and much higher aggregate throughput than was previously achievable.

8 Figure 7 illustrates an example of a bridged IEEE 802 network that can be configured with bridge-style
9 interconnection. The bridges A and B, and the IEEE 802.3 LAN configurations to which they attach, are
10 typical of the older style of bridged IEEE 802 network in which a bridge interconnects a small number of
11 access domains, each containing many end stations, as is similar with K, L, T and U. The IEEE 802.3
12 connections to M and those between S and T and S and U form IEEE 802 backbone networks. On the other
13 hand, the bridges T, and U function as bridges that combine IEEE 802.3 and IEEE 802.16™ networks. S and
14 M are bridges on an IEEE 802 backbone network, handling a number of network attachments. T and U are
15 bridges that support multiple end stations, with connection to an IEEE 802 backbone network. B and K also
16 provide access to an IEEE 802 backbone network. The end station shown connected to S by a point-to-point
17 link could be a server system. The wireless interfaces shown in Figure 7 are specified in each of the listed
18 standards. For example, a discussion of the 802.11 architecture is given in Annex B.2.

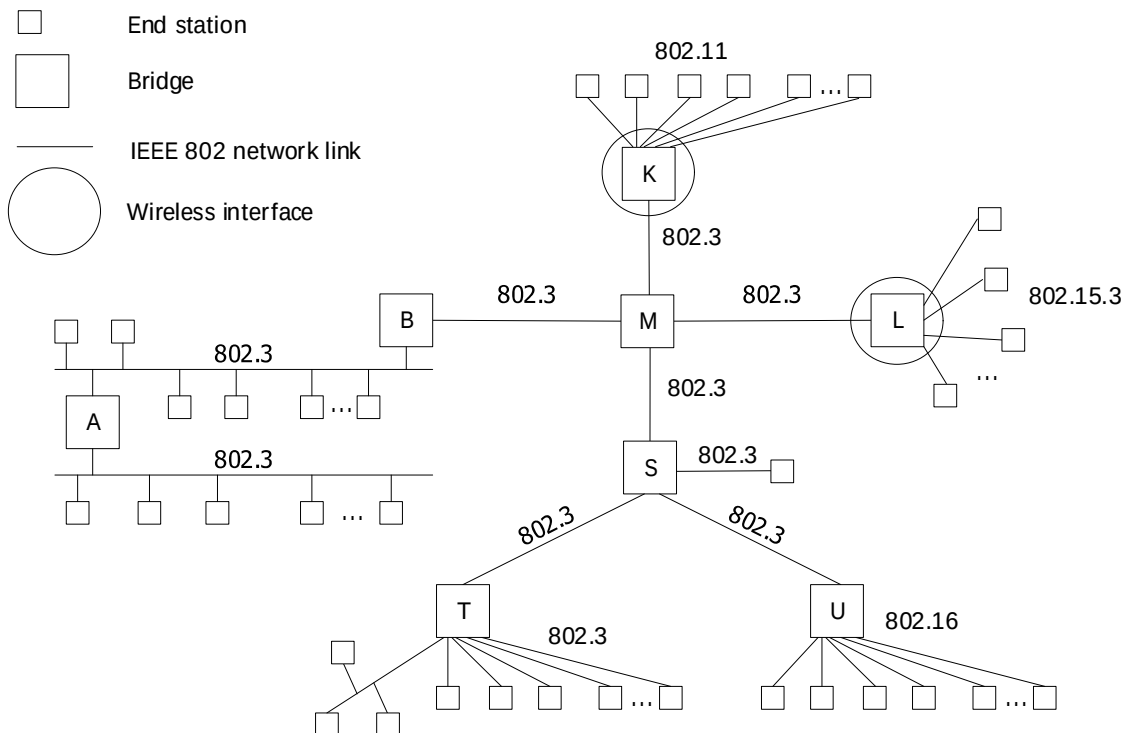


Figure 7—An example of a bridged IEEE 802 network

19 5.3.3 Network-layer interconnection: Routers

20 Routers are interconnection devices that operate as IEEE 802 end stations. These process Network layer
21 protocols that operate directly above the LLC sublayer, as shown in Figure 3, with forwarding decisions
22 based on Network layer addresses. Details of this kind of interconnection lie outside the scope of IEEE 802
23 standards, but the various standard and proprietary Network-layer protocols involved represent a substantial
24 part of the user traffic on many IEEE 802 networks. In particular, IEEE 802 networks are often

¹ interconnected by routers for the Internet Protocol (IP) and its related routing and management protocols,
² either directly to other IEEE 802 networks or by means of WAN connections.

6. General requirements for an IEEE 802 network

6.1 Services supported

With the descriptions in Clause 5 as a basis, a bridgeable IEEE 802 network can be characterized as a communication resource that provides sufficient capabilities to support the MAC service specified in IEEE Std 802.1AC, between two or more MSAPs. In particular, this requires the ability to convey LLC sublayer data from one MSAP to n other MSAPs, where n can be any number from 1 to the number of all of the other MSAPs on the network. A bridgeable IEEE 802 network is required, at a minimum, to support the MAC Internal Sublayer Service specified in IEEE Std 802.1AC.

NOTE—Some networks not specified in IEEE 802 standards meet these requirements and are therefore bridgeable IEEE 802 networks.

6.2 Error ratios

The error performance of IEEE 802 networks is as follows:

- a) For wired or optical fiber physical media: Within an access domain, the probability that a transmitted MAC frame (excluding any preamble) is not reported correctly at the PHY service interface of an intended receiving peer MAC entity, due only to operation of the PHY, shall be less than 8×10^{-8} per octet of MAC frame length.
- b) For wired physical media with MAC frames shorter than 2048 octets: The probability that an MAC service data unit (MSDU) delivered at an MSAP contains an undetected error, due to operation of the MAC service provider, shall be less than 5×10^{-14} per octet of MSDU length.
- c) For wireless physical media, the error performance within an access domain is variable over time.

NOTE—For example, the worst-case probability of losing a maximum-length IEEE 802.3 MAC frame at the PHY is to be less than 1.21×10^{-4} , or approximately 1 in 8250. The worst-case probability that a similar MAC frame, which contains an MSDU of 1500 octets, is delivered with an undetected error is to be less than 7.5×10^{-11} , or approximately 1 in 13 300 000 000.

1 7. IEEE 802 network management

2 7.1 General

3 The provision of an adequate means of remote management is an important factor in the design of today's
4 network equipment. Such management mechanisms fall into two broad categories: those that provide
5 general-purpose management capability, allowing control and monitoring for a wide variety of purposes,
6 and those that provide specific capabilities aimed at a particular aspect of management. These aspects of
7 management are discussed in 7.2 and 7.3, respectively.

8 7.2 General-purpose IEEE 802 network management

9 This subclause introduces the functions of management to assist in the identification of the requirements
10 placed on IEEE 802 network equipment for support of management facilities, and it identifies
11 general-purpose management standards that may be used as the basis of developing management
12 specifications for such equipment.

13 7.2.1 Management functions

14 Management functions relate to users' needs for facilities that support the planning, organization,
15 supervision, control, protection, and security of communications resources. These facilities may be
16 categorized as supporting the functional areas of configuration, fault, performance, security, and accounting
17 management. These can be summarized as follows:

- 18 — Configuration management provides for the identification of communications resources,
19 initialization, reset and shut-down, the supply of operational parameters, and the establishment and
20 discovery of the relationships between resources.
- 21 — Fault management provides for fault prevention, detection, diagnosis, and correction.
- 22 — Performance management provides for evaluation of the behavior of communications resources and
23 of the effectiveness of communication activities.
- 24 — Security management provides for the protection of resources.
- 25 — Accounting management provides for the identification and distribution of costs and the setting of
26 charges.

27 Management facilities in IEEE 802 network equipment address some or all of these areas, as appropriate to
28 the needs of that equipment and the environment in which it is to be operated.

29 7.2.2 Management architecture

30 The management facilities specified in IEEE 802 standards are based on the concept of managed objects that
31 model the semantics of management operations. Operations on a managed object supply information
32 concerning, or facilitate control over, the managed object and thereby, indirectly, the process or entity
33 associated with that object.

34 Operations on a managed object can be initiated by mechanisms local to the equipment being managed (e.g.,
35 via a control panel built into the equipment) or can be initiated from a remote management system by means
36 of a general-purpose management protocol carried using the data services provided by the IEEE 802
37 network to which the equipment being managed is connected.

1 The Simple Network Management Protocol (SNMP), as described in IETF RFC 3411 [B14], and Network
2 Configuration Protocol (NETCONF), as described in RFC 6241 [B17], are examples of general-purpose
3 management protocol that can be used for the management of IEEE 802 network equipment.

4 **7.2.3 Managed object definitions**

5 In order for an IEEE 802 standard to specify management facilities, it is necessary for it to specify managed
6 objects that model the operations that can be performed on the communications resources specified in the
7 standard. The components of a managed object definition are as follows:

- 8 a) A definition of the functionality provided by the managed object, and the relationship between this
9 functionality and the resource to which it relates.
- 10 b) A definition of the syntax that is used to convey management operations, and their arguments and
11 results, in a management protocol.
- 12 c) An address that allows the management protocol to specifically communicate with the managed
13 object in question. In IEEE 802 this is done with either an object identifier (OID), as described in
14 Clause 10, or a Uniform Resource Name (URN), as described in Clause 11.

15 The functionality of a managed object can be described in a manner that is independent of the protocol that
16 is used; this abstract definition can then be used in conjunction with a definition of the syntactic elements
17 required in order to produce a complete definition of the object for use with specific management protocols.

18 SNMP is used in many cases together with the structure of management information known as SMIV2 (IETF
19 RFC 2578, IETF RFC 2579 [B12], and IETF RFC 2580 [B13]), which uses a set of macros based on a
20 subset of ASN.1 for defining managed objects. YANG (IETF RFC 7950 [B18]) is a data modeling language
21 used to model configuration data, state data, remote procedure calls, and notifications for network
22 management protocols. The YANG objects are modeled in IEEE 802 standards using the Object
23 Management Group® (OMG®) similar to those of Unified Modeling Language™ (UML®) diagrams.

24 IEEE 802 networks can support management with SNMP MIBs or YANG to describe management objects.

25 The choice of notational tools for defining managed objects depends on the available management protocols
26 the standard supports.

27 **7.3 Special-purpose IEEE 802 network management standards**

28 Special-purpose protocols relating to the management functionality of IEEE 802 stations can be developed
29 when the use of a general-purpose management protocol is inappropriate. Examples of special-purpose
30 management protocols that can be found in the family of IEEE 802 standards include the Connectivity Fault
31 Management Protocol specified in IEEE Std 802.1Q; the Operations, Administration, and Maintenance
32 (OAM) Protocol specified in IEEE Std 802.3; and the Link Layer Discovery Protocol (LLDP) in IEEE
33 Std 802.1AB™.

1 8. MAC addresses

2 8.1 Terms and notational conventions

3 In this standard, the term *MAC address* is used to refer to a 48-bit or 64-bit number that is used to identify
4 the source or destination MAC entities. More specifically, the MAC address identifies the MAC SAP pro-
5 vided by the MAC entity for data transfer. In many IEEE 802 standards, the term *MAC address* refers only
6 to a 48-bit MAC address. In some IEEE 802 standards, the term *extended address* is used to refer to a 64-bit
7 MAC address.

8 If interoperability through bridges is required for a standard, then 48-bit MAC addressing is required. IEEE
9 802 standards that do not require bridged connectivity may use 64-bit MAC addressing. To avoid exhausting
10 the 48-bit global address space, 64-bit MAC addressing or the local address space, as described in 8.4, may
11 be used.

12 Hexadecimal representation is a sequence of octet values in which the values of the individual octets are dis-
13 played in order from left to right, with each octet value represented as a 2-digit hexadecimal numeral and
14 with the resulting pairs of hexadecimal digits separated by hyphens. The order of the hexadecimal digits in
15 each pair, as well as the mapping between the hexadecimal digits and the bits of the octet value, is derived
16 by interpreting the bits of the octet value as a binary numeral using the normal mathematical rules for digit
17 significance.

18 Bit-reversed representation is a sequence of octet values in which the values of the individual octets are dis-
19 played in order from left to right, with each octet value represented as a 2-digit hexadecimal numeral and
20 with the resulting pairs of hexadecimal digits separated by colons. The order of the hexadecimal digits in
21 each pair, as well as the mapping between the hexadecimal digits and the bits of the octet value, is derived
22 by reversing the order of the bits in the octet value and interpreting the resulting bit sequence as a binary
23 numeral using the normal mathematical rules for digit significance.

24 NOTE 1—The bit-reversed representation is of historical interest only and is no longer applicable to any active IEEE
25 802 standard.

26 NOTE 2—Many representations of MAC addresses outside of IEEE 802 standards use hexadecimal digits ordered as in
27 the hexadecimal representation but with colons separating octet numerals.

28 See 8.2.2 for a comparative example of bit-reversed and hexadecimal representation.

29 8.2 Universal addresses

30 8.2.1 Concept and overview

31 The concept of universal addressing is based on the idea that all potential members of a network need to
32 have a unique identifier. The advantage of a universal address is that a station with such a MAC address can
33 be attached to any IEEE 802 network in the world with an assurance that the MAC address is unique, if all
34 stations adhere to the rules and the security of the network prevents malicious spoofing of MAC addresses.

35 NOTE—Some network standards that are not IEEE 802 standards also use MAC addresses that are compliant with this
36 standard.

37 A universal address is a MAC address that is globally unique. Two different lengths of universal addresses
38 have been specified by the IEEE Registration Authority (RA): 48-bit Extended Unique Identifier (EUI-48)
39 and 64-bit Extended Unique Identifier (EUI-64).

8.2.2 Assignment of universal addresses

The IEEE RA has the responsibility for the administration of IEEE 802 network universal addresses.¹⁷ The IEEE RA is recognized by ISO/IEC as the registration authority for universal addresses for the ISO/IEC 8802 series of standards. The responsibility for defining the procedures is discharged by the IEEE Registration Authority Committee, which is a standing committee of the IEEE Standards Association Board of Governors.

The IEEE RA assigns blocks of universal addresses to assignee applicants (i.e., EUI-48s and EUI-64s) in various address block sizes. Each block assigns a common value (leading bits) that is common to all addresses in the assignment as described in Table 1.¹⁸

Table 1—IEEE RA assignment summary

IEEE RA assignment	Number of assigned bits	Block size of EUI-48	Block size of EUI-64	Used for company or organization identifier?
Company ID (CID)	24	0 (zero)	0 (zero)	yes (CID)
MAC Address Block Large (MA-L)	24	2 ²⁴	2 ⁴⁰	yes [Organizationally Unique Identifier (OUI)]
MAC Address Block Medium (MA-M)	28	2 ²⁰	2 ³⁶	no
MAC Address Block Small (MA-S)	36	2 ¹²	2 ²⁸	yes (OUI-36 only)

NOTE 1—The terms *OUI* and *OUI-36* were previously used by the IEEE RA to refer both to the globally unique 24-bit OUI and 36-bit identifiers and to the “products” that included the identifier and a block of addresses. The OUI “product” was replaced by the *MA-L* that, in addition to the address block, includes an assignment of an OUI; likewise, the OUI-36 was replaced by the *MA-S* that includes the assignment of an address block and an OUI-36. The use of these terms is not always consistent within IEEE standards.

NOTE 2—A CID assignment is used to identify a company or organization. It is not used to create universal addresses. For more information see 8.4.

The standard representation of MA-L, MA-M, and MA-S is to use the hexadecimal representation.

Figure 8 illustrates the structure of the first three octets of a MAC address. These octets have the same structure for all IEEE 802 MAC addresses, so this structure applies to all address block assignments (MA-S, MA-M and MA-L), and also to 48-bit or 64-bit MAC addresses, including universal and local addresses as well as individual and group addresses. The least significant bit (LSB) of the first octet is the individual/group (I/G) address bit, also known as the M bit in an OUI. The next-to-LSB of the first octet for the MAC address is the universal/local (U/L) address bit, also known as the X bit in an OUI. In an OUI, EUI-48 or EUI-64, both the U/L and the I/G bits, as shown in Figure 8, have the value zero.

The I/G address bit is used to identify the destination MAC address as an individual MAC address or a group MAC address. If the I/G address bit is zero, it indicates that the MAC address field is an individual MAC address. If this bit is one, the MAC address is a group MAC address that identifies zero or more (or all) stations connected to the IEEE 802 network. The all-stations broadcast MAC address is a special group MAC address of all ones.

¹⁷Interested applicants should contact the IEEE RA <https://standards.ieee.org/regauth>

¹⁸More information on MA-L, MA-M, and MA-S assignment can be found on the IEEE RA web site, <https://standards.ieee.org/develop/regauth/>.

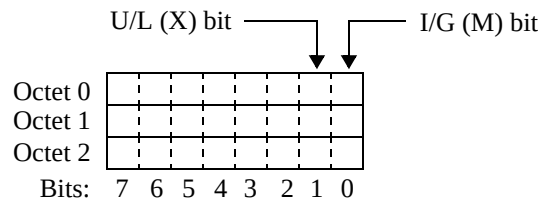


Figure 8—First three octets of a MAC address

1 The U/L bit indicates whether the MAC address is universal or has been assigned by a local administrator.
2 Universal addresses have the U/L bit set to zero. If the U/L bit is set to one, the address is locally adminis-
3 tered, as described in 8.4.

4 A universal address consists of two parts: the leading bits (24, 28, or 36) are assigned by the IEEE RA to an
5 assignee with both the U/L and the I/G bit set to zero and the remaining bits from the assigned block of
6 addresses. An example of an EUI-48 is shown in Figure 9. For MA-M and MA-S, the final 4 bits of the
7 assigned number are in a nibble that is not adjacent to the other bits in the assigned number when displayed
8 with LSB on the left and most significant bit (MSB) on the right. For example, when using an MA-S to cre-
9 ate an EUI-48, the MA-S value is contained in octets 0, 1, 2, 3 and the most significant four bits of octet 4,
10 and the value assigned by the assignee is contained in the least significant 4 bits of octet 4 and in octet 5.

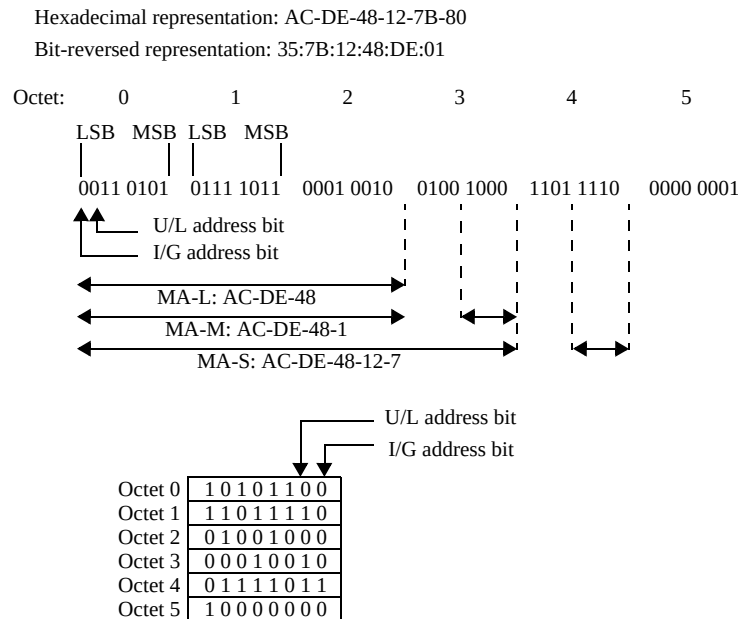


Figure 9—Example EUI-48

11 NOTE 3—The octet string AC-DE-48-12-7B-80 is used in this standard because it is clear when a bit pattern is reversed.
12 This octet string could be in use and is not a reserved value. While AC-DE-48 is used as the same first 3 octets for the
13 examples of MA-L, MA-M, and MA-S, the first 3 octets are different for valid assigned IEEE RA values.

14 An example of an EUI-64 is illustrated in Figure 10.

15 NOTE 4—The upper, bit-stream representation of the EUI-48 in Figure 9 and the EUI-64 in Figure 10 shows the LSB of
16 each octet first; this corresponds to the data-communications convention for representing bit-serial transmission in left-
17 to-right order, applied to the model for transmission of EUI-48 fields (see 5.2.3) and EUI-64 fields. See also 8.6 for fur-

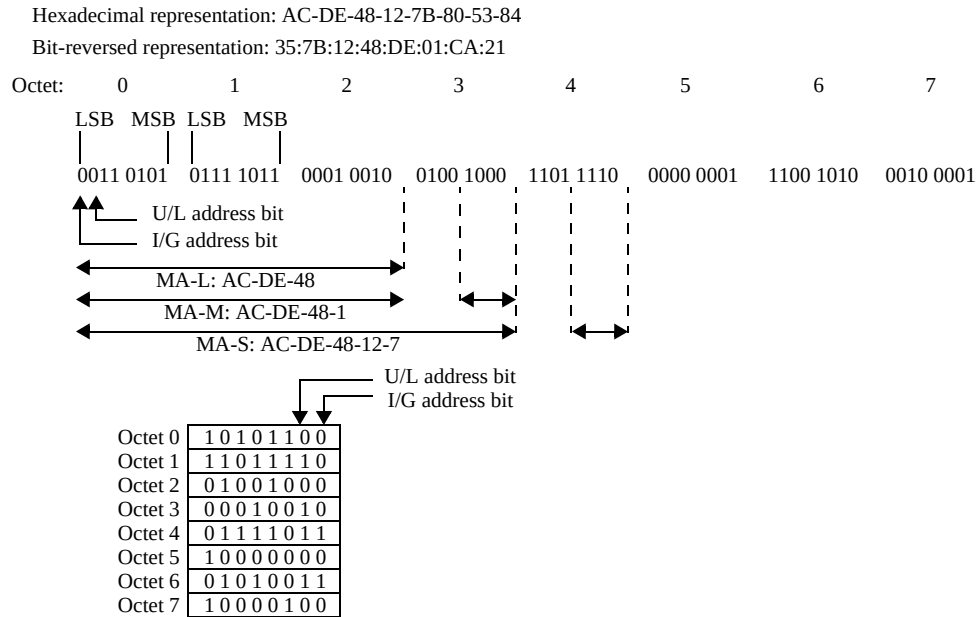


Figure 10—Example EUI-64

1 the discussion of bit-ordering issues. The lower, octet-sequence representation shows the bits within each octet in the
2 usual order for binary numerals; the order of octet transmission is from the top downward.

3 8.2.3 Assignment by organizations

4 The IEEE RA does not assign an MA-L, MA-M, or MA-S to any organization having previously received an
5 address block assignment, unless specific requirements are met. The requirements are designed to assure
6 that all addresses in an assigned block are used for network attachments. Requirements for an additional
7 allocation include the organization certifying it has exhausted or will soon exhaust its previous assignment.

8 It is important to note that universal addresses created from MA-Ls, MA-Ms, or MA-Ss should not be used
9 for purposes that would lead to skipping large numbers of them (for example, as product identifiers for the
10 purpose of aiding company inventory procedures). The IEEE RA asks that organizations implement internal
11 policies that minimize the misuse of addresses, that would unnecessarily exhaust an address block prema-
12 turely. There are sufficient identifiers to satisfy the intended needs for a long time, even as uses of IEEE 802
13 network stations grow – however, no address space is infinite.

14 The method that an assignee uses to ensure that no two of its stations carry the same universal address is not
15 specified in this standard. However, the users of networks worldwide expect to have unique addresses. The
16 ultimate responsibility for assuring that user expectations and requirements are met, therefore, lies with the
17 organization offering such stations.

18 8.2.4 Uniqueness of address assignment

19 It is recommended that each MAC entity on an IEEE 802 network have its own unique MAC address.

20 NOTE—Some implementations have used a single MAC address to identify more than one of the system's MAC enti-
21 ties. This approach does meet the requirements of IEEE 802.1Q™ MAC bridging.

22 8.3 Interworking with 48-bit and 64-bit MAC addresses

23 In response to concerns that the EUI-48 space could be exhausted by the breadth of products requiring
24 unique identifiers, 64-bit MAC addresses were introduced. Initially, new IEEE standards projects that did

1 not require backward compatibility with EUI-48 were requested to use 64-bit MAC addresses. This led to
2 some IEEE 802 standards adopting 64-bit MAC addressing, which cannot be bridged onto IEEE 802 net-
3 works that use 48-bit MAC addressing. Truncating a 64-bit MAC address into a 48-bit field can lead to two
4 stations having the same 48-bit value. To avoid this, traffic between a 64-bit MAC addressed network and a
5 48-bit MAC addressed network needs to be routed at a layer above the DLL..

6 Bridging for an IEEE 802 network with 64-bit MAC addresses is currently not specified.

7 **8.4 Local MAC addresses**

8 **8.4.1 Concept and overview**

9 The U/L bit of a local MAC address is set to one, indicating that the remaining bits (i.e., all bits except the
10 U/L bit and the I/G bit, which is set as described in 8.2.2) are locally administered. Local MAC addresses are
11 not presumed globally unique across all IEEE 802 networks. The locally administered bits of local MAC
12 addresses are arbitrarily assignable under the condition that local MAC addresses are unique within a LAN
13 (which may be a bridged LAN or virtual bridged LAN). In a virtual bridged LAN wherein the bridges use
14 Independent VLAN Learning, the uniqueness condition applies to each VLAN rather than to the entire vir-
15 tual bridged LAN. Any failure of such uniqueness invalidates the fundamental premises of IEEE 802 net-
16 work operation and can lead to disruption. Therefore, administrators should ensure that the probability of
17 local MAC address non-uniqueness is acceptably small.

18 While a local administrator may assign addresses throughout the local range, the optional Structured Local
19 Address Plan (SLAP) specifies different assignment approaches in four specified regions of the local MAC
20 address space.

21 Unlike universal addresses, which are persistent to the MAC entity, local MAC addresses are not necessarily
22 persistent. The local MAC address assigned to a MAC entity, including any subsequent change to that
23 assignment, is entirely within the scope of the local administration.

24 **8.4.2 Local MAC address assignment protocols**

25 An address assignment protocol assigning local MAC addresses to devices on a LAN should ensure unique-
26 ness of those addresses, per the description of F.1.2 of IEEE Std 802.1Q. That standard's Annex F also iden-
27 tifies risks of non-uniqueness.

28 When multiple address assignment protocols operate on a LAN without centralized administration, address
29 duplication is possible, even if each protocol alone is designed to avoid duplication, unless such protocols
30 assign addresses from disjoint address pools. Administrators who deploy multiple protocols on a LAN in
31 accordance with the SLAP enable the unique assignment of local MAC addresses within the LAN as long as
32 each protocol maintains unique assignments within its own address subspace.

33 **8.4.3 Structured Local Address Plan (SLAP)**

34 The SLAP specifies use of local MAC address space. Under the SLAP, the use is specified differently in
35 four quadrants of local MAC address space.

36 The least and second least significant bits of the initial octet of a MAC address are designated the M bit and
37 X bit, respectively, as shown in Figure 11. The third and fourth least significant bits of the initial octet in the
38 local MAC address are designated the Y bit and Z bit, respectively, as illustrated for a 48-bit address in
39 Figure 11 (see NOTE 4 of 8.2.2).

40 NOTE 1—The specific address used in Figure 11 and Figure 12 is not a reserved value.

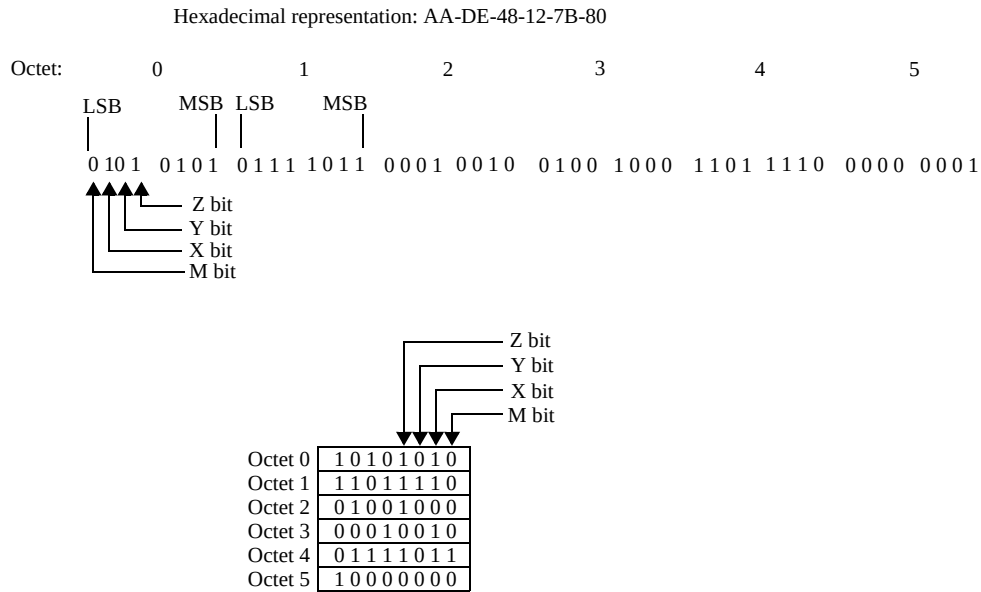


Figure 11—M, X, Y and Z bits of local MAC address

1 NOTE 2—The bit-stream representation of the addresses in Figure 11 and Figure 12 show the LSB of each octet first;
2 this corresponds to the data-communications convention for representing bit-serial transmission in left-to-right order.
3 See also 8.6 for further discussion of bit-ordering issues.

4 A local MAC address exists in one of four SLAP quadrants, each identified by a different combination of the
5 Y and Z bits, as indicated in Table 2. That table also indicates the SLAP local identifier type specified for
6 each SLAP quadrant. The SLAP local identifier types are specified in 8.4.4.

Table 2—SLAP quadrants, addresses and bit settings

SLAP quadrant name	M bit	X bit	Y bit	Z bit	SLAP local identifier type	SLAP local addresses	Number of bits (including I/G and U/L) assigned by IEEE RA or IEEE Std 802
01	I/G	1	0	1	Extended Local	ELI-48, ELI-64	24 (CID)
11	I/G	1	1	1	Standard Assigned	SAI-48, SAI-64	4
00	I/G	1	0	0	Administratively Assigned	AAI-48, AAI-64	4
10	I/G	1	1	0	<i>Reserved</i>	<i>Reserved</i>	—

7 For compliance to the SLAP, local MAC addresses shall be assigned only as valid Extended Local Identifier
8 s (ELIs), Standards Assigned Identifiers (SAIs) or Administratively Assigned Identifiers (AAIs).

9 While IEEE Std 802 assigns four SAI bits, additional SAI bits may also be assigned by IEEE 802 standards.

8.4.4 SLAP local identifier types

8.4.4.1 Extended Local Identifier (ELI)

A SLAP identifier of type “Extended Local” is known as an Extended Local Identifier (ELI). ELIs are in SLAP quadrant “01”. The X, Y and Z bits of an ELI are specified in Table 2. An ELI may be used as a local MAC address; such an address is known as an ELI address.

The IEEE RA uniquely assigns a 24-bit identifier known as the Company ID (CID)¹⁹ to identify a company, organization, entity, protocol, etc., as described in the IEEE RA tutorial [B2].

An ELI is based on an assigned CID. Two different lengths of ELI are specified: ELI-48 is a 48-bit ELI, and ELI-64 is a 64-bit ELI, as described in Table 3.

Table 3—IEEE RA CID and ELI summary

IEEE RA assignment	Number of bits (including I/G and U/L) assigned by IEEE RA	Address block size, ELI-48	Address block size, ELI 64
CID	24	2^{24}	2^{40}

The structure of the CID is identical to that of the OUI, which is illustrated in Figure 8. However, the CID is not used to create universal addresses. Each CID assignment provided by the IEEE RA has the X bit set to one, so that any MAC address created by extension of the CID, as shown in Figure 12, exists in locally administered address space. Changing the X bit of an OUI assigned by the IEEE RA is not authorized by the IEEE RA, does not result in a valid CID, can invalidly duplicate a valid CID assignment, and shall not be used as the basis of an ELI. Likewise, changing the X bit of an CID assigned by the IEEE RA is not authorized by the IEEE RA, does not result in a valid OUI, can invalidly duplicate a valid OUI assignment, and shall not be used as the basis of an EUI.

While each CID assignment provided by the IEEE RA has the M bit set to zero, the corresponding bit of the ELI address represents the I/G bit and is set as described in 8.2.2.

An ELI-48 or ELI-64 created as an extension of the CID consists of two parts: the leading 24 bits are assigned as the CID, with the I/G bit assignable as described in 8.2.2, and the remaining bits are specified as an extension by the CID assignee or by a protocol designated by the CID assignee.

Several CIDs are reserved as Administrator CIDs and not assigned exclusively. The local administrator is an implicitly authorized assignee of the Administrator CIDs and may, within the SLAP, create and assign an ELI as an extension of an Administrator CID. Administrator CIDs are specified in Table 4.

An example of an ELI-48 created by extension of a CID is shown in Figure 12 (see NOTE 4 of 8.2.2). An ELI-64 is created with the same method but using two additional octets.

NOTE—The specific CID and ELI-48 used in this example are not reserved values.

Figure 12 also illustrates the location of the Y bit and Z bit in a CID.

¹⁹For more information on CIDs, follow the tutorial link from <https://standards.ieee.org/regauth> to locate the appropriate tutorial.

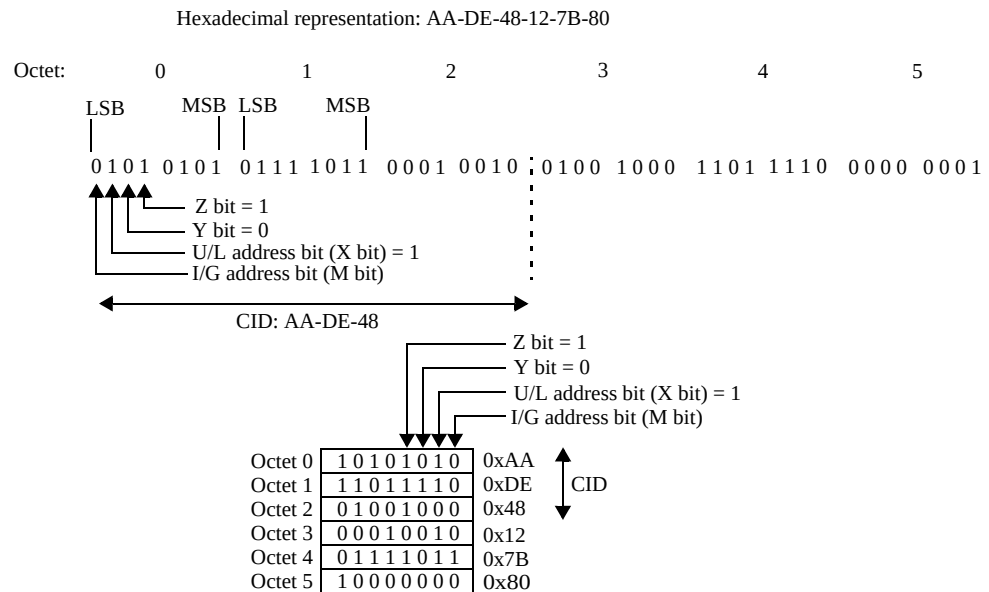


Figure 12—Example ELI-48 created as an extension of CID

Table 4—Administrator CIDs

Administrator CIDs
0x3AA3F8
0xCA30BF
0x4A07D6
0xFA94F1

1 IEEE 802 standards support the use of ELI-48 and ELI-64 based on CID only for CID values that specify
2 zero for the Y bit and one for the Z bit. The IEEE RA assigns CIDs only with zero for the Y bit and one for
3 the Z bit.

4 In some cases, an ELI assignment protocol may assign the 24-bit or 40-bit extension of the ELI to convey
5 specific information. Such information may be interpreted by receivers and bridges that recognize the CID
6 and are cognizant of the protocol identified by the CID. The functionality of receivers and bridges that do
7 not recognize the protocol is not affected. Such address formats, and their interpretation, are outside the
8 scope of this standard but may be specified by the entity to which the specific CID is assigned by the IEEE
9 RA.

10 Note, in contrast to such uses of local addresses, IEEE RA policies (detailed in [B2]) intended to prevent
11 premature exhaustion of the universal address space do not allow for similar subdivision of a universal
12 address block. Such subdivision does not provide assurance that addresses in the block have been used for
13 an identifiable physical instance per EUI-48 identifier

1 8.4.4.2 Standard Assigned Identifier (SAI)

2 A SLAP identifier of type “Standard Assigned” is known as a Standard Assigned Identifier (SAI). SAIs are
3 in SLAP quadrant “11”. The X, Y and Z bits of an SAI are specified in Table 2. An SAI may be used as local
4 MAC address; such an address is known as an SAI address. The specification of this quadrant is reserved for
5 a future IEEE 802 standard (see E.3).

6 An SAI is assigned by a protocol specified in an IEEE 802 standard.

7 Two different lengths of an SAI are specified: SAI-48 is a 48-bit SAI, and SAI-64 is a 64-bit SAI.

8 Multiple protocols for assigning SAI may be specified within various IEEE 802 standards. Coexistence of
9 such protocols may be supported by restricting each to assignments within a subspace of SAI space. In some
10 cases, an SAI assignment protocol may assign the SAI to convey specific information. Such information
11 may be interpreted by receivers and bridges that recognize the specific SAI assignment protocol, as identi-
12 fied by the subspace of the SAI. The functionality of receivers and bridges that do not recognize the protocol
13 is not affected. SAI address formats, and their interpretation, are outside the scope of this standard but may
14 be specified in other IEEE 802 standards.

15 8.4.4.3 Administratively Assigned Identifier (AAI)

16 A SLAP identifier of type “Administratively Assigned” is known as an Administratively Assigned Identifier
17 (AAI). AAIs are in SLAP quadrant “00”. The X, Y and Z bits of an AAI are specified in Table 2. An AAI
18 may be used as local MAC address; such an address is known as an AAI address.

19 Administrators who wish to assign local MAC addresses in an arbitrary fashion (for example, randomly) and
20 yet maintain compatibility with other assignment protocols operating under the SLAP on the same LAN
21 may assign a local MAC address as an AAI.

22 Two different lengths of an AAI are specified: AAI-48 is a 48-bit AAI, and AAI-64 is a 64-bit AAI, as
23 described in Table 5.

Table 5—AAI summary

IEEE 802 assignment	Number of bits (including I/G and U/L) assigned by IEEE Std 802	Address block size, ELI-48	Address block size, ELI 64
Bits M, X, Y, Z	4	2^{44}	2^{60}

24 Per IETF RFC 2464 [B11], a multicast IPv6 packet uses a Ethernet destination address whose first 2 octets
25 are the value 0x3333. Such addresses lie within the AAI quadrant of the multicast local MAC address space.
26 Therefore, administrators who wish to support IPv6 should not assign AAIs beginning with 0x3333.

27 8.4.4.4 SLAP quadrant “10”

28 Under the SLAP, the local identifiers in SLAP quadrant “10” are reserved for future use.

29 While SLAP quadrant “10” identifiers remain reserved, they may be administratively used and assigned in
30 accordance with the considerations specified for AAI usage, without effect on SLAP assignments. However,
31 administrators should be cognizant of possible future specifications regarding SLAP quadrant “10” that
32 would render administrative assignment incompatible with the SLAP.

1 8.4.5 Network Unique Identifier (NUI)

2 Network Unique Identifier (NUI) is an identifier that is unique within the IEEE 802 LAN (which may be a
3 bridged LAN or virtual bridged LAN). An NUI-48 is a 48-bit NUI that is an EUI-48, ELI-48, SAI-48 or
4 AAI-48. An NUI-64 is a 64-bit NUI that is an EUI-64, ELI-64, SAI-64 or AAI-64.

5 8.5 Group MAC addresses

6 The IEEE RA's assignment of blocks of universal individual MAC addresses is described in 8.2. In some
7 cases, the assignee of an address block may need to make use of group addresses as well as individual ones.
8 Such addresses are included in the IEEE RA assignment. An IEEE RA tutorial [B2] states that "The
9 assignee of an OUI or OUI-36 is exclusively authorized to assign group MAC addresses, with I/G=1, by
10 extending a modified version of the assigned OUI or OUI-36 in which the M bit is set to 1. Such addresses
11 are not EUIs and do not globally identify hardware instances, even though U/L=0."

12 In some cases, group MAC addresses are designated for use in standardized protocols; these are known as
13 standardized group MAC addresses²⁰. These may be created by the procedure described in [B2], based on an
14 OUI or OUI-36 assigned for use in a standard. For example, many standardized group MAC addresses are
15 derived from an OUI that has been assigned to the IEEE 802.1 Working Group. In some cases, a group
16 address has been assigned by the IEEE RA for use in a standard. The administration of such standardized
17 group MAC addresses is described by the IEEE RA²¹. The IEEE RA also provides a list of currently
18 assigned values; some of the items listed are simply informative, since the assignments are specified by the
19 assignee of the OUI and not by the IEEE RA. The IEEE RA also lists some standardized group addresses
20 that, because the U/L bit is set to 1, are not assigned uniquely to the standard.

21 8.6 Bit-ordering within octets

22 Clause 5 describes the reference models for IEEE Std 802 networks. Regarding interoperability at the MAC
23 Service Access Point (MSAP), IEEE Std 802 network standards specify how octets from the LLC are trans-
24 mitted and received. Though most IEEE 802 network Physical Layers encode multiple bits or multiple
25 octets of the MAC frame for transmission on the medium, a few IEEE 802 network Physical Layers have a
26 one-to-one mapping of a bit in the MAC frame to an encoded bit on the medium.

27 In this case of bit-serial transmission, some MAC standards have specified serial transmission of the bits of
28 an octet LSB first and other MAC standards specifying transmission of the MSB first.

29 For MAC frames transmitted in bit-serial fashion, IEEE 802 standards shall specify LSB-first transmission
30 for each octet of a source or destination address field. For MAC frames transmitted in bit-serial fashion,
31 IEEE 802 standards should specify LSB-first transmission for each octet of the MAC frame.

²⁰These were previously referred to as standard group MAC addresses.

²¹See "Standardized Group MAC addresses" at: <https://standards.ieee.org/products-programs/regauth/grpmac/>

9. Protocol identifiers and context-dependent identifiers

9.1 Introduction

A key function of the LLC sublayer is to support multiplexing and demultiplexing of multiple Network layer protocols over an IEEE 802 network.

Within the Network layer, entities can exchange data by a mutually agreed upon protocol mechanism. A pair of entities that do not support a common protocol cannot communicate with each other. For multiple Network layer protocols to operate within an IEEE 802 network, the transmitting and receiving higher layer protocol discrimination entities (HLPDEs) of the LLC sublayer cooperate to identify the Network layer protocol to be invoked for each service data unit delivered by the lower layer.

A network-layer protocol is identified within the LLC sublayer by means of a protocol identifier of a specific protocol type, associated with the protocol. Three specific types of protocol identifier are supported:

a) E-Type: The E-Type protocol identifier is an EtherType, which is a two-octet identifier, in the range from 06-00 through FF-FF²², that is uniquely assigned to a protocol. Assignments are made and recorded by the IEEE Registration Authority²³. Two EtherType values, known as the Local Experimental EtherTypes, do not reflect global protocol assignments but instead are assigned for use by local administrators who decide on their local mapping to protocols.

NOTE 1—While every E-Type protocol identifier is an EtherType, not all EtherTypes are E-Type protocol identifiers. For example, some EtherType values are assigned to indicate specific Layer 2 functionality rather than a network-layer protocol; in these cases, a network-layer PDU is typically encapsulated and carried later in the MAC frame.

b) L-Type: The L-Type protocol identifier is an LSAP address, which is a one-octet identifier that is uniquely assigned to a protocol. LSAP address value assignments are made and recorded by the IEEE Registration Authority.

NOTE 2—While every L-Type protocol identifier is an LSAP address, not all LSAP addresses are L-Type protocol identifiers. For example, some LSAP address values are assigned to indicate specific Layer 2 functionality rather than a network-layer protocol; in these cases, a network-layer PDU is typically encapsulated and carried later in the MAC frame.

c) O-Type: The O-Type protocol identifier is created under the authority of an OUI, OUI-36, or CID assignee by appending bits to the OUI, OUI-36, or CID assignment. The O-Type identifier allows the OUI, OUI-36, or CID assignee to derive globally-unique protocol identifiers without an external registration authority.

Because each protocol identifier type is a different length, the protocol identifier type of a protocol identifier follows from its length. The types are also distinguishable by numeric value. The largest valid L-Type value is 0xFE (254). Valid E-Type values are within the range 0x0600 (1536) to 0xFFFF (65 535). The O-Type value is always greater than 0xFFFF.

In IEEE 802 networks, the protocol identifier is encoded into a protocol identification field (PIF) that is incorporated as the initial octets of the LPDU, prepended to the higher-layer protocol data unit, as shown in Figure 13. The LPDU transferred between LLC entities is encapsulated in the MDSU and is not exposed at the MAC sublayer. Some exceptions to this are specified in IEEE 802 standards; for example, the first two octets of the LPDU are exposed to the MAC sublayer of IEEE Std 802.3.

²²By convention, EtherTypes are represented as two hexadecimal numbers followed by a dash followed by two hexadecimal numbers with no prefix. In this standard, hexadecimal numbers other than EtherTypes are prefixed with 0x.

²³More information can be found at <https://standards.ieee.org/products-programs/regauth/> and <https://regauth.standards.ieee.org/standards-ra-web/pub/view.html#registries>.

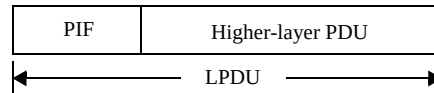


Figure 13—LPDU including prepended PIF

1 Two forms of encoding a protocol identification field are specified. With either of these two encoding forms,
2 the encoding includes sufficient information for the receiving HLPDE to: a) identify the protocol
3 identification field; b) determine the protocol identifier type; and c) identify the protocol identifier. The
4 HLPDE is then enabled to strip the PIF from the data payload and forward the resulting payload to the
5 network-layer protocol that is associated with the protocol identifier.

6 Three PIF encoding forms are specified, each of which allows the HLPDE to parse the PIF for any of the
7 three protocol identifier types. These are:

- 8 1) Type 1 PIF encoding, which is reserved;
- 9 2) Type 2 PIF encoding, which does not use a Length/Type field; and
- 10 3) Type 3 PIF encoding, which makes use of a Length/Type field.

11 While the two PIF encoding forms are each capable of supporting all protocol identifier types, no provision
12 is made herein for the HLPDE to ascertain which of the two encoding forms was applied at the source.
13 Without such information, the HLPDE cannot parse the data payload to identify the PIF. This standard
14 presumes that the HLPDE is aware of the encoding form used.

15 9.2 EtherTypes and E-Type protocol identifiers

16 9.2.1 Format, function, and administration

17 EtherType values are assigned by the IEEE RA²⁴. An EtherType is a sequence of 2 octets, interpreted as a
18 16-bit numeric value with the first octet containing the most significant 8 bits and the second octet
19 containing the least significant 8 bits. Values in the 0–1535 range are not available for use.

20 Some EtherTypes are assigned as E-Type protocol identifiers and are associated with higher-layer protocols,
21 typically network-layer protocols. Examples of such EtherTypes are 08-00 and 86-DD, which are used to
22 identify IPv4 and IPv6, respectively.

23 Some EtherTypes are not assigned as E-Type protocol identifiers but are instead used within Layer 2.
24 Examples of such EtherType are the OUI Extended EtherType 88-B7 and the LLC Encapsulation EtherType
25 88-70. Specifications associated with an assigned EtherType describe the method to parse the remainder of the
26 data field to extract the protocol identifier.

27 9.2.2 Public EtherType assignments subset

28 The IEEE Registration Authority (RA) provides a public listing of EtherType assignments²⁵. Many of these
29 are for private or proprietary purposes. However, others are incorporated into well-known standards. In
30 some cases, the IEEE RA Public Listing for an EtherType identifies an assignee without explicitly
31 identifying the standards in which the use of that EtherType is specified. For ready reference by users and
32 developers of such standards, Annex F identifies some well-known EtherTypes and the protocols they

²⁴More information on EtherTypes can be found on the IEEE RA web site, <https://standards.ieee.org/products-programs/regauth/ether-type/>, and <https://regauth.standards.ieee.org/standards-ra-web/pub/view.html#registries>.

²⁵The EtherType public listing is the public view of the EtherType registry managed by the Registration Authority (see <https://standards.ieee.org/regauth/>).

1 identify. This subset is derived by combining the EtherTypes listed in the ietf-ethertypes YANG module
2 specified in IETF RFC 8519 [B19] with the subset of EtherTypes specified by IEEE 802 Standards (e.g.,
3 IEEE 802.1Q, 802.3, etc.) and as provided by participants that developed this standard. Information on
4 EtherTypes (included in Annex F or not) can be found on the IEEE SA Registration Authority web site:
5 <https://standards.ieee.org/products-programs/regauth/ethertype/> and [https://regauth.standards.ieee.org/](https://regauth.standards.ieee.org/standards-ra-web/pub/view.html#registries)
6 [standards-ra-web/pub/view.html#registries](https://regauth.standards.ieee.org/standards-ra-web/pub/view.html#registries). The subset in Table F.1 and in F.3 is provided solely for the
7 convenience of users of this standard and does not constitute an endorsement by IEEE of the listed
8 protocols.

9 The EtherType public listing includes the following fields, specified by the EtherType assignee:

- 10 — **Assignment** — The hexadecimal representation of the EtherType.
- 11 — **Assignment Type** — The type is EtherType²⁶.
- 12 — **Company Name** — The registrant of the Assignment.
- 13 — **Company Address** — The address of the registrant.
- 14 — **Protocol** — A brief protocol description, as provided by the registrant.

15 This Standard includes the following fields in Table F.1 for use by the YANG module:

- 16 a) **Friendly Name** — A short alphanumeric name for the Assignment that is unique within the YANG
17 module in F.2 and is used to enumerate the entry.
- 18 b) **Short Description** — A short description of the assigned protocol per its typical usage.
- 19 c) **Reference** — A reference to a standard associated with the EtherType assignment.

20 A YANG model representation can be found in F.3.2.

21 9.2.3 EtherType sub-protocol encoding

22 The EtherType identifier space is a finite resource. When the IEEE Registration Authority assigns an
23 EtherType to an organization, it specifies that the usage should be extensible to alternative variations of the
24 protocol and to new versions. This protects the resource against premature exhaustion due to repeat
25 assignment requests from a single user. Such usage also benefits the assignee, since attaining an assignment
26 requires time, effort, and funds.

27 In order to allow for a single EtherType to multiplex various sub-protocols and versions, a protocol subtype
28 and a protocol version identifier should be used. Figure 14 is an example of sub-protocol encoding that
29 follows the EtherType field. As shown, the PIF is followed by additional fields that, together with the PIF,
30 form the sub-protocol information field (SPIF). While the contents of the PIF are sufficient to identify the
31 protocol sufficiently for the HLPDE to direct the MAC frame to the correct higher-layer protocol, the
32 contents of the protocol subtype and protocol version identifier are intended to be used within the higher-
33 layer protocol to direct the MAC frame to the correct sub-protocol. The lengths of the protocol subtype and
34 the protocol version identifier fields, as well as their order of appearance within the MAC frame, are not
35 constrained by this standard but are determined by the user. The IEEE 802 network has no visibility into this
36 structure.

37 9.2.4 Local Experimental EtherTypes

38 In order to allow users to conveniently operate E-Type protocol identification without a unique assignment,
39 two EtherType values, known as the Local Experimental EtherTypes, are assigned for use within a locally
40 administered network. The values of the Local Experimental EtherTypes are listed in Table 6.

²⁶EtherType is the only assignment type for the records in the EtherType public listing.

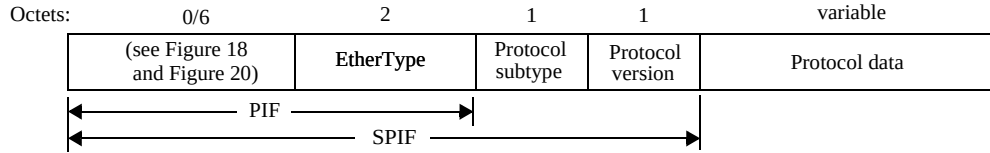


Figure 14—Example of sub-protocol encoding

Table 6—Assigned EtherType values

Name	Value
Local Experimental EtherType 1	88-B5
Local Experimental EtherType 2	88-B6

1 Within that network, a local administrator is free to use a Local Experimental EtherType and to assign
2 subtypes for protocol development purposes. However, by virtue of the way these EtherTypes are intended
3 to be used, the following practical and administrative constraints apply to their use:

- 4 a) Since the format for protocols using the Local Experimental EtherTypes does not contain a means to
5 identify the administrative domain, it might not be possible to identify the protocol of a MAC frame
6 if protocols developed within different administrative domains using Local Experimental
7 EtherTypes are used in the same network. Hence, the use of these EtherTypes to identify protocols
8 can only be achieved reliably if all uses of the EtherTypes are within the control of a single
9 administrative domain. Therefore, these EtherTypes shall not be used in protocols or products that
10 are to be released for use in the wider networking community, as freeware, shareware, or any part of
11 a company's commercial product offering. Products shall be transitioned to a product EtherType
12 before it is deployed in an environment outside the developing organization's administrative control,
13 for example, when deployed with a customer or any other connected environments for testing.
- 14 b) Local Experimental EtherType shall not be permanently assigned for use with a given protocol or
15 protocols.
- 16 c) End stations that bound any administrative domain should be configured to prevent MAC frames
17 containing a Local Experimental EtherType from passing either into or out of a domain in which its
18 contents can be misinterpreted. For example, the default configuration of any firewall should be to
19 not pass this EtherType.

20 A Local Experimental EtherType is processed by the HLPDE in the same manner as other E-Type protocol
21 identifiers, using either Type 2 PIF encoding or Type 3 PIF encoding. However, in order to allow for a
22 single Local Experimental EtherType to multiplex various experimental protocols, sub-protocols, and
23 versions within the same experimental network, a protocol subtype and a protocol version identifier shall be
24 used in conjunction with the Local Experimental EtherType value, as illustrated in Figure 14.

25 9.3 LSAP addresses and L-Type protocol identifiers

26 LSAP addresses values are assigned by the IEEE RA. An LSAP addresses is a sequence of 8 bits, interpreted
27 as a numeric value. The least significant bit is set to 0 for individual identifiers. All LSAP addresses are
28 individual identifiers.

29 Some LSAP addresses are assigned as L-Type protocol identifiers and associated with higher-layer
30 protocols. An example is 0x42, which is used to identify the bridge protocol data unit of IEEE Std 802.1Q.

¹ Some LSAP addresses are not assigned as L-Type protocol identifiers but are instead used within Layer 2.

² An example of such an LSAP address is 0xAA which is used in SNAP encoding.

³ LSAP address 0xFE is the basis of an extensible identifier format, as specified in ISO/IEC TR 9577:1999.

⁴ One use of that extensible protocol identification is the IS-IS protocol of IEEE Std 802.1Q.

⁵ The IEEE Registration Authority (RA) provides a public listing of LSAP addresses²⁷.

6 9.4 O-Type protocol identifiers

⁷ The O-Type protocol identifier, while not directly assigned by a registration authority, is nevertheless

⁸intended to allow a globally-unique association to a protocol. The O-Type protocol identifier is created

9 under the authority of an OUI, OUI-36, or CID assignee by appending bits to the OUI, CID, or OUI-36

10 assignment. The assignee is exclusively authorized to create O-Type protocol identifiers using their OUI,
11 OUI-36 or CID.

¹² An O-Type protocol identifier created by the assignee of an OUI or CID is illustrated in Figure 15.

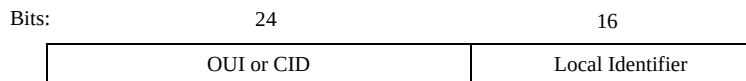


Figure 15—Protocol identifier composed of an OUI or CID

¹³ An O-Type protocol identifier created by the assignee of an OUI-36 is illustrated in Figure 16.

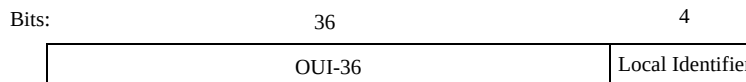


Figure 16—Protocol identifier composed of an OUI-36

14 9.5 PIF Encoding

15 The encoding of a protocol identifier does not change the meaning of the protocol identifier or its association

16 to a protocol. For example, the protocol identified by a particular E-Type EtherType is identical, regardless

17 of its PIF encoding. The same is true of L-Type and O-Type identifiers. If a bridge transforms the PIF

¹⁸ encoding of a MAC frame while relaying, the receiving end station is nevertheless be able to ascertain the

¹⁹ destination protocol as long as it knows the final PIF encoding form.

20 9.5.1 Type 2 PIF encoding

21 9.5.1.1 Type 2 PIF encoding of an L-Type protocol identifier

²² Type 2 PIF encoding of an L-Type protocol identifier entails embedding the protocol identifier as illustrated

23 in Figure 17.

²⁴ For this encoding, the protocol identifier is duplicated in the PIF.

²⁷The LSAP address public listing (<https://standards.ieee.org/products-programs/regauth/lc/public/>) is the public view of the LSAP address registry managed by the IEEE Registration Authority

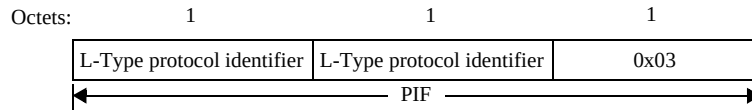


Figure 17—Type 2 PIF encoding of an L-Type protocol identifier

1 NOTE—The special case of L-Type protocol identifier value of 0xAA is disallowed in this encoding.

2 9.5.1.2 Type 2 PIF encoding of an E-Type protocol identifier

3 Type 2 PIF encoding of an E-Type protocol identifier entails embedding the protocol identifier as illustrated
4 in Figure 18.

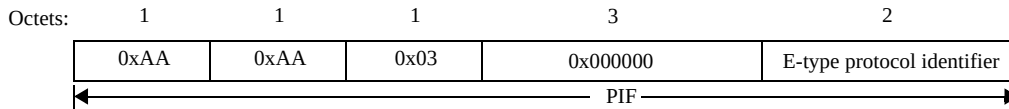


Figure 18—Type 2 PIF encoding of an E-Type protocol identifier

5 The one-octet value 0xAA is never assigned as the L-Type protocol identifier of a network-layer protocol.
6 This allows the HLPDE to distinguish the PIF with respect to the Type 2 PIF encoding of an L-Type
7 protocol identifier, 9.5.1.1.

8 9.5.1.3 Type 2 PIF encoding of an O-Type protocol identifier

9 Type 2 PIF encoding of an O-Type protocol identifier entails embedding the protocol identifier as illustrated
10 in Figure 19.

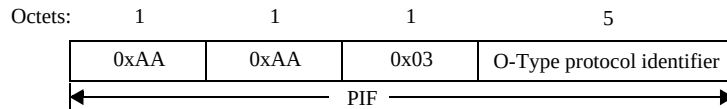


Figure 19—Type 2 PIF encoding of an O-Type protocol identifier

11 The O-Type protocol identifier shall not be set to begin with 0x000000. This allows the HLPDE to
12 distinguish the PIF with respect to the Type 2 PIF encoding of an E-Type protocol identifier.

13 9.5.1.4 SNAP encoding

14 Both the Type 2 PIF encoding of an E-Type protocol identifier and the Type 2 PIF encoding of an O-Type
15 protocol identifier are also known as Subnetwork Access Protocol (SNAP) encoding. SNAP encoding of an
16 EtherType per Figure 18 was first described in RFC 1042 and is known as the RFC 1042 form of SNAP.

17 9.5.2 Type 3 PIF encoding

18 9.5.2.1 Type 3 PIF encoding of an E-Type protocol identifier

19 Type 3 PIF encoding of an E-Type protocol identifier entails embedding the protocol identifier as illustrated
20 in Figure 20.

21 The PIF contains only the EtherType.

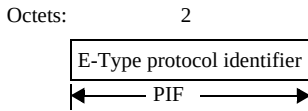


Figure 20—Type 3 PIF encoding of an E-Type protocol identifier

1 NOTE—The EtherType is uniquely distinguishable from any possible value of the Length field, 9.5.2.2.

2 9.5.2.2 Type 3 PIF encoding of an L-Type protocol identifier

3 Type 3 PIF encoding of an L-Type protocol identifier entails embedding the protocol identifier as illustrated
4 in Figure 21.

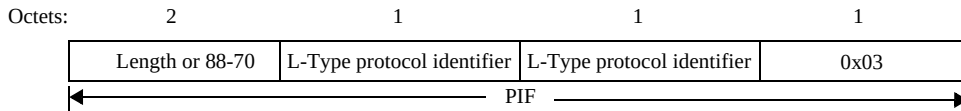


Figure 21—Type 3 PIF encoding of an L-Type protocol identifier

5 The initial field is typically a Length, which takes a value no greater than 0x05DC. Since the minimum
6 EtherType value is 06-00, the HLPDE can distinguish this encoding with respect to the Type 3 PIF encoding
7 of an E-Type protocol identifier. When using a Length, the value of the Length field assigned by the LLC
8 indicates the length of the LLC service data unit in octets, plus 3, but never exceeding 0x05DC. Some MAC
9 sublayers (in particular, that of IEEE Std 802.3) specify that the LLC service data unit may be padded to
10 meet a minimum length, with the Length field unchanged. In this case, the length and the Length field are
11 temporarily inconsistent during transmission; however, the Length field is then used to remove the padding
12 prior to delivery to the LLC.

13 In lieu of a Length, Type 3 PIF encoding of an L-Type protocol identifier alternatively uses the LLC
14 Encapsulation EtherType (value 88-70), which is never used as an E-Type protocol identifier and does not
15 indicate a length. This allows, for example, Type 3 PIF encoding of an L-Type protocol identifier even when
16 the LLC service data unit is too long to be expressed in the limited range of the Length field.

17 The LLC Encapsulation EtherType does not support depadding of padded short MAC frames. Likewise,
18 Type 3 PIF encoding of an E-Type protocol identifier does not provide a Length for depadding. In either
19 case, the higher-layer protocol might need to provide a depadding service for short MAC frames. If the LLC
20 service data unit is sufficiently long so that MAC padding is not added, then MAC uses neither the Length
21 field nor the LLC Encapsulation EtherType as indicative of a Length value.

22 9.5.2.3 Type 3 PIF encoding of an O-Type protocol identifier

23 Type 3 PIF encoding of an O-Type protocol identifier entails embedding the protocol identifier in a PIF as
24 illustrated in Figure 22.

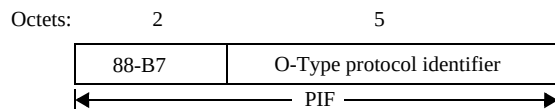


Figure 22—Type 3 PIF encoding of an O-Type protocol identifier

1 The initial field is the OUI Extended EtherType which is never used as an E-Type protocol identifier. This
2 allows the HLPDE to distinguish this encoding with respect to the Type 3 PIF encodings of the E-Type and
3 L-Type protocol identifier.

4 **9.5.3 Encoding type and PIF length**

5 Type 3 PIF encoding is more efficient than Type 2 PIF encoding for carrying the E-Type protocol identifier
6 due to a smaller PIF: 2 octets vs. 8 octets. E-Type protocol identifiers are typically the most common in use.
7 Type 3 PIF encoding is also more efficient than Type 2 PIF encoding for carrying the O-Type protocol
8 identifier: 7 octets vs. 8 octets. Type 2 PIF encoding is more efficient than Type 3 PIF encoding for carrying
9 the L-Type protocol identifier: 3 octets vs. 5 octets.

10 **9.6 Context-dependent identifiers**

11 An IEEE RA tutorial [B2] explains the creation of context dependent identifiers. Just as the OUI is extended
12 to create EUI-48 and EUI-64 identifiers, or a CID can be extended to create a locally administered MAC
13 address, other extended identifiers can be created from an OUI or CID assignment. Such extended identifiers
14 are referred to as context-dependent identifiers. These identifiers are not necessarily globally unique, but are
15 intended to only be unique within a well specified context.

16 In some cases, the context of a context-dependent identifier is the IEEE 802 LAN. Since this is the same
17 context in which local identifiers operate, the SLAP of Clause 8 provides a basis to assign unique context-
18 dependent identifiers, such as NUI-48 and NUI- 64, within that context.

10. Allocation of OID values in IEEE 802 standards

10.1 General

From time to time, various IEEE 802 standards have a requirement to allocate OID values. The most common example is for defining management information base (MIB) objects for SNMP, but other examples exist. MIB modules describe the structure of the management data of a device subsystem and use a hierarchical name space based on OIDs to identify variables. This clause specifies a simple and consistent OID hierarchy, based on the use of the OID value that has been assigned by ISO to identify the IEEE 802 series of standards. This hierarchy should be used by all current and future IEEE 802 Working Groups and can be used flexibly to meet the needs of the standards developed by those working groups. This establishes a consistent practice within IEEE 802 for the development and allocation of OIDs. Consistency of OID allocation facilitates implementation and operation of IEEE 802-compliant equipment.

10.2 OIDs and ISO standards

An OID is an ASN.1 data type, specified in ITU-T Recommendation X.660, that is used as a means of defining unique identifiers for objects. Values of the OID data type can then be used to name the objects to which they relate.

The OID data type consists of a sequence of one or more non-negative integers, often referred to as arcs, that specify a hierarchy, or tree, of OID values. The first arc in the sequence identifies the registration authority responsible for allocating the values of the second and subsequent arcs. For example:

iso (1)

indicates that an initial arc value of 1 identifies ISO as the registration authority. Subsequent arcs in the sequence are determined by ISO or are allocated by registration authorities subordinate to ISO.

Under the iso arc, a second arc has been allocated to identify organizations recognized by ISO, such as the IEEE; hence, the two-integer sequence

iso (1) iso-identified-organization (3)

Under the iso-identified-organization arc, a subsequent arc has been allocated to identify the IEEE; hence, the three-integer sequence

iso (1) iso-identified-organization (3) ieee (111)

indicates that the fourth integer identifies a particular registry within the IEEE and that the allocation of the fourth and subsequent arcs is the responsibility of the IEEE. Under the ieee arc, the IEEE RA has specified an arc for the numbered series of IEEE standards; hence, the four-integer sequence

iso (1) iso-identified-organization (3) ieee (111)
standards-association-numbered-series-standards (2)

indicates that the fifth integer is used to identify a particular IEEE numbered series standard. The actual number corresponding to the numbered series standard is used in the fifth arc; hence, the following identifies the IEEE 802 series of standards:

iso (1) iso-identified-organization (3) ieee (111)
standards-association-numbered-series-standards (2) ieee-802 (802)

1 The responsibility for allocating the subsequent arcs under iso (1) iso-identified-organization (3) ieee (111)
2 standards-association-numbered-series-standards (2) ieee-802 (802) lies with IEEE 802.

3 As the standard number 802 is used to identify a member of the family of IEEE 802 standards, this particular
4 sequence of integer values can form the basis of an OID hierarchy for use by the individual standards in the
5 IEEE 802 family. The act of assigning a number to a standard has the effect of automatically assigning an
6 OID arc to that standard; therefore, no further administrative effort is needed before that standard can
7 allocate OID values under that point in the tree, using the subsequent arcs.

8 **10.3 The OID hierarchy for IEEE 802 standards**

9 The OID value assigned to the family of IEEE 802 standards is:

10 iso (1) iso-identified-organization (3) ieee (111)
11 standards-association-numbered-series-standards (2) ieee-802 (802)

12 The next arc under iso (1) iso-identified-organization (3) ieee (111) standards-association-numbered-series-
13 standards (2) ieee-802 (802) is used to differentiate between members of the family of IEEE 802 standards,
14 by using it as a working group designator, as follows:

15 iso (1) iso-identified-organization (3) ieee (111)
16 standards-association-numbered-series-standards (2) ieee-802 (802) ieee802dotX (X)

17 where X is the working group number of the IEEE 802 Working Group responsible for that standard. These
18 arcs are assigned for use in all current and future IEEE 802.X standards.

19 For example, under this hierarchy, the value used for standards developed by the IEEE 802.3 Working
20 Group is:

21 iso (1) iso-identified-organization (3) ieee (111)
22 standards-association-numbered-series-standards (2) ieee-802 (802) ieee802dot3 (3)

23 and the value used for IEEE 802.11™ standards is:

24 iso (1) iso-identified-organization (3) ieee (111)
25 standards-association-numbered-series-standards (2) ieee-802 (802) ieee802dot11 (11)

26 The working group concerned is free to decide how further arcs are allocated within their standards, in a
27 manner that makes sense for their particular needs.

28 It is the responsibility of each working group to ensure that any values that are allocated to the fifth and
29 subsequent arcs are documented, in a manner that ensures that the same OID value cannot be assigned to
30 two different objects. In the IEEE 802.1 Working Group, this has been achieved in the past by placing tables
31 of OID allocations in an annex within the standard concerned²⁸; in the IEEE 802.3 Working Group, a master
32 spreadsheet of allocated OID values is maintained by the chair and posted on the working group's website.
33 For future allocations, adopting a master spreadsheet approach is appropriate.

34 It is important that the allocation scheme for the fifth and subsequent arcs is constructed in a manner that
35 leaves appropriate “escapes” for uses that cannot be foreseen. The simple expedient of allocating a “type of
36 allocation” value as the fifth arc is sufficient to ensure that such an escape is always available.

²⁸More information on IEEE 802.1 OIDs can be found on the working group web site, <https://www.ieee802.org/1/pages/OIDS.html>.

10.4 The OID hierarchy under iso(1) std(0) iso8802(8802)

The 2001 revision of this standard documented the use of iso(1) std(0) iso8802(8802) as the root arc under which IEEE 802 standards would develop their OID hierarchies. The use of this root arc is deprecated.

10.5 Migration from previous OID allocations

The OID hierarchy described in this clause need not have any effect upon existing IEEE 802 standards that have already solved this problem by using a specific allocation obtained elsewhere (for example, from ANSI).

With the hierarchy as specified in this clause, as each new working group is created in IEEE 802, its base OID arc is also created automatically; therefore, no administrative effort is required on the part of the working group, other than to determine how the fifth and subsequent arcs are used in its standards.

For those working groups that have already made use of other allocation schemes (e.g., IEEE 802.3 and IEEE 802.1), it may be considered appropriate to migrate existing allocations to the hierarchy specified in this clause. In considering this, the following should be borne in mind:

- While it might be perceived as “tidy” to have all IEEE 802 OIDs allocated under a single arc of the OID tree, this is not a requirement for any other reason; one OID value is no better or no worse than any other from a technical point of view (with the possible exception that the encoded length can vary with the number of arcs to be encoded), as long as any given OID identifies a single object.
- If migration is desired, there is no requirement to remove the old OID values²⁹. Indeed, this is not permitted for objects in SNMP MIB modules that are not obsolete, as specified in IETF RFC 2578, nor is it permitted to associate such objects with more than one OID value. Instead, new definitions are required to be created and registered under the desired OID tree³⁰.

²⁹There is no general requirement that an object should have only a single identifier; if it has more than one, then it can be “reached” by following more than one set of branches of the naming tree, just as a map can provide more than one path to a destination.

³⁰This appears to contradict the earlier statement and footnote that indicate that it does not matter if multiple OIDs point at the same object; however, this is a specific requirement imposed on MIB objects for SNMP by the relevant IETF standards, and not a general rule.

11. Allocation of Uniform Resource Name (URN) values in IEEE 802 standards

11.1 Introduction

From time to time, some IEEE 802 standards have a requirement to allocate Uniform Resource Name (URN) values—the most common example being for the purpose of defining data models using the YANG data modeling language specified in IETF RFC 6020 [B16] and IETF RFC 7950 [B18], but other examples exist. This clause specifies a simple and consistent URN hierarchy, based on the use of the base URN value that has been assigned by the Internet Assigned Numbers Authority (IANA) for use in IEEE standards. All current and future IEEE working groups can use this hierarchy flexibly to meet the needs of the standards specified by those working groups. This hierarchy provides a consistent practice within IEEE 802 for the development and allocation of URNs. Consistency of URN allocation facilitates implementation and operation of IEEE 802 compliant equipment.

NOTE—While the focus of this Clause is on the use of URN values in IEEE 802 standards, the base URN value identified in 11.2 and the hierarchy of values that follows forms a basis for the assignment of URN values in all IEEE standards, not just those developed by IEEE 802.

11.2 The IEEE Namespace ID and Namespace Specific String

URN values used in IEEE standards use the following Namespace ID (NID) value assigned to the IEEE (see IETF RFC 3406 and IETF RFC 8069):

ieee

The Namespace Specific String (NSS) of all URNs that use the IEEE NID shall use the following structure:

urn:ieee:{IEEEresource}:{ResourceSpecificString}

The strings used as values of IEEEresource and ResourceSpecificString are case-insensitive.

There are potential uses of URNs in the IEEE outside of standards use. Only standards use is considered in this standard; therefore, the IEEEresource is always as follows:

std

Hence, all URN values assigned for use in the context of IEEE standards are of the following form:

urn:ieee:std:{ResourceSpecificString}

NOTE—The mechanism for allocation of URN values used by the IEEE is fully conformant with IETF RFC 3406 and is documented in IETF RFC 8069.

11.3 ResourceSpecificString values in IEEE 802 standards

ResourceSpecificString values identify the IEEE standard that has assigned the URN value, and the particular resource specified by that standard that the URN value identifies. The structure of ResourceSpecificString is as follows:

{IEEE standard designation}:{resourceType}:{resourceIdentifier}

1 {IEEE standard designation} is the standard designation assigned to the base standard that specifies the
2 URN value. For example, in the case of IEEE Std 802.1Q, the standard designation is 802.1Q; in the case of
3 IEEE Std 802.11, the standard designation is 802.11. Where URN values are assigned in amendments or
4 corrigenda to a base standard, the base standard's IEEE standard designation shall be used, not the IEEE
5 standard designation of the amendment or corrigendum. The IEEE standard designation shall not include
6 any colons. The form of standard designation numbers is as specified in the IEEE SA Project Numbering
7 Policy.³¹

8 {resourceType} identifies the type of resource to which the URN value applies. A single value of
9 resourceType is specified for use across all IEEE 802 standards as follows:

10 yang

11 The yang resourceType shall be used to create any URN in YANG modules specified in IEEE 802
12 standards.

13 Should further resourceType values be required for consistent use across multiple IEEE 802 standards, they
14 would be specified via future amendments to this standard. Further resourceType values that are specific to
15 a designated IEEE 802 standard can be specified within that standard.

16 The {resourceIdentifier} identifies a specific resource, in the context of the designated IEEE standard and
17 the resourceType. All resourceIdentifier values are specified within the designated standard.

18 For example, in IEEE Std 802.1Q, a URN value for use in a YANG module would take the following form:

19 urn:ieee:std:802.1Q:yang:{resourceIdentifier}

20 Or in IEEE Std 802.11, a URN value for use in a YANG module would take the following form:

21 urn:ieee:std:802.11:yang:{resourceIdentifier}

³¹The current IEEE SA Numbering Policy Document is found at <https://mentor.ieee.org/myproject/Public/mytools/init/parnum.pdf>.
Some IEEE projects use a numbering method that predates this policy

¹ Annex A

² (informative)

³ Bibliography

⁴ Bibliographic references are resources that provide additional or helpful material but do not need to be
⁵ understood or used to implement this standard. Reference to these resources is made for informational use
⁶ only.

⁷ [B1] IEEE Project Authorization Request (PAR) P802.1CQ, Draft Standard for Local and Metropolitan
⁸ Area Networks: Multicast and Local Address Assignment, February 2016.³²

⁹ [B2] IEEE Registration Authority Tutorial, “Guidelines for Use of Extended Unique Identifier (EUI),
¹⁰ Organizationally Unique Identifier (OUI) and Company ID (CID)”.³³

¹¹ [B3] IEEE Std 802.1AS™-2020, IEEE Standard for Local and metropolitan area networks—Timing and
¹² Synchronization for Time-Sensitive Applications in Bridged Local Area Networks.

¹³ [B4] IEEE Std 802.1BA™, IEEE Standard for Local and metropolitan area networks—Audio Video
¹⁴ Bridging (AVB) Systems.

¹⁵ [B5] IEEE Std 802.1CB™-2017, IEEE Standard for Local and metropolitan area networks—Frame
¹⁶ Replication and Elimination for Reliability.

¹⁷ [B6] IEEE Std 802.1CM™, IEEE Standard for Local and metropolitan area networks—Time-Sensitive
¹⁸ Networking for Fronthaul.

¹⁹ [B7] IEEE Std 802.1CS™-2020, IEEE Standard for Local and metropolitan area networks—Link-local
²⁰ Registration Protocol.

²¹ [B8] IEEE Std 802.3™-2022, IEEE Standard for Ethernet.

²² [B9] IETF RFC 1042, A Standard for the Transmission of IP Datagrams over IEEE 802 Networks. Postel, J.,
²³ and J. Reynolds, Feb. 1988.³⁴

²⁴ [B10] IETF RFC 1390, Transmission of IP and ARP over FDDI Networks. Katz, D., Jan. 1993.

²⁵ [B11] IETF RFC 2464, Transmission of IPv6 Packets over Ethernet Networks.

²⁶ [B12] IETF RFC 2579, STD 58, Textual Conventions for SMIv2. McCloghrie, K., D. Perkins,
²⁷ J. Schoenwaelder, J. Case, M. Rose, and S. Waldbusser, Apr. 1999.

²⁸ [B13] IETF RFC 2580, STD 58, Conformance Statements for SMIv2. McCloghrie, K., D. Perkins,
²⁹ J. Schoenwaelder, J. Case, M. Rose, and S. Waldbusser, Apr. 1999.

³⁰ [B14] IETF RFC 3411, STD 62, An Architecture for Describing Simple Network Management Protocol
³¹ (SNMP) Management Frameworks.

³²IEEE publications are available from The Institute of Electrical and Electronics Engineers (<https://standards.ieee.org/>)

³³The tutorial is available at <https://standards.ieee.org/regauth>. Follow the tutorial link and search for the tutorial title.

³⁴ETF documents (i.e., RFCs) are available the Internet Engineering Task Force (<https://www.rfc-archive.org/>).

¹ [B15] IETF RFC 5677, IEEE 802.21 Mobility Services Framework Design (MSFD). Melia, T., G. Bajko,
² S. Das, N. Golmie, and J. C. Zuniga, Dec. 2009.

³ [B16] IETF RFC 6020, YANG—A Data Modeling Language for the Network Configuration Protocol
⁴ (NETCONF), October 2010.

⁵ [B17] IETF RFC 6241, Network Configuration Protocol (NETCONF), June 2011.

⁶ [B18] IETF RFC 7950, The YANG 1.1 Data Modeling Language, August 2016.

⁷ [B19] IETF RFC 8519, YANG Data Model for Network Access Control Lists (ACLs), March 2019.

⁸ [B20] ISO/IEC 7498-1:1994, Information technology—Open Systems Interconnection—Basic Reference
⁹ Model: The Basic Model.

1 **Annex B**

2 (informative)

3 **Reference Models for IEEE 802 standards**

4 **B.1 IEEE 802.3 RMs**

- 5 IEEE Std 802.3 offers multiple options, each of which has a different RM.
- 6 The basic RM for IEEE 802.3 stations without optional sublayers is illustrated in Figure B.1.

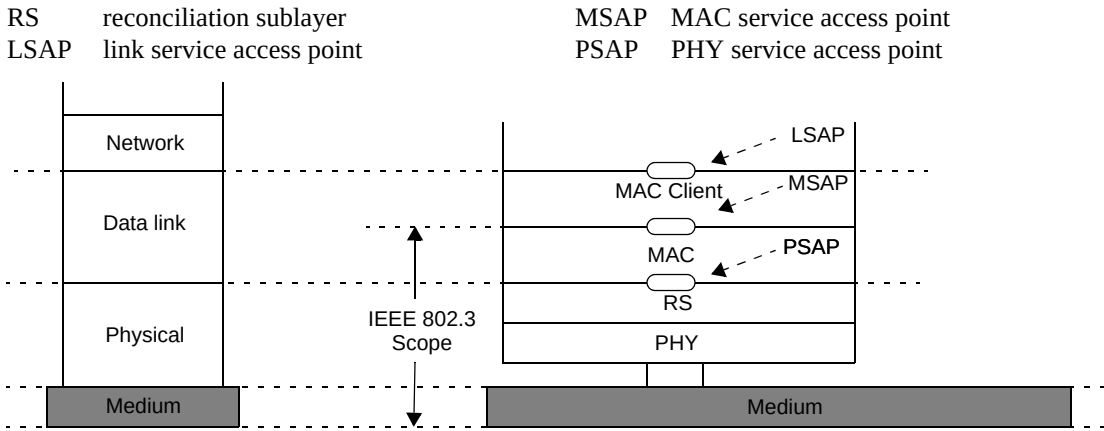


Figure B.1—Basic RM for IEEE 802.3 stations

- 7 The RM for IEEE Std 802.3 is illustrated in Figure B.2.

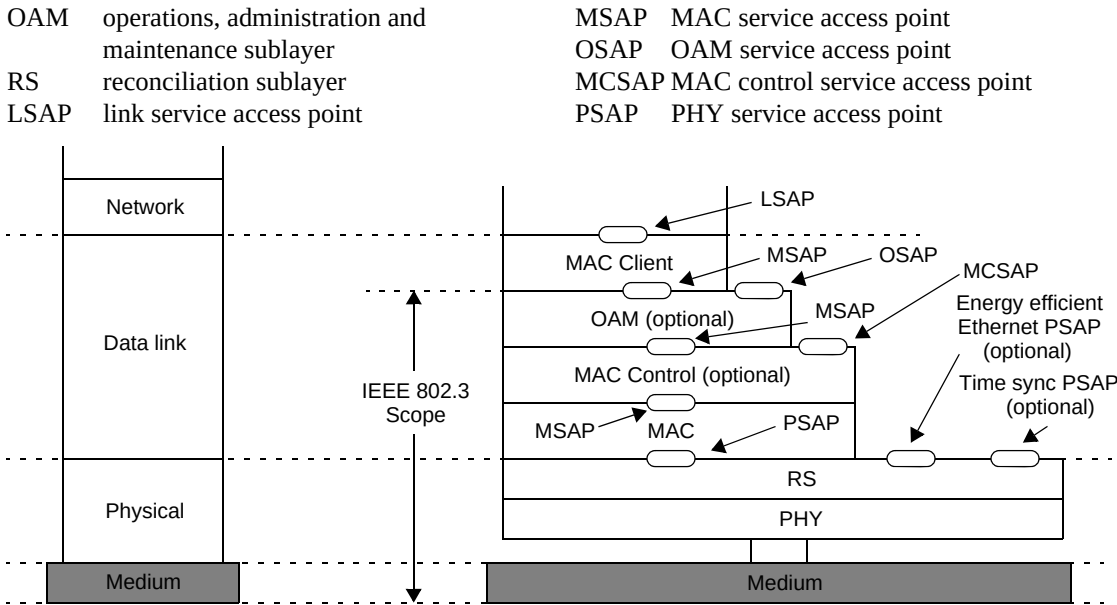


Figure B.2—The RM for IEEE 802.3 point-to-point stations

1 The RM for IEEE 802.3 Ethernet passive optical networks (EPON) optical line terminal (OLT) is illustrated
2 in Figure B.3.

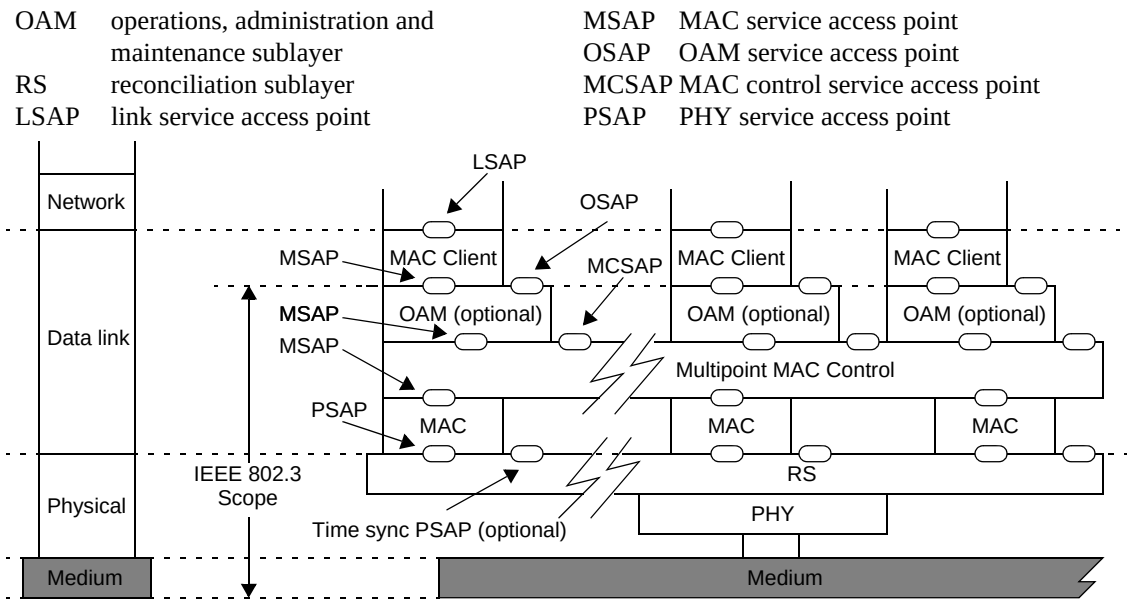


Figure B.3—IEEE 802.3 RM for point-to-multipoint OLT

3 The RM for IEEE 802.3 EPON optical network unit (ONU) is illustrated in Figure B.4.

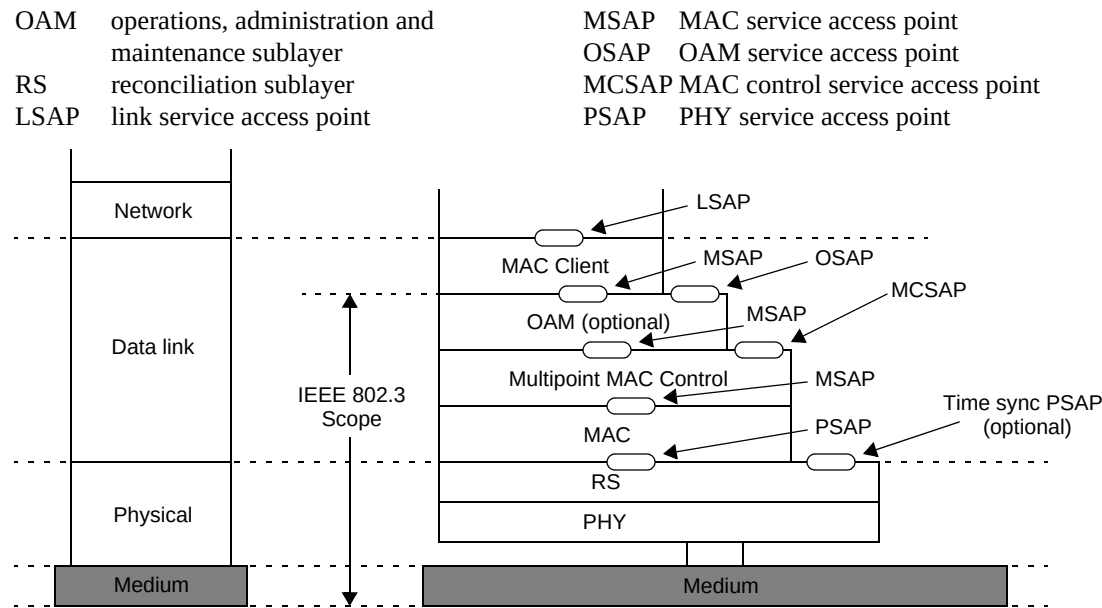


Figure B.4—The RM for IEEE 802.3 point-to-multipoint ONU

4 IEEE Std 802.3 was amended in 2004 to introduce the concept of subscriber access network.³⁵ The purpose
5 of Ethernet in the first mile (EFM), as well as its distinction from traditional Ethernet networks, is that it
6 specifies functionality required for the subscriber access network, i.e., public network access. Network

³⁵The amendment was IEEE Std 802.3ah™-2004, which is now part of the current IEE Std 802.3.

1 design considerations for public access that may differ from traditional Ethernet LANs include the OAM
2 function and the regulatory requirements.

3 **B.2 IEEE 802.11 RM**

4 The IEEE 802.11 RM is based on the functional station (STA) model, as shown in Figure B.5.

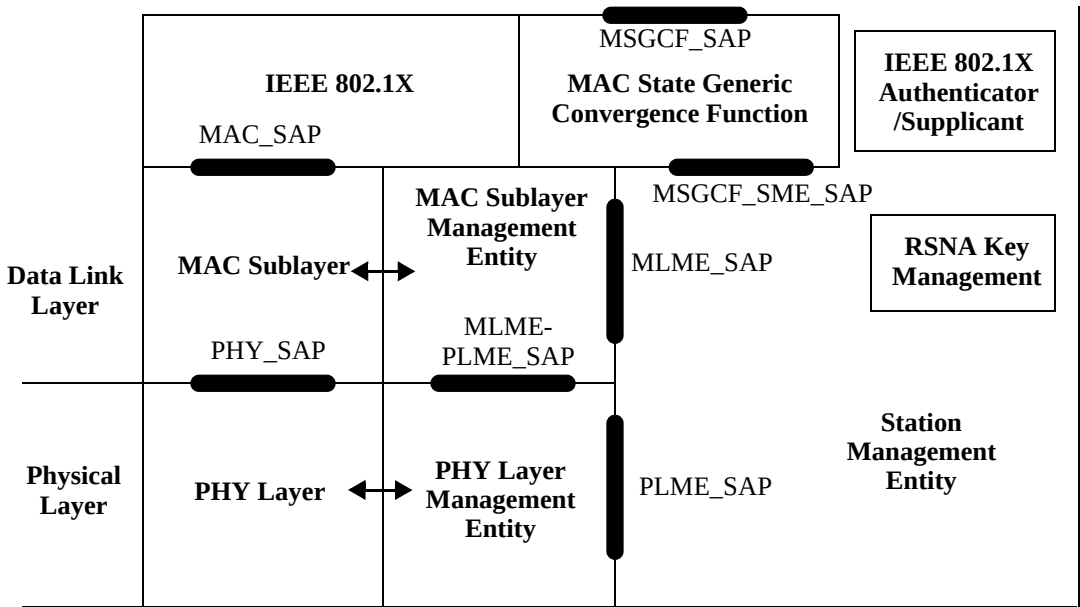


Figure B.5—IEEE 802.11 STA RM

5 The interconnections between IEEE 802.11 STAs follow four general connection models.

6 The first interconnection model provides several types of peer-to-peer, direct, pair-wise communication
7 between STAs, each applicable in differing use scenarios. In these direct communications the STAs in each
8 pair have symmetrical operations, with each STA matching the functional STA model, although they can
9 take on different behavioral roles to establish and maintain the interconnection link.

10 The second interconnection model, the infrastructure model, supports multiple STAs, collected into one or
11 more wireless access domains, called basic service sets (BSSs). These access domains (BSSs) are
12 interconnected via the distribution system (DS) and can interwork with other IEEE 802 networks via a
13 portal.

14 Each access domain in the infrastructure model is established by an access point (AP), which extends the
15 basic STA model to include repeating and forwarding functions that allow communications between non-AP
16 STAs that do not directly interconnect. The AP, acting in cooperation with the DS, is a forwarding entity
17 that enables communications between non-AP STAs within the access domain (intra-BSS relay) and also to
18 different IEEE 802.11 wireless access domains established by other APs connected to the same DS (inter-
19 BSS relay) and/or via a portal to non-IEEE 802.11 networks.

20 The third interconnection model, is a mesh model consisting of autonomous STAs. Inside the mesh, STAs
21 establish peer-to-peer wireless links with neighbor STAs to mutually exchange messages. Further, using the
22 mesh's multi-hop capability, messages can be transferred between STAs that are not in direct
23 communication with each other over a single instance of the wireless medium. From the data delivery point

1 of view, it appears as if all STAs in a mesh are directly connected at the MAC layer even if the STAs are not
2 within range of each other. A mesh might have an interface to the distribution system, through a Mesh Gate,
3 and thereby can enable communication to non-AP STAs in infrastructure access domains, and/or via a portal
4 to non-IEEE 802.11 networks.

5 The fourth interconnection model, is the general link (GLK) model consisting of GLK STAs connected by
6 IEEE 802.11 general links that are suitable to be used as links inside an IEEE 802.1Q bridged network. A
7 GLK STA coordinates with a GLK convergence function to provide an instance of Internal Sublayer
8 Service, as specified in IEEE Std 802.1AC-2016, to an IEEE 802.1Q bridge for each peer GLK STA with
9 which it is communicating. GLK STAs also provide link metrics for the use of external path selection
10 protocols such as spanning tree protocol. GLK operation does not involve a DS. Instead, the general links
11 formed with GLK operation are a point-to-point connection between pairs of instances of Internal Sublayer
12 Service SAPs.

13 Figure B.6 illustrates the infrastructure model for APs, the distribution system and a portal. The arrows
14 indicate the intra-BSS and inter-BSS relay functions for MSDUs as well as interconnection to other
15 IEEE 802 networks.

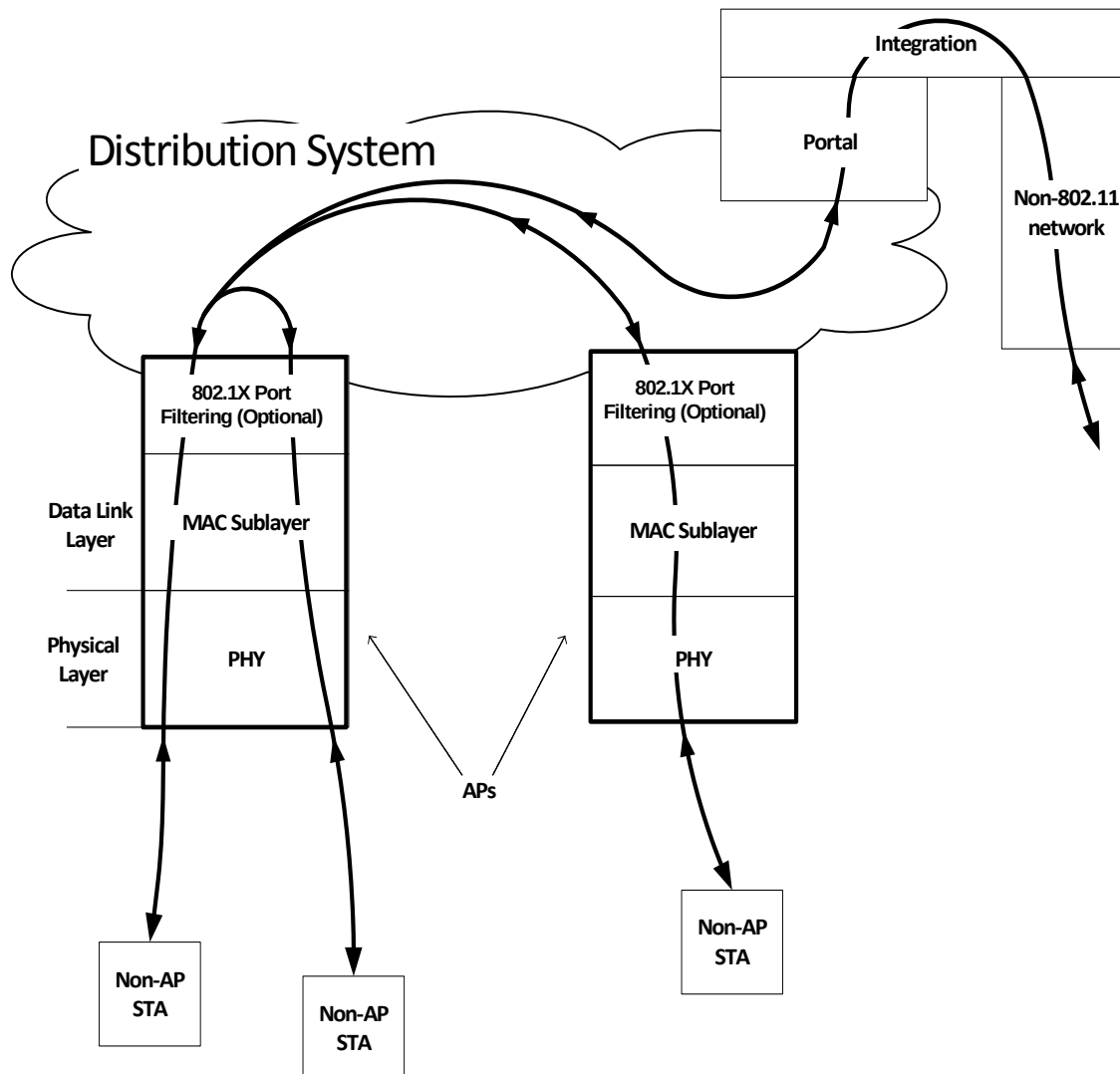


Figure B.6—IEEE 802.11 infrastructure model

1 Figure B.7 illustrates the infrastructure model for GLK APs and GLK non-APs STA s, the general links
2 shown connect the IEEE 802.1Q MAC relay entities shown with each other or the LLC sublayer of the end
3 stations shown.

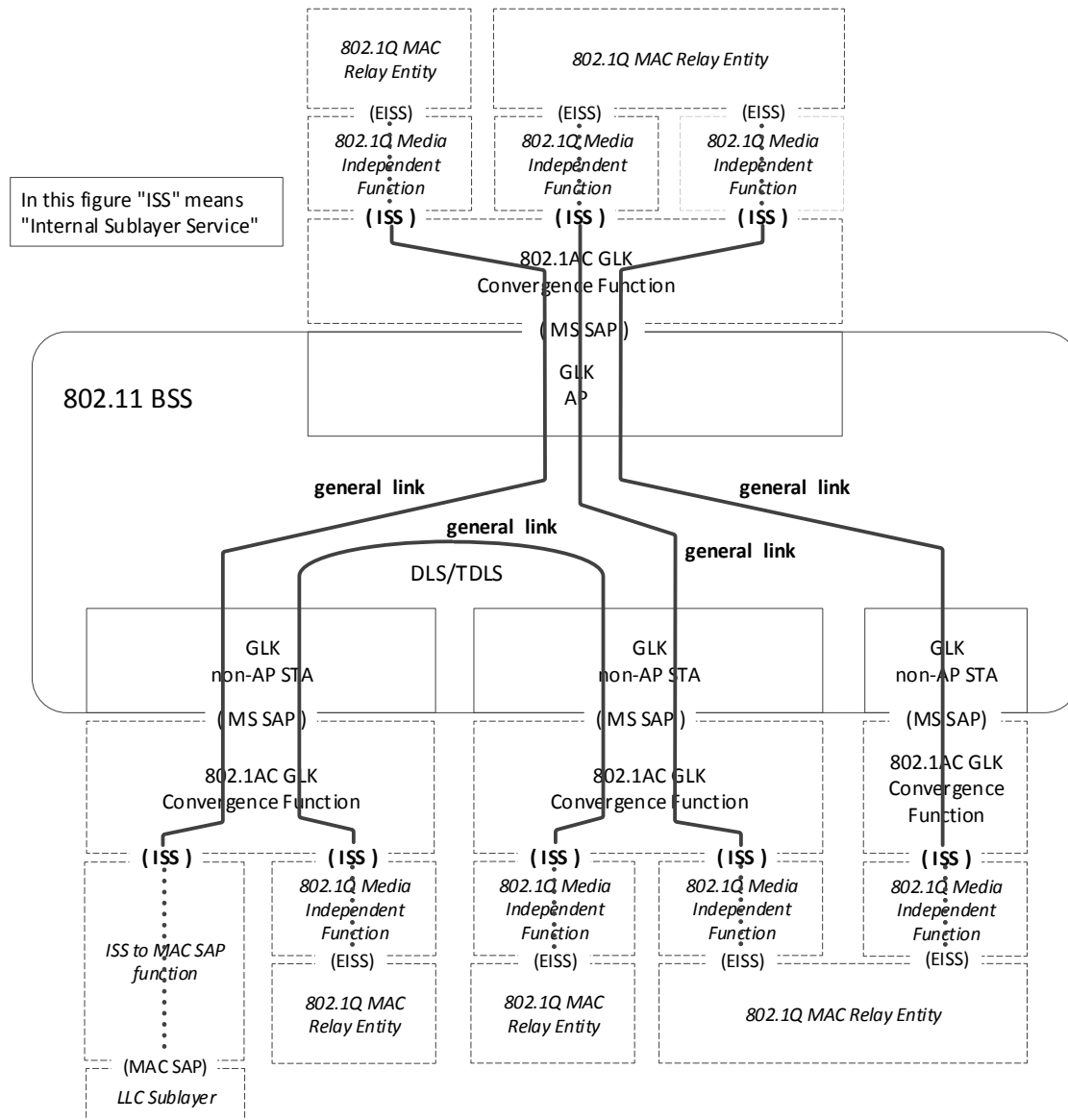


Figure B.7—Example of infrastructure BSS with general links

4 B.3 IEEE 802.15™ RMs

5 B.3.1 IEEE 802.15.3™ RM

⁶ The RM for IEEE Std 802.15.3 is illustrated in Figure B.8.

7 The PHY SAP and physical layer management entity (PLME) SAP are not specified in IEEE Std 802.15.3
8 as they are rarely, if ever, exposed in a typical implementation. The PHY management objects and attributes

1 are accessed through the MAC sublayer management entity (MLME) SAP with the generic management
2 primitives used to access the MAC management objects and attributes.

3 The MAC SAP and MLME SAP are specified as logical interfaces to access the services provided by
4 IEEE 802.15.3 end stations.

5 The PLME and MLME are logical entities that control the PHY and MAC, respectively, based on request
6 from the higher layers.

7 The frame convergence sublayer (FCSL) is used to allow multiple protocols to simultaneously
8 access the services of an IEEE 802.15.3 PAN. IEEE Std 802.15.3 specifies an FCSL for connection to the
9 ISO/IEC 8802-2.

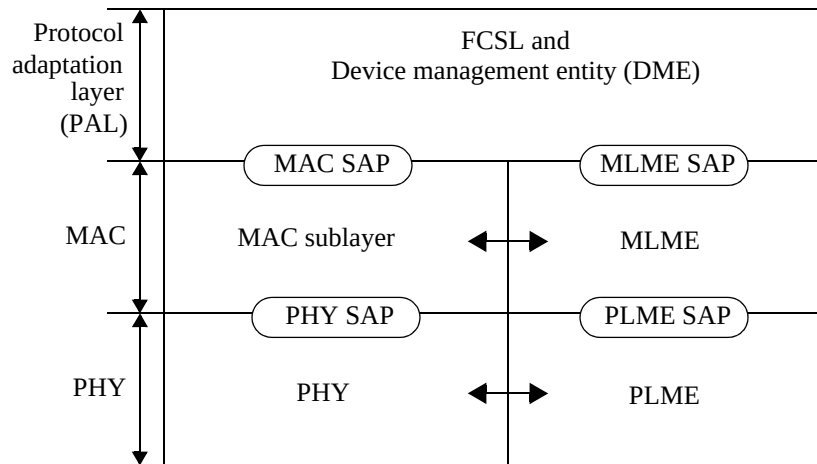


Figure B.8—IEEE 802.15.3 RM

1 B.3.2 IEEE 802.15.4™ RM

- 2 The RM for IEEE Std 802.15.4 is illustrated in Figure B.9.
- 3 The upper layers shown in Figure B.9 consist of a network layer (which provides network configuration,
4 manipulation, and message routing) and an application layer (which provides the intended function of the
5 device). The upper layers are not specified in IEEE Std 802.15.4.

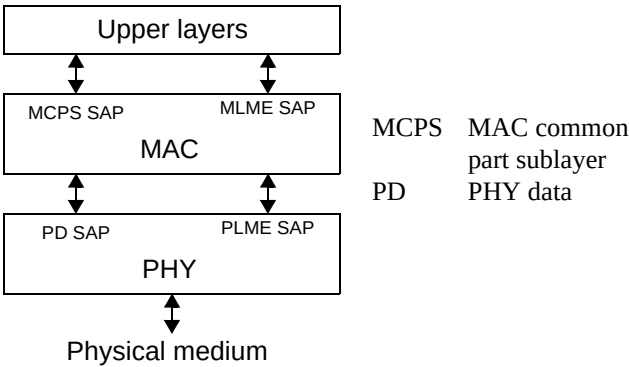


Figure B.9—IEEE 802.15.4 RM

6 B.3.3 IEEE 802.15.6™ RM

- 7 The RM for IEEE 802.15.6 hub or node are shown in Figure B.10.

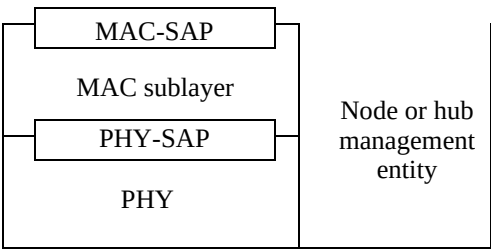


Figure B.10—IEEE 802.15.6 RM

8 B.3.4 IEEE 802.15.7™ RM

- 9 The RM for IEEE Std 802.15.7™ is shown in Figure B.11.
- 10 The MAC sublayer provides the following two services, accessed through two SAPs:
- 11 — The MAC data service, accessed through the MAC common part sublayer (MCPS) data SAP (MCPS-SAP)
 - 12 — The MAC management service, accessed through the MLME-SAP
- 13 In addition to these external interfaces, an implicit interface also exists between the MLME and the MCPS
- 14 that allows the MLME to use the MAC data service.
- 15
- 16 The PHY provides two services, accessed through two SAPs:
- 17 — The PHY data service, accessed through the PHY data SAP (PD-SAP)
 - 18 — The PHY management service, accessed through the PLME's SAP (PLME-SAP).

1 The optical SAP (OPTICAL-SAP) provides an interface between the PHY and the optical channel and is not
2 specified in the standard.

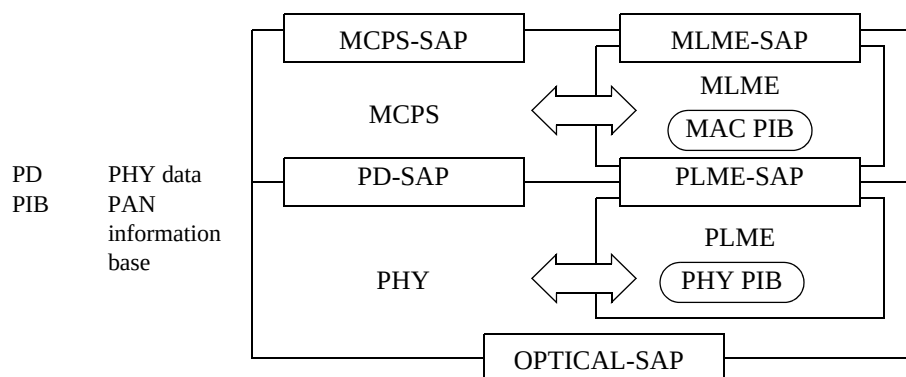


Figure B.11—IEEE 802.15.7 RM

3 B.4 IEEE 802.16™ RM

4 B.4.1 Protocol RM

5 Figure B.12 illustrates the protocol RM for IEEE Std 802.16.

6 The service-specific convergence sublayer (CS) provides any transformation or mapping of external
7 network data, received through the CS SAP, into MSDUs received by the MCPS through the MAC SAP.
8 This includes classifying external network service data units and associating them to the proper MAC
9 service flow identifier and connection identifier. Multiple CS specifications are provided for interfacing with
10 various protocols.

11 The MAC CPS provides the core MAC functionality of system access, bandwidth allocation, connection
12 establishment, and connection maintenance. Quality of service is applied to the transmission and scheduling
13 of data over the PHY.

14 The security sublayer in the MAC provides authentication, secure key exchange, and encryption.

15 Data, PHY control, and statistics are transferred between the MAC CPS and the PHY via the PHY SAP
16 (which is implementation specific).

17 The PHY definition includes multiple specifications, each appropriate to a particular frequency range and
18 application.

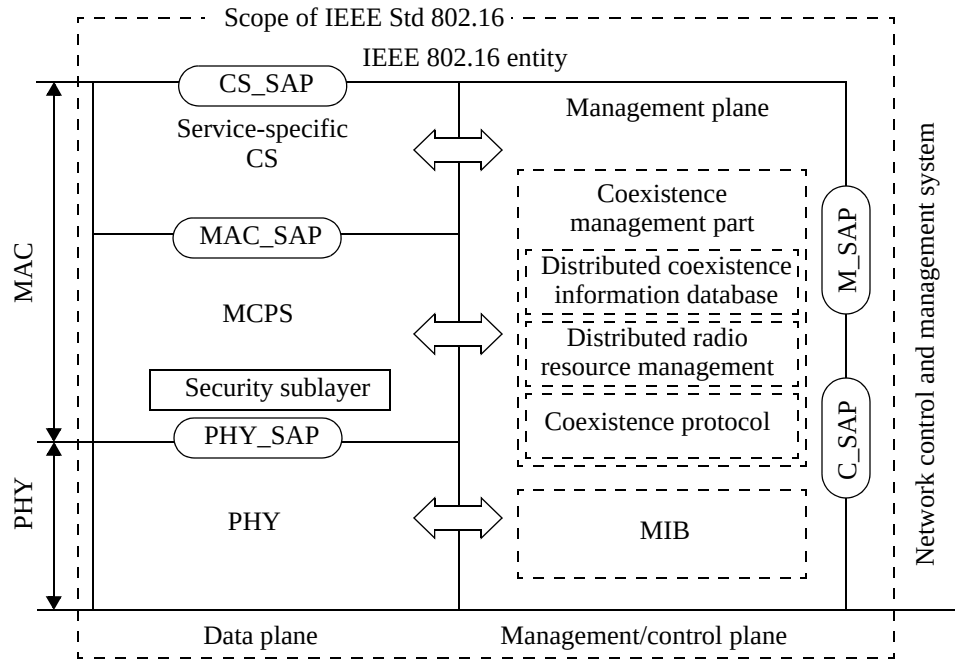


Figure B.12—IEEE 802.16 protocol RM

1 B.4.2 Network RM

2 Figure B.13 describes a simplified network RM for IEEE Std 802.16.

3 The network control and management system (NCMS) abstraction allows the PHY/MAC layers specified in
4 IEEE Std 802.16 to be independent of the network architecture, the transport network, and the protocols
5 used at the backend and, therefore, allows greater flexibility.

6 NCMS logically exists at base station (BS) side and subscriber station/mobile subscriber station (SS/MS)
7 side of the radio interface, termed *NCMS(BS)* and *NCMS(SS/MS)*, respectively. Any necessary inter-BS
8 coordination is handled through the *NCMS(BS)*.

9 The control service access point (C-SAP) and management service access point (M-SAP) expose the control
10 plane and management plane functions to upper layers. The NCMS uses the C-SAP and M-SAP to interface
11 with the IEEE 802.16 entity. In order to provide correct MAC operation, NCMS is present within each SS/
12 MS. The NCMS is a layer independent entity that may be viewed as a management entity or control entity.
13 General system management entities can perform functions through NCMS, and standard management
14 protocols can be implemented in the NCMS.

15 An IEEE 802.16 entity is the logical entity in an SS/MS or BS that comprises the PHY and MAC layers of
16 the data plane and the management/control plane. The IEEE 802.16 end stations can include SS, MS or BS.
17 Multiple SS or MS may be attached to a BS.

18 SS or MS communicate to the BS over the U interface using a primary management connection, a basic
19 connection, or a secondary management connection.

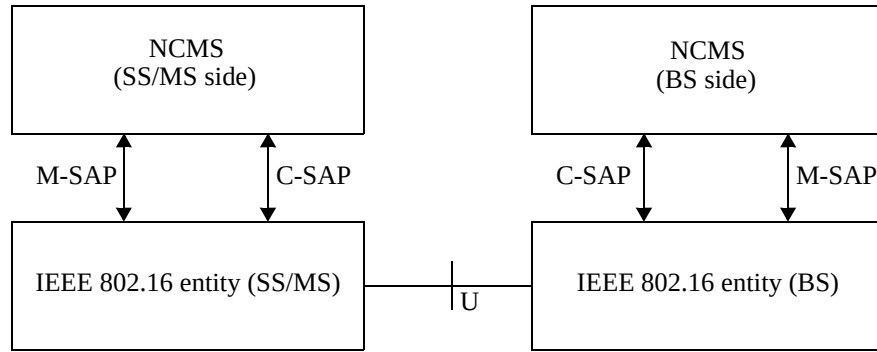


Figure B.13—IEEE 802.16 network RM

1 B.5 IEEE 802.21™ RM

2 Figure B.14 shows an implementation view of a dual-mode IEEE 802.11/IEEE 802.16 station with
3 IEEE 802.21 MIH functionality. The MIHF provides the required services to perform handovers between
4 IEEE 802.11 and IEEE 802.16 access technologies. Also, the MIHF becomes a higher layer when it requires
5 data transport services to communicate with an IEEE 802.21 MIH peer. For Layer 2 transport of MIH data,
6 services are provided by the IEEE 802.16 CS_SAP or the IEEE 802.11 LSAP. For Layer 3 transport,
7 services are provided as described in IETF RFC 5677 [B15].

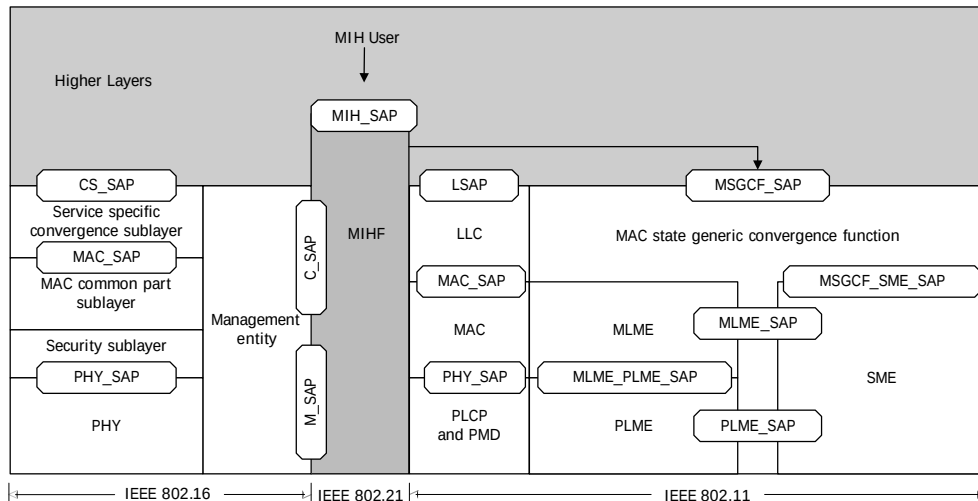


Figure B.14—Example of dual-mode IEEE 802.11 and IEEE 802.16 end station with IEEE 802.21 end-station RM

1 **B.6 IEEE 802.22™ RM**

2 The RM of IEEE Std 802.22 is depicted in Figure B.15. A unique characteristic of this architecture is its
3 cognitive components, which are used to allow for dynamic frequency selection and avoid interference to
4 incumbents on a real-time basis.

AAA	authentication, authorization and accounting	MAC SAP	MAC sublayer service access point
C-SAP	control service access point	PHY SAP	PHY service access point
CS SAP	convergence sublayer service access point	SM-SSF SAP	spectrum manager, spectrum sensing function service access point
M-SAP	management service access point	SM-GL SAP	spectrum manager, geolocation service access point
NCMS	network control and management system		

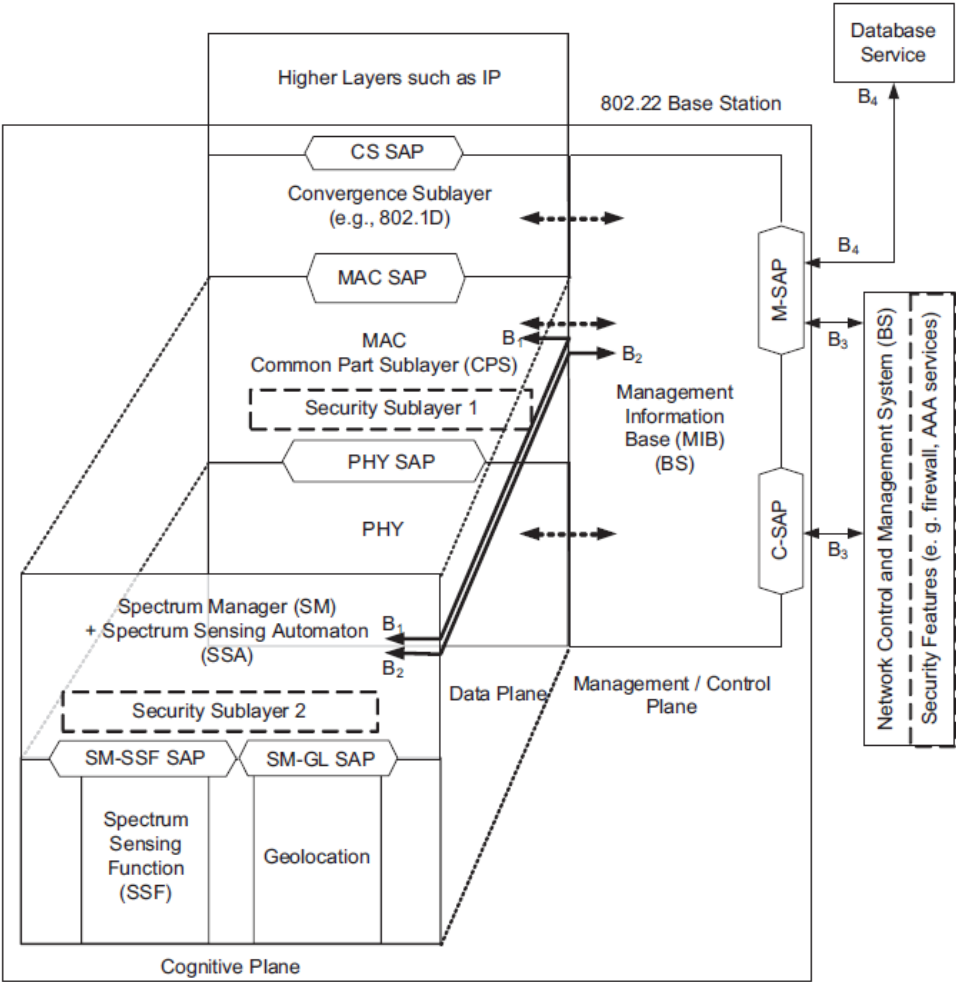


Figure B.15—IEEE 802.22 RM for the BS and CPE

1 **B.6.1 Data plane**

2 The service-specific CS provides the transformation or mapping of external network data that is received
3 through the CS SAP into MSDUs and data that is received by the MAC CPS through the MAC SAP.
4 Multiple CS specifications are provided for interfacing with various protocols.

5 The MAC CPS provides the core MAC functionality of system access, connection establishment, and
6 connection maintenance. The data that the MAC layer receives from the various CSs through the MAC SAP
7 is classified according to the particular MAC connections.

8 The security sublayer 1 provides mechanisms for authentication, secure key exchange, encryption, etc.

9 Data, PHY control, and radio statistics are transferred between the MAC CPS and the PHY via the PHY
10 SAP.

11 **B.6.2 Management/control plane**

12 The management/control plane contains the MIB. SNMP is used to communicate with the MIB database,
13 and some of its primitives can be used to manage the network entities, e.g., BS, customer-premises
14 equipment (CPE), bridges, routers. The MIB at the CPE is a subset of MIB at the BS.

15 **B.6.3 Cognitive plane**

16 The SM maintains spectrum availability information, manages channel lists, manages quiet periods
17 scheduling, implements self-coexistence mechanisms, and processes requests from the MAC/PHY. The SM
18 is the central point at the BS where all the information on the spectrum availability resulting from the
19 database service and the spectrum sensing function (SSF) is gathered. Based on this combined information,
20 local regulations, and predefined SM policies, the SM provides the necessary configuration information to
21 the BS MAC, which then remotely configures all the registered CPEs. Connection B2 is used to configure
22 the SM at the BS, to transmit the available television channel list to the SM, and to report the RF
23 environment information via the MIB objects. Connection B1 is used by the SM to initiate a channel move,
24 to configure the SSA at the CPE (e.g., backup/candidate channel list) and to gather information from the
25 CPEs (e.g., local sensing information, local geolocation information).

26 The spectrum sensing automaton (SSA) is present at the BS and at the CPEs and independently implements
27 specific procedures for sensing the RF environment at initialization of the BS and before the registration of a
28 CPE with the BS. The SSA at the CPE also includes features to allow proper operation when the CPE is not
29 under the control of a BS. At any other time, the SSA at the CPE is under the control of the SM. The SSA at
30 the BS is also active when the BS is not transmitting to conduct out-of-band sensing. The SSA located at the
31 BS can also carry out sensing to clear channels when the BS is not transmitting.

32 The SSF implements spectrum sensing algorithms while the geolocation module provides the information to
33 determine the location of the IEEE 802.22 end station (BS or CPE).

34 The role of the security sublayer 2 is to provide enhanced protection to the incumbents as well as necessary
35 protection to the IEEE 802.22 stations.

¹ **Annex C**

² (informative)

³ **Examples of bit ordering for addresses**

⁴ This annex was deleted during the development of IEEE Std 802-202x.

¹ Annex D

² (informative)

³ List of IEEE 802 standards

⁴ This annex contains a list of approved IEEE 802 standards. The list was current when this standard was
⁵ completed.

⁶ IEEE Std 802.1AB™, IEEE Standard for Local and metropolitan area networks—Station and Medium
⁷ Access Control Connectivity Discovery.^{36, 37}

⁸ IEEE Std 802.1AC™, IEEE Standard for Local and metropolitan area networks—Media Access Control
⁹ (MAC) Service Definition.

¹⁰ IEEE Std 802.1AE™, IEEE Standard for Local and metropolitan area networks: Media Access Control
¹¹ (MAC) Security.

¹² IEEE Std 802.1AR™, IEEE Standard for Local and metropolitan area networks—Secure Device Identity.

¹³ IEEE Std 802.1AS™, IEEE Standard for Local and metropolitan area networks—Timing and
¹⁴ Synchronization for Time-Sensitive Applications in Bridged Local Area Networks.

¹⁵ IEEE Std 802.1AX™, IEEE Standard for Local and metropolitan area networks—Link Aggregation.

¹⁶ IEEE Std 802.1BA™, IEEE Standard for Local and metropolitan area networks—Audio Video Bridging
¹⁷ (AVB) Systems.

¹⁸ IEEE Std 802.1BR™, IEEE Standard for Local and metropolitan area networks—Virtual Bridged Local
¹⁹ Area Networks – Bridge Port Extension.

²⁰ IEEE Std 802.1CB™, IEEE Standard for Local and metropolitan area networks—Frame Replication and
²¹ Elimination for Reliability.

²² IEEE Std 802.1CF™, IEEE Recommended Practice for Network Reference Model and Functional
²³ Description of IEEE 802® Access Network

²⁴ IEEE Std 802.1CM™, IEEE Standard for Local and metropolitan area networks—Time-Sensitive
²⁵ Networking for Fronthaul.

²⁶ IEEE Std 802.1CS™, IEEE Standard for Local and metropolitan area networks—Link-local Registration
²⁷ Protocol.

²⁸ IEEE Std 802.1Q™, IEEE Standard for Local and metropolitan area networks—Bridges and Bridged
²⁹ Networks.

³⁰ IEEE Std 802.1X™, IEEE Standard for Local and metropolitan area networks—Port-Based Network
³¹ Access Control.

³⁶The IEEE standards and products referred to in Annex D are trademarks owned by The Institute of Electrical and Electronics Engineers, Incorporated.

³⁷IEEE publications are available from The Institute of Electrical and Electronics Engineers (<https://standards.ieee.org/>).

- 1 IEEE Std 802.3™, IEEE Standard for Ethernet.
- 2 IEEE Std 802.3.1™, IEEE Standard for Management Information Base (MIB) Definitions for Ethernet.
- 3 IEEE Std 802.3.2™, IEEE Standard for Ethernet - YANG Data Model Definitions.
- 4 IEEE Std 802.11™, IEEE Standard for Information technology—Telecommunications and information
5 exchange between systems—Local and metropolitan area networks—Specific Requirements—Part 11:
6 Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications.
- 7 IEEE Std 802.15.3™, IEEE Standard for High Data Rate Wireless Multimedia Networks.
- 8 IEEE Std 802.15.4™, IEEE Standard for Low-Rate Wireless Networks.
- 9 IEEE Std 802.15.6™, IEEE Standard for Local and metropolitan area networks—Part 15.6: Wireless Body
10 Area Networks.
- 11 IEEE Std 802.15.7™, IEEE Standard for Local and metropolitan area networks—Part 15.7: Short-Range
12 Optical Wireless Communications.
- 13 IEEE 802.15.8™, IEEE Standard for Wireless Medium Access Control (MAC) and Physical Layer (PHY)
14 Specifications for Peer Aware Communications (PAC).
- 15 IEEE 802.15.9™, IEEE Standard for Transport of Key Management Protocol (KMP) Datagrams.
- 16 IEEE 802.15.10™, IEEE Recommended Practice for Routing Packets in IEEE 802.15.4 Dynamically
17 Changing Wireless Networks.
- 18 IEEE 802.15.22.3™, IEEE Standard for Spectrum Characterization and Occupancy Sensing.
- 19 IEEE Std 802.16™, IEEE Standard for Air Interface for Broadband Wireless Access Systems.
- 20 IEEE 802.19.1™, IEEE Standard for Information technology--Telecommunications and information
21 exchange between systems--Local and metropolitan area networks--Specific requirements--Part 19:
22 Wireless Network Coexistence Methods.
- 23 IEEE 802.19.3™, IEEE Recommended Practice for Local and Metropolitan Area Networks--Part 19:
24 Coexistence Methods for IEEE 802.11 and IEEE 802.15.4 Based Systems Operating in the Sub-1 GHz
25 Frequency Bands
- 26 IEEE Std 802.21™, IEEE Standard for Local and metropolitan area networks—Media Independent Services
27 Framework.
- 28 IEEE Std 802.21.1™, IEEE Standard for Local and metropolitan area networks--Part 21.1: Media
29 Independent Services.
- 30 IEEE Std 802.22™, IEEE Standard for Information Technology—Telecommunications and information
31 exchange between systems Wireless Regional Area Networks (WRAN)—Specific requirements Part 22:
32 Cognitive Wireless RAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications:
33 Policies and Procedures for Operation in the TV Bands.

1 Annex E

2 (informative)

3 History

4 E.1 Universal addresses

5 The universal administration of MAC addresses began with the Xerox Corporation administering Block
6 Identifiers (Block IDs) for Ethernet addresses. Block IDs, subsequently referred to as OUI by the IEEE RA,
7 were assigned by the Ethernet Administration Office. The Block IDs were 24 bits in length, and an
8 organization developed addresses by assigning the remaining 24 bits. For example, the address as
9 represented by the 6 octets P-Q-R-S-T-U comprises the Block ID, P-Q-R, and the locally assigned octets S-
10 T-U.

11 The IEEE RA, because of the work in IEEE 802 on standardizing networking technologies, assumed the
12 responsibility of defining and carrying out procedures for the universal administration of these addresses.
13 The IEEE RA has also been designated by ISO/IEC to act as a registration authority for the ISO/IEC 8802
14 series of standards. The responsibility for defining the procedures is discharged by the IEEE Registration
15 Authority Committee, which is a standing committee of the IEEE Standards Association Board of
16 Governors.

17 The IEEE RA has the responsibility for the administration of universal addresses. There are agreements and
18 processes in place for IEEE 802 standards to also become ISO/IEC standards. Most of these standards are in
19 the ISO/IEC 8802 series. The IEEE was designated by ISO/IEC to act as the registration authority for the
20 ISO/IEC 8802 series of standards. IEEE established the IEEE Registration Authority to manage the
21 administration of universal addresses (as well as other registries specified in IEEE standards. When the
22 IEEE Standards Association was created, 1998, responsibility for management of the IEEE RA was
23 assigned to the IEEE SA. The IEEE SA Board of Governors in turn delegates responsibility for technical
24 oversight of the IEEE RA to a subcommittee, the IEEE Registration Authority Committee (RAC).

25 E.2 IEEE RA address block products

26 When the IEEE RA took over administration of universal addresses, blocks of addresses were allocated by
27 assigning an OUI to companies and organizations that requested them. When the Internet began to grow
28 exponentially, it seemed as if the currently allocated address space using 24-bit OUIs would run out quickly.
29 The IEEE RA addressed one part of this concern by introducing 64-bit addressing and recommending this
30 addressing scheme for new standards that did not require 48-bit addressing for backwards compatibility.

31 In addition, the IEEE RA looked for ways to make the original OUI space last longer. Many times, a
32 company or organization would be allocated an OUI, but would not use a significant portion of the 2^{24}
33 (16 777 216) EUI-48 addresses or 2^{40} (1 099 511 627 776) EUI-64 addresses available in the address block.
34 The addresses would be “lost”, never being assigned. To avoid this situation, the IEEE RA created the OUI-
35 36, which could be used as an identifier as well as for creating universal addresses (up to 4096 EUI-48s or
36 268 435 456 EUI-64s).

37 Based on customer requests, beginning on January 1, 2014, the IEEE RA added a 28-bit identifier (MA-M)
38 and renamed the products to be MA-L (24 bits, previously OUI), MA-M (28 bits), and MA-S (36 bits, also
39 referred to as OUI-36). The MA-L assignment includes the assignment of an OUI, whereas the MA-M and
40 MA-S do not. The MA-M assignment is derived from an OUI that is assigned to IEEE.

1 The MA-S assignment is derived from an OUI that is assigned to IEEE and encompasses both the Individual
2 Address Block and the OUI-36 assignments offered prior to January 1, 2014. An MA-S assignment includes
3 an OUI-36 that is specified in some standards for identification of a company or organization and used in
4 creation of extended identifiers.

5 **E.3 Local MAC addresses**

6 Local addresses were included in the initial series of IEEE 802 standards (published in 1985). IEEE Std
7 802.5-1985 included an annex for hierarchical addressing (dividing an address into a ring number and a
8 node number both being locally administered). Pre-standard Ethernet did not include local addresses. A few
9 Block IDs assigned by Xerox Corporation prior to the definition of local addresses (circa 1981) had the X bit
10 equal to one. The vendors assigned such Block IDs participated in remediation of problems created by the
11 definition of local addresses for all IEEE 802 networks.

12 The amendment IEEE Std 802c-2017 (now included in IEEE Std 802) on local MAC address usage
13 introduced the SLAP, ELI, AAI, and SAI. Prior to that amendment, IEEE Std 802 provided little normative
14 content regarding the use of local MAC address space beyond the description of the U/L bit. A brief
15 subclause on local MAC addresses was introduced in the revision IEEE Std 802-2014, stating that local
16 MAC addresses “need to be unique on a LAN or bridged LAN unless the bridges support VLANs with
17 independent learning.” That revision also introduced the Company ID (CID), referring to the IEEE RA for
18 details. It did not specify the creation of local MAC addresses based on CID, but it did hint at the possibility,
19 stating that “A CID assignment has the X bit (the U/L address bit in a MAC address) set to one, which would
20 place any address created with a CID in the locally administered address space.”

21 The IEEE RA opened its CID registry on 1 January 2014. Later that year, the IEEE introduced an expanded
22 version of its tutorial “Guidelines for use of the 24-bit Organizationally Unique Identifiers (OUI)” as
23 “Guidelines for Use Organizationally Unique Identifier (OUI) and Company ID (CID)”³⁸, including
24 explanatory material regarding CIDs. Using language similar to that of IEEE Std 802-2014, the tutorial also
25 suggested the possibility of building a local MAC address from a CID.

26 All CIDs publicly listed by the IEEE RA are assigned with the Y and Z bits equal to zero and one,
27 respectively. Local MAC addresses based on CIDs are in SLAP quadrant “01”, in accordance with the
28 specification of the ELI, 8.4.4.1.

³⁸The tutorial has been updated again to include EUIs, [B2]

1 Annex F

2 (informative)

3 EtherType Listing Subset

4 F.1 Introduction

5 This Annex lists the subset of EtherType assignments described in 9.2.2 in tabular form, Table F.1, and in
6 the form of a YANG module, F.3. This subset is provided solely for the convenience of the users of this
7 standard and does not constitute an endorsement by IEEE of the listed protocols.

8 F.2 Tabular format

9 A subset of EtherType assignments by the IEEE RA is given in Table F.1. Each Friendly Name in Table F.1
10 is unique and is used as an identifier in the YANG module. The Short Description identifies the protocol,
11 protocol message, or protocol field that uses the assignment as specified in the Reference, or the EtherType
12 assignment itself as named in the Reference. Where the Reference specifies more than one name or use
13 (distinguished for example by sub-type) these are included in the Short Description field.

14 NOTE—The fields “Friendly Name” and “Short Description” in Table F.1 may include trademarks that are owned by
15 their respective trademark owners. The information in these fields is provided solely for the convenience of users of this
16 standard and does not constitute an endorsement by IEEE of those products or the companies producing those products.

Table F.1—EtherType listing subset*

EtherType Assignment (HEX)	Friendly Name	Short Description	Reference
08-00	ipv4	Internet Protocol version 4 (IPv4)	IETF RFC 894
08-06	arp	Address Resolution Protocol (ARP)	IETF RFC 826, IETF RFC 7042
08-42	wol	Wake-on-LAN	IEEE Std 802
22-E2	misp	MAC Status Protocol (MSP)	IEEE Std 802.1Q
22-E7	cnm	Congestion Notification Message (CNM)	IEEE Std 802.1Q
22-E9	cn-tag	Congestion Notification Tag (CN-TAG)	IEEE Std 802.1Q
22-EA	msrp	Multiple Stream Reservation Protocol (MSRP)	IEEE Std 802.1Q
22-F3	trill	Transparent Interconnection of Lots of Links	IETF RFC 6325
60-03	decnet	DECnet DNA Routing	DECnet DIGITAL Network Architecture - Ethernet Data Link Architectural Specification v1.0.0

Table F.1—EtherType listing subset* (continued)

EtherType Assignment (HEX)	Friendly Name	Short Description	Reference
80-35	rarp	Reverse Address Resolution Protocol	IETF RFC 903
80-9B	appletalk	Appletalk (Ethernalk)	Inside Appletalk, Second Edition
80-F3	aarp	Appletalk Address Resolution Protocol	Inside Appletalk, Second Edition
81-00	c-tag	Customer VLAN Tag (C-TAG)	IEEE Std 802.1Q
81-37	ipx	Internetwork Packet Exchange (IPX)	Internetwork Packet Exchange - Novell, Inc.
82-04	qnx	QNX Qnet	QNX - Quantum Software Systems, Ltd.
86-DD	ipv6	Internet Protocol Version 6 (IPv6)	IETF RFC 2464
88-08	efc	Multipoint Control Protocol (MPCP)	IEEE Std 802.3
88-09	esp	Ethernet Slow Protocol	IEEE Std 802.3
88-19	cobranet	CobraNet CobraNet	Programmer's Reference, Version 2.5
88-47	mpls-unicast	Multiprotocol Label Switching (MPLS) unicast traffic	IETF RFC 3031
88-48	mpls-multicast	Multiprotocol Label Switching (MPLS) multicast	IETF RFC 3031
88-63	pppoe-discovery	Point-to-Point Protocol over Ethernet (PPPoE) Discovery Stage	IETF RFC 2516
88-64	pppoe-session	Point-to-Point Protocol over Ethernet (PPPoE) Session Stage	IETF RFC 2516
88-6D	intel-ans	Intel Advanced Networking Services Probe Packets	Intel® Advanced Network Services (Intel® ANS) Advanced Settings for Teams
88-70	llc-encaps	LLC Encapsulation	IEEE Std 802.1AC
88-7B	homeplug	Homeplug	INT51X1 datasheet
88-8E	eap	Port Access Entity (PAE) EtherType, Extensible Authentication Protocol over LANs (EAPOL)	IEEE Std 802.1X
88-92	profinet	PROFINET	IEC 61158-6-10
88-9A	hyperscsi	Small Computer System Interface (SCSI) over Ethernet.	An Ethernet Based Data Storage Protocol for Home Network
88-A2	aoe	Advanced Technology Attachment (ATA) over Ethernet	ATA over Ethernet (AoE)

Table F.1—EtherType listing subset* (continued)

EtherType Assignment (HEX)	Friendly Name	Short Description	Reference
88-A4	ethercat	Ethernet for Control Automation Technology (EtherCAT)	IEC 61158-4-12
88-A8	s-tag	Service VLAN Tag (S-TAG) or Backbone VLAN Tag (B-TAG)	IEEE Std 802.1Q
88-AB	ethernet-powerlink	Ethernet Powerlink	IEC 61158-4-13
88-B5	exp1	Local experimental EtherType 1	IEEE Std 802
88-B6	exp2	Local experimental EtherType 2	IEEE Std 802
88-B7	oui-ext	OUI Extended EtherType	IEEE Std 802
88-B8	goose	IEC 61850 Generic Object Oriented Substation Event (GOOSE)	IEC 61850-8-1
88-B9	gse	IEC 61850 Generic Substation Events (GSE) management services	IEC 61850-8-1
88-BA	sv	IEC 61850 Sampled Value Transmission (SV)	IEC 61850-8-2
88-C7	pre-auth	RSNA Pre-Authentication	IEEE Std 802.11
88-CC	lldp	Link Layer Discovery Protocol (LLDP)	IEEE Std 802.1AB
88-CD	sercos	Sercos Interface	IEC 61158-4-19
88-DC	wsmpp	WAVE Short Message Protocol (WSMP)	IEEE Std 1609
88-E1	homeplug-av-mme	HomePlug AV Mobile Management Entity (MME)	HomePlug AV Specification
88-E3	mrp	Media Redundancy Protocol	IEC 62439-2
88-E5	macsec	MACsec EtherType	IEEE Std 802.1AE
88-E7	i-tag	Backbone Service Instance Tag	IEEE Std 802.1Q
88-F5	mvrp	Multiple VLAN Registration Protocol (MVRP)	IEEE Std 802.1Q
88-F6	mmrp	Multiple MAC Registration Protocol (MMRP)	IEEE Std 802.1Q
88-F7	ptp	Precision Time Protocol	IEEE Std 1588
89-02	cfm	IEEE 802.1Q Connectivity Fault Management (CFM) PDU Encapsulation EtherType	IEEE Std 802.1Q
89-06	fcoe	Fibre Channel over Ethernet (FCoE)	T11 FC-BB-5
89-0D	wlan-mgmt	802.11 Management Protocol	IEEE Std 802.11

Table F.1—EtherType listing subset* (continued)

EtherType Assignment (HEX)	Friendly Name	Short Description	Reference
89-10	encap	Backbone Service Encapsulated Addresses	IEEE Std 802.1Q
89-14	fip	FCoE Initialization Protocol	T11 FC-BB-5
89-15	roce	Remote Direct Memory Access (RDMA) over Converged Ethernet (RoCEv1)	InfiniBand™ Architecture Specification
89-17	mis	Media Independent Service (MIS) Protocol	IEEE Std 802.21
89-1D	tte	TTEthernet Protocol Control Frame (TTE)	SAE AS6802
89-29	mirp	Multiple I-SID Registration Protocol (MIRP)	IEEE Std 802.1Q
89-2F	hsr	High-availability Seamless Redundancy (HSR)	IEC 62439-3
89-3F	e-tag	Bridge Port Extension Tag (E-TAG)	IEEE Std 802.1BR
89-40	ecp	Edge Control Protocol	IEEE Std 802.1Q
89-4B	f-tag	Flow Filtering Tag (F-TAG) IEEE	Std 802.1Q
89-52	drpc	Distributed Relay Control Protocol (DRCP)	IEEE Std 802.1AX
89-A2	cim	Congestion Isolation Message (CIM)	IEEE Std 802.1Q
C9-D1	llc-legacy	LLC Encapsulation (obsolete)	IEEE Std 802.1AC
F1-C1	r-tag	Frame Replication and Elimination for Reliability (FRER) Redundancy Tag (R-TAG)	IEEE Std 802.1CB

*Hexadecimal values in the Assignment field are provided from the public listing, while the information in the other fields (i.e., Friendly Name, Short Description, and Reference) is specified herein.

1 F.3 YANG module for EtherType subset

2 F.3.1 YANG Framework

3 The YANG module representation of the EtherType subset (as specified in Table F.1) is provided in this
4 subclause.

5 Changes to the ieee802-ethertypes.yang module, adding or revising entries, are made by amending or
6 revising this standard and will add a new revision statement to the module. YANG augmentation should not
7 be used to extend the module.

8 NOTE — The ietf-ethertypes.yang module (as specified in rfc8519) is currently used by the ietf-packet-fields.yang mod-
9 ule (as specified in rfc8519) and the ietf-detnet.yang module. Moving forward it is anticipated that the YANG module

1 (ieee802-ethertype.yang) specified in F.3.2 will supersede ietf-ethertypes.yang, which would result in ietf-ether-
2 types.yang being deprecated.

3 F.3.2 Definition for ieee802-ethertype YANG module^{39,40}

```
4 module ieee802-ethertype {  
5  
6   namespace "urn:ieee:std:802.1Q:yang:ieee802-ethertype";  
7   prefix "ieee-ethertype";  
8  
9   organization  
10      "IEEE 802.1 Working Group";  
11  
12   contact  
13      "WG-URL: http://ieee802.org/1/  
14      WG-EMail: stds-802-1@ieee.org  
15  
16      Contact: IEEE 802.1 Working Group Chair  
17      Postal: C/O IEEE 802.1 Working Group  
18              IEEE Standards Association  
19              445 Hoes Lane  
20              Piscataway  
21              NJ 08854  
22              USA  
23  
24      E-mail: stds-802-1-chairs@ieee.org";  
25  
26   description  
27      "This module contains a subset of commonly used 802 network Ether-  
28 Types.  
29  
30      Copyright (C) IEEE (2023).  
31  
32      This version of this YANG module is part of the IEEE Std 802;  
33      see the standard itself for full legal notices.";  
34  
35   revision "2023-04-17" {  
36     description  
37        "Initial revision.";  
38     reference  
39        "IEEE Std 802f, Overview and Architecture -  
40        YANG Data Model for EtherTypes";  
41   }  
42  
43   typedef ethertype {  
44     type enumeration {  
45       enum ipv4 {  
46         value 2048;  
47         description
```

³⁹Copyright release for YANG: Users of this standard may freely reproduce the YANG modules contained in this standard so that they can be used for their intended purpose.

⁴⁰An ASCII version of the YANG module is attached to the PDF of this standard and can also be obtained from the IEEE 802.1 website at <https://1.ieee802.org/yang-modules/>.


```
1         "08-00 Internet Protocol version 4 (IPv4)";
2     reference
3         "Organization: Xerox, US
4         Reference: IETF RFC 894";
5     }
6     enum arp {
7         value 2054;
8         description
9             "08-06 Address Resolution Protocol (ARP)";
10    reference
11        "Organization: Symbolics, Inc.
12        Reference: IETF RFC 826, IETF RFC 7042";
13    }
14    enum wol {
15        value 2114;
16        description
17            "08-42 Wake-on-LAN";
18        reference
19            "Organization: None
20            Reference: IEEE Std 802";
21    }
22    enum msp {
23        value 8930;
24        description
25            "22-E2 MAC Status Protocol (MSP)";
26    reference
27        "Organization: IEEE 802.1 Working Group
28        Reference: IEEE Std 802.1Q";
29    }
30    enum cnm {
31        value 8935;
32        description
33            "22-E7 Congestion Notification Message (CNM)";
34    reference
35        "Organization: IEEE 802.1 Working Group
36        Reference: IEEE Std 802.1Q";
37    }
38    enum cn-tag {
39        value 8937;
40        description
41            "22-E9 Congestion Notification Tag (CN-TAG)";
42    reference
43        "Organization: IEEE 802.1 Working Group
44        Reference: IEEE Std 802.1Q";
45    }
46    enum msrp {
47        value 8938;
48        description
49            "22-EA Multiple Stream Reservation Protocol (MSRP)";
50    reference
51        "Organization: IEEE 802.1 Working Group
52        Reference: IEEE Std 802.1Q";
53    }
54    enum trill {
```

```
1      value 8947;
2      description
3          "22-F3 Transparent Interconnection of Lots of Links";
4      reference
5          "Organization: IETF TRILL Working Group
6          Reference: IETF RFC 6325";
7  }
8  enum decnet {
9      value 24579;
10     description
11         "60-03 DECnet DNA Routing";
12     reference
13         "Organization: DEC
14         Reference: DECnet DIGITAL Network Architecture - Ethernet
15         Data Link Architectural Specification v1.0.0";
16 }
17 enum rarp {
18     value 32821;
19     description
20         "80-35 Reverse Address Resolution Protocol";
21     reference
22         "Organization: Private
23         Reference: IETF RFC 903";
24 }
25 enum appletalk {
26     value 32923;
27     description
28         "80-9B Appletalk (Ethertalk)";
29     reference
30         "Organization: Private
31         jReference: Inside Appletalk, Second Edition";
32 }
33 enum aarp {
34     value 33011;
35     description
36 "      80-F3 Appletalk Address Resolution Protocol";
37     reference
38         "Organization: Private
39         Reference: Inside Appletalk, Second Edition";
40 }
41 enum c-tag {
42     value 33024;
43     description
44         "81-00 Customer VLAN Tag (C-TAG)";
45     reference
46         "Organization: IEEE 802.1 Working Group
47         Reference: IEEE Std 802.1Q";
48 }
49 enum ipx {
50     value 33079;
51     description
52         "81-37 Internetwork Packet Exchange (IPX)";
53     reference
54         "Organization: Novell, Inc.
```

```
1      Reference: Internetwork Packet Exchange - Novell, Inc.";
2  }
3  enum qnx {
4      value 33284;
5      description
6          "82-04 QNX Qnet";
7      reference
8          "Organization: Quantum Software Systems, Ltd.
9          Reference: QNX - Quantum Software Systems, Ltd.";
10 }
11 enum ipv6 {
12     value 34525;
13     description
14         "86-DD Internet Protocol Version 6 (IPv6)";
15     reference
16         "Organization: USC/ISI
17         Reference: IETF RFC 2464";
18 }
19 enum efc {
20     value 34824;
21     description
22         "88-08 Multipoint Control Protocol (MPCP)";
23     reference
24         "Organization: IEEE 802.3 Working Group
25         Reference: IEEE Std 802.3";
26 }
27 enum esp {
28     value 34825;
29     description
30         "88-09 Ethernet Slow Protocol";
31     reference
32         "Organization: IEEE 802.3 Working Group
33         Reference: IEEE Std 802.3";
34 }
35 enum cobranet {
36     value 34841;
37     description
38         "88-19 CobraNet";
39     reference
40         "Organization: Peak Audio
41         Reference: CobraNet Programmer's Reference, Version 2.5";
42 }
43 enum mpls-unicast {
44     value 34887;
45     description
46         "88-47 Multiprotocol Label Switching (MPLS) unicast
47         traffic";
48     reference
49         "Organization: Cisco Systems
50         Reference: IETF RFC 3031";
51 }
52 enum mpls-multicast {
53     value 34888;
54     description
```

```
1      "88-48 Multiprotocol Label Switching (MPLS) multicast";
2  reference
3      "Organization: Cisco Systems
4      Reference: IETF RFC 3031";
5  }
6  enum pppoe-discovery {
7      value 34915;
8      description
9          "88-63 Point-to-Point Protocol over Ethernet (PPPoE)
10         Discovery Stage";
11     reference
12         "Organization: UUNET Technologies, Inc.
13         Reference: IETF RFC 2516";
14 }
15 enum pppoe-session {
16     value 34916;
17     description
18         "88-64 Point-to-Point Protocol over Ethernet (PPPoE)
19         Session Stage";
20     reference
21         "Organization: UUNET Technologies, Inc.
22         Reference: IETF RFC 2516";
23 }
24 enum intel-ans {
25     value 34925;
26     description
27         "88-6D Intel Advanced Networking Services Probe Packets";
28     reference
29         "Organization: Intel Corporation
30         Reference: Intel® Advanced Network Services (Intel® ANS)
31     }
32 enum llc-encaps {
33     value 34928;
34     description
35         "88-70 LLC Encapsulation";
36     reference
37         "Organization: IEEE 802.1 Working Group
38         Reference: IEEE Std 802.1AC";
39 }
40 enum homeplug {
41     value 34939;
42     description
43         "88-7B Homeplug";
44     reference
45         "Organization: Intellon Corporation
46         Reference: INT51X1 datasheet";
47 }
48 enum eapol {
49     value 34958;
50     description
51         "88-8E Port Access Entity (PAE) EtherType, Extensible
52         Authentication Protocol over LANs (EAPOL)";
53     reference
54         "Organization: IEEE 802.1 Working Group
```

```
1      Reference: IEEE Std 802.1X";
2  }
3  enum profinet {
4      value 34962;
5      description
6          "88-92 PROFINET";
7      reference
8          "Organization: PROFIBUS International
9          Reference: IEC 61158-6-10";
10 }
11 enum hyperscsi {
12     value 34970;
13     description
14         "88-9A Small Computer System Interface (SCSI) over
15         Ethernet.";
16     reference
17         "Organization: Data Storage Institute
18         Reference: An Ethernet Based Data Storage Protocol for Home
19         Network";
20 }
21 enum aoe {
22     value 34978;
23     description
24         "88-A2 Advanced Technology Attachment (ATA) over Ethernet.";
25     reference
26         "Organization: Coraid Inc
27         Reference: AoE (ATA over Ethernet)";
28 }
29 enum ethercat {
30     value 34980;
31     description
32         "88-A4 Ethernet for Control Automation Technology
33         (EtherCAT)";
34     reference
35         "Organization: Beckhoff Automation GmbH & Co KG
36         Reference: IEC 61158-4-12";
37 }
38 enum s-tag {
39     value 34984;
40     description
41         "88-A8 Service VLAN Tag (S-TAG) or Backbone VLAN Tag
42         (B-TAG)";
43     reference
44         "Organization: IEEE 802.1 Working Group
45         Reference: IEEE Std 802.1Q";
46 }
47 enum ethernet-powerlink {
48     value 34987;
49     description
50         "88-AB Ethernet Powerlink";
51     reference
52         "Organization: Ethernet Powerlink Standardization Group
53         (EPSG)
54         Reference: IEC 61158-4-13";
```

```
1   }
2   enum exp1 {
3       value 34997;
4       description
5           "88-B5 Local experimental EtherType 1";
6       reference
7           "Organization: IEEE 802.1 Working Group
8           Reference: IEEE Std 802";
9   }
10  enum exp2 {
11      value 34998;
12      description
13          "88-B6 Local experimental EtherType 2";
14      reference
15          "Organization: IEEE 802.1 Working Group
16          Reference: IEEE Std 802";
17  }
18  enum oui-ext {
19      value 34999;
20      description
21          "88-B7 OUI Extended EtherType";
22      reference
23          "Organization: IEEE 802.1 Working Group
24          Reference: IEEE Std 802";
25  }
26  enum goose {
27      value 35000;
28      description
29          "88-B8 IEC 61850 Generic Object Oriented Substation Event
30          (GOOSE)";
31      reference
32          "Organization: IEC TC57
33          Reference: IEC 61850-8-1";
34  }
35  enum gse {
36      value 35001;
37      description
38          "88-B9 IEC 61850 Generic Substation Events (GSE) management
39          services";
40      reference
41          "Organization: IEC TC57
42          Reference: IEC 61850-8-1";
43  }
44  enum sv {
45      value 35002;
46      description
47          "88-BA IEC 61850 Sampled Value Transmission (SV)";
48      reference
49          "Organization: IEC TC57
50          Reference: IEC 61850-8-2";
51  }
52  enum pre-auth {
53      value 35015;
54      description
```

```
1      "88-C7 RSNA Pre-Authentication";
2      reference
3      "Organization: IEEE 802.11 Working Group
4      Reference: IEEE Std 802.11";
5  }
6  enum lldp {
7      value 35020;
8      description
9      "88-CC Link Layer Discovery Protocol (LLDP)";
10     reference
11     "Organization: IEEE 802.1 Working Group
12     Reference: IEEE Std 802.1AB";
13 }
14 enum sercos {
15     value 35021;
16     description
17     "88-CD Sercos Interface";
18     reference
19     "Organization: sercos international e.V.
20     Reference: IEC 61158-4-19";
21 }
22 enum wsmpp {
23     value 35036;
24     description
25     "88-DC WAVE Short Message Protocol (WSMP)";
26     reference
27     "Organization: IEEE P1609 WG
28     Reference: IEEE Std 1609";
29 }
30 enum homeplug-av-mme {
31     value 35041;
32     description
33     "88-E1 HomePlug AV Mobile Management Entity (MME)";
34     reference
35     "Organization: HomePlug Powerline Alliance, Inc.
36     Reference: HomePlug AV Specification";
37 }
38 enum mrp {
39     value 35043;
40     description
41     "88-E3 Media Redundancy Protocol";
42     reference
43     "Organization: Siemens AG
44     Reference: IEC 62439-2";
45 }
46 enum macsec {
47     value 35045;
48     description
49     "88-E5 MACsec EtherType";
50     reference
51     "Organization: IEEE 802 LAN/MAN Standards Committee
52     Reference: IEEE Std 802.1AE";
53 }
54 enum i-tag {
```

```
1      value 35047;
2      description
3          "88-E7 Backbone Service Instance Tag";
4      reference
5          "Organization: IEEE 802.1 Working Group
6          Reference: IEEE Std 802.1Q";
7  }
8  enum mvrp {
9      value 35061;
10     description
11         "88-F5 Multiple VLAN Registration Protocol (MVRP)";
12     reference
13         "Organization: IEEE 802.1 Working Group
14         Reference: IEEE Std 802.1Q";
15 }
16 enum mmrp {
17     value 35062;
18     description
19         "88-F6 Multiple MAC Registration Protocol (MMRP)";
20     reference
21         "Organization: IEEE 802.1 Working Group
22         Reference: IEEE Std 802.1Q";
23 }
24 enum ptp {
25     value 35063;
26     description
27         "88-F7 Precision Time Protocol";
28     reference
29         "Organization: IEEE I&M Society TC9
30         Reference: IEEE Std 1588";
31 }
32 enum cfm {
33     value 35074;
34     description
35         "89-02 IEEE 802.1Q Connectivity Fault Management (CFM) PDU
36         Encapsulation EtherType";
37     reference
38         "Organization: IEEE 802.1 Working Group
39         Reference: IEEE 802.1Q";
40 }
41 enum fcoe {
42     value 35078;
43     description
44         "89-06 Fibre Channel over Ethernet (FCoE)";
45     reference
46         "Organization: Cisco Systems, Inc
47         Reference: T11 FC-BB-5";
48 }
49 enum wlan-mgmt {
50     value 35085;
51     description
52         "89-0D 802.11 Management Protocol";
53     reference
54         "Organization: IEEE 802.11 Working Group
```



```
1         Reference: IEEE Std 802.11";
2     }
3     enum encap {
4         value 35088;
5         description
6             "89-10 Backbone Service Encapsulated Addresses";
7         reference
8             "Organization: IEEE 802.1 Working Group
9             Reference: IEEE Std 802.1Q";
10    }
11    enum fip {
12        value 35092;
13        description
14            "89-14 FCoE Initialization Protocol";
15        reference
16            "Organization: Brocade Communications Systems LLC
17            Reference: T11 FC-BB-5";
18    }
19    enum roce {
20        value 35093;
21        description
22            "89-15 Remote Direct Memory Access (RDMA) over Converged
23            Ethernet (RoCE)";
24        reference
25            "Organization: Mellanox Technologies, Inc.
26            Reference: IBTA Specification";
27    }
28    enum mis {
29        value 35095;
30        description
31            "89-17 Media Independent Service (MIS) Protocol";
32        reference
33            "Organization: IEEE 802.21 Working Group
34            Reference: IEEE Std 802.21";
35    }
36    enum tte {
37        value 35101;
38        description
39            "89-1D Time-Triggered Ethernet (TTE) Protocol Control
40            Frame";
41        reference
42            "Organization: TTTech Computertechnik AG
43            Reference: SAE AS6802";
44    }
45    enum mirp {
46        value 35113;
47        description
48            "89-29 Multiple I-SID Registration Protocol (MIRP)";
49        reference
50            "Organization: IEEE 802.1 Working Group
51            Reference: IEEE Std 802.1Q";
52    }
53    enum hsr {
54        value 35119;
```

```
1      description
2      "89-2F High-availability Seamless Redundancy (HSR)";
3      reference
4      "Organization: International Electrotechnical Commission
5      Reference: IEC 62439-3";
6  }
7  enum e-tag {
8      value 35135;
9      description
10     "89-3F Bridge Port Extension Tag (E-TAG)";
11     reference
12     "Organization: IEEE 802.1 Working Group
13     Reference: IEEE Std 802.1BR";
14 }
15 enum ecp {
16     value 35136;
17     description
18     "89-40 Edge Control Protocol";
19     reference
20     "Organization: IEEE 802.1 Working Group
21     Reference: IEEE Std 802.1Q";
22 }
23 enum f-tag {
24     value 35147;
25     description
26     "89-4B Flow Filtering Tag (F-TAG)";
27     reference
28     "Organization: IEEE 802.1 Working Group
29     Reference: IEEE Std 802.1Q";
30 }
31 enum drcp {
32     value 35154;
33     description
34     "89-52 Distributed Relay Control Protocol (DRCP)";
35     reference
36     "Organization: IEEE 802.1 Working Group
37     Reference: IEEE Std 802.1AX";
38 }
39 enum cim {
40     value 35234;
41     description
42     "89-A2 Congestion Isolation Message (CIM)";
43     reference
44     "Organization: IEEE 802.1 Working Group
45     Reference: IEEE Std 802.1Q";
46 }
47 enum llc-legacy {
48     value 51665;
49     description
50     "C9-D1 LLC Encapsulation (obsolete)";
51     reference
52     "Organization: IEEE 802.1 Working Group
53     Reference: IEEE Std 802.1AC";
54 }
```

```
1      enum mpp {
2          value 57915;
3          description
4              "E2-3B MAC Privacy protection Protocol";
5          reference
6              "Organization:
7              Reference: IEEE Std 802.1AE";
8      }
9      enum r-tag {
10         value 61889;
11         description
12             "F1-C1 Frame Replication and Elimination for Reliability
13             (FRER) Redundancy Tag (R-TAG)";
14         reference
15             "Organization: IEEE 802.1 Working Group
16             Reference: IEEE Std 802.1CB";
17     }
18 }
19 description
20     "IEEE Std 802 EtherTypes subset.";
21 }
22
23 }
24
```

¹ **Annex G**

² (informative)

³ **Wake-on-LAN**

⁴ Wake-on-LAN (WoL) is a common protocol to wake up devices from a very low power mode. It can be
⁵ implemented over IEEE 802 networks as a MAC frame using the EtherType 08-42. WoL is not standardized
⁶ in an IEEE 802 standard.